



EVIDENCE ROOM

AP Agent OS — Professional

Sixteen agents. One payment boundary. The default paid SKU.

Proof before permission.

The operating system for the AP agent layer. Specify, charter, govern, and measure sixteen named roles without replacing the stack you already run.

Hard hold. ERP remains the system of record for accounting. Named humans remain the system of record for cash. Vendor-bank change and close attestation stay human. Duplicate & Anomaly flags are hypotheses, not fraud verdicts. This book is a method, not hosted software.

How to use this book

Buy Friday. Begin Monday. Do not charter from memory.

1. Score the free diagnostic (36 questions). Honour vetoes.
2. Map one path (usually PO-goods). Code exceptions from the closed list.
3. Charter Wave 1 at L0 or L1: Intake, Validation, Matching, Triage, Duplicate & Anomaly, Orchestrator.
4. Write the payment sentence on the wall. Agent 12 annotates; humans release.
5. Shadow before any write path. Promote only with an evidence pack.

Every page should answer at least one of: What should I do? How? Who owns it? What can go wrong? How do I control it? How do I measure it? What evidence proves it works?

Northline Industrials is a *fictional* manufacturer used so tables can be concrete (~15,000 invoices/month, Dynamics 365, Cleveland SSC). Figures are illustrative, not a client result.

Customer-facing statistics are limited to the research ledger. Do not paste vendor cost-per-invoice figures into a business case as targets.

START HERE — 12 steps

Product: Evidence Room — AP Agent OS

Time: One working day if you already know last quarter's volumes; two if you must gather facts.

Output: A named Wave 1, named owners, a payment sentence, and a shadow-mode plan.

Not an output: A savings target, a fraud programme, a compliance certificate, or permission to pay.

Do these in order. Skipping to a favourite agent is how undocumented workforces get built.

Before step 1

Confirm which licence you hold. Use only the folders you are licensed to use. If you have the Free Diagnostic only, stop after step 4 and use the starter language in 01_FREE_AP_AI_READINESS/BUSINESS_CASE_STARTER.md to decide whether to licence Starter or Professional.

Have at hand: last-quarter invoice volume, exception mix (even if messy), the last payment-run procedure, the names of the AP Manager, Controls Lead, and whoever signs the payment file.

Do not upload production invoices, vendor bank details, or employee records to Evidence Room. This toolkit does not need them.

Step 1 — Read the contract of the product (15 minutes)

Read 00_READ_ME/README.md and the short disclaimer in 10_LEGAL_AND_LICENSING/DISCLAIMER.md (or the on-page short form if you only have a commercial pack).

Write this sentence at the top of your working note:

We are designing an agent layer across the AP stack we already run. Payment authorisation stays human. Proof before permission.

If a sponsor wants that sentence rewritten into a guarantee, stop. This product will not write the guarantee.

Step 2 — Diagnose the layer you have (45–60 minutes)

Open 01_FREE_AP_AI_READINESS/DIAGNOSTIC_GUIDE.md.

Answer all 36 questions across six dimensions: process, data, controls, technology, talent, economics. Score each 0–4 using the rubric in that file. Use evidence you can reopen (a report, a SOP, a ticket), not recollection.

Compute dimension averages. Do **not** let a high technology score cancel a 0 on payment control or vendor-bank dual control.

Place the organisation on the L1–L5 maturity model in 01_FREE_AP_AI_READINESS/MATURITY_MODEL.md.

Step 3 — Write the one-sentence band (10 minutes)

Use this form. Fill the blanks. Do not add a savings clause.

We are at **L[n]** — **[name]**. Agents are **[absent / informal prompts / named but ungated / chartered at L0–L1 / evidence-gated]**. Payment release is **[written as human / assumed / unclear]**. First wave is **[six names]** at **L0/L1**.

Northline's fictional Friday sentence: *We are at L2 — Informal Activity. Agents are informal prompts in capture and a matching copilot. Payment release is written as human. First wave is Intake, Validation, Matching, Triage, Duplicate & Anomaly, Orchestrator at L0/L1.*

Step 4 — Name the human holds (20 minutes)

Copy the human-held list from 02_AP_AGENT_STARTER/STARTER_GUIDE.md (or Professional: 03_AP_AGENT_OS_PRO/Agent_Library/HUMAN_VS_AGENT_DECISION_FRAMEWORK.md).

Tick each item that is already true in your policy **and** in the system. A policy that says dual approval while a demo script auto-releases is not a hold. It is a gap.

Minimum holds that must be true before any agent is live:

1. Payment authorisation and bank-file transmission — human.
2. Vendor bank-detail create or change — human dual control.
3. New vendor approval — human.
4. DOA and tolerance-table edits — human.
5. Period-end attestation — human.
6. Clearing a high-value duplicate or anomaly flag — named control owner.

If any of 1–3 is false, Wave 1 is still allowed at L0 only, and Agent 12 is not opened.

Step 5 — Classify ten exceptions and one payment run (70 minutes)

Print or export:

- the ten most recent parked or rejected invoices
- the last payment proposal (or equivalent)

For each line write: decision class (cash / master data / reversible accounting / documentary / routing / external commitment / judgement / investigation signal / attestation / design change), modifiers (value, new vendor, open flag, tax, multi-entity), who decided, and what an agent would have been allowed to do at L1.

Record, in one column, what a vendor demo would have auto-done. Those cells are the control conversation, not a shopping list.

Starter has a table for this. Professional has the full framework and worked Northline examples.

Step 6 — Charter Wave 1 only (90 minutes)

Charter six agents, no more:

ID	Agent	Default level	Typical owner
01	Invoice Intake	L1 (or L0 if extract quality is unknown)	AP Operations Lead
02	Invoice Validation	L1	AP Quality Lead
03	Matching	L1	AP Match Lead
04	Exception Triage	L1	Exception Desk Lead
10	Duplicate & Anomaly	L0	AP Controls Lead
16	Orchestrator	L1	AP Process Owner

Use the charter template in Starter or 03_AP_AGENT_OS_PRO/Agent_Library/AGENT_CHARTER_TEMPLATE.md. For each agent name: object types, entities, evidence required, exclusions, failure detection, rollback.

Do not charter Agents 12 or 13 above L1. Do not discuss L3.

If you cannot name a human owner, leave the agent unchartered.

Step 7 — Freeze the evidence standard (20 minutes)

A recommendation is eligible for a human decision only when the packet contains:

- source (image, XML, EDI, or portal artefact)
- extract or equivalent structured fields
- match worksheet or coded fail reason
- owner and SLA clock
- autonomy context
- flag outcome if Agent 10 fired
- link the Orchestrator can reopen

Incomplete packets do not travel. Perfect packets still do not pay.

Write this list into the Orchestrator charter. That is the product mechanic for **proof before permission**.

Step 8 — Stand a two-week L0 window (15 minutes to schedule; two weeks to run)

Shadow only. Output is not the system of record. Operators work as they do today. The agent writes a parallel record.

Minimum L0 evidence before anyone talks about L1 as the system of record:

- coverage on the in-scope path
- required fields present
- a dual-review sample of disagreements
- zero silent scope expansion
- a cost ledger started (licence + model + exception minutes), even if the numbers are small
- language check: no fraud, compliance, accuracy-as-property, or savings claims in agent output

Gates are in Professional: AUTONOMY_PROGRESSION.md. Starter users: stay at shadow until those gates are written down locally.

Step 9 — Put cost and quality on one page (30 minutes)

Do not build an ROI slide.

Open 01_FREE_AP_AI_READINESS/BUSINESS_CASE_STARTER.md and, if licensed, 03_AP_AGENT_OS_PRO/Business_Case/COST_PER_INVOICE_MODEL.md.

Record:

- invoices per month (your number)
- fully loaded AP cost you already use internally
- exception rate you can defend
- current cost-per-invoice **range** if you have one; otherwise leave blank
- estimated layer cost (tools + implementation + model) as a **budget**, not a payback

Industry figures, when used, must be labelled as vendor-originated citations of independent research. See the business-case files. Do not invent a midpoint and call it yours.

Step 10 — Schedule governance before go-live (20 minutes)

Book four meetings now:

When	Meeting	Purpose
Week 0	Kickoff with AP, Controls, Transformation	Confirm Wave 1, owners, payment sentence
Weekly	Exception mix (30 minutes)	Codes, aging by owner, language slips
Day 30	Control review	Internal Audit observes packets; no operator role
Day 90	Recertify or demote	Autonomy register; retire what is expensive and wrong

Team licence holders: use 04_AP_AGENT_OS_TEAM/STEERING_COMMITTEE_TEMPLATES.md.

Step 11 — Write the Monday note (20 minutes)

Send one page. Suggested subject: *AP agent layer — Wave 1 at observe/recommend*.

Include: maturity band, six agent names, six owners, the payment sentence, the two-week shadow plan, the 30-day review, and one refusal (“we will not auto-release payments”).

Do not include a savings range. Do not include a fraud-detection claim. Do not include Northline’s numbers as if they were yours.

Executive templates: 04_AP_AGENT_OS_TEAM/EXECUTIVE_COMMUNICATION_TEMPLATES.md (Team) or the short form in 03_AP_AGENT_OS_PRO/Business_Case/BOARD_CFO_SUMMARY.md (Professional).

Step 12 — Stop conditions (read every week)

Stop commissioning, and demote or pause, if any of these is true:

- A payment was released on an agent recommendation without the human path.
- Vendor bank details were changed from an agent pack without dual control.
- Agent 10 output used the word “fraud” in an operator-facing ticket.
- An agent processed object types outside its charter.

- Owners are vacant and someone “just kept it running.”
- A vendor or integrator proposed skipping evidence gates to hit a date.

Stopping is a successful control outcome.

After the 12 steps

If this is true	Do this
Diagnostic only	Decide whether Starter or Professional is the next licence
Starter, Wave 1 named	Run the two-week shadow; buy Professional before promoting
Professional	Open the sixteen specs; stay on Wave 1 until gates pass
Team	Run the workshop in 04_AP_AGENT_OS_TEAM/WORKSHOP_AGENDA.md
Stack is unusual or political	Apply for Custom Blueprint (05_CUSTOM_BLUEPRINT/)

Document control

Field	Value
Toolkit	Evidence Room — AP Agent OS
Object	12-step START HERE
Companion	README.md, diagnostic, Starter guide
Status	Edition 1.0.0

AP Agent OS — 16-Agent Stack Overview

Product: Evidence Room — AP Agent OS (digital toolkit, not software)

Audience: Finance Transformation lead, AP Manager, Controller, Internal Audit, IT/ERP owner

Use: Specify, charter, and govern a portfolio of AP agents. This document is the architectural brief. Individual agent files are the working specifications.

Principle: Agents earn responsibility. The default is never full autonomy. Payment authorisation always remains human.

1. What this stack is — and is not

AP Agent OS is an operating design for accounts payable work. It defines sixteen named roles, the handoffs between them, the evidence each must produce, and the human decisions that never leave the control environment.

It is not an ERP module, a bot catalogue, or a promise that invoices will post themselves. It does not replace the general ledger, the bank, the delegation of authority, or the people who sign payment runs.

A Finance Transformation lead should be able to take this overview on Monday, assign owners, and open the individual agent charters the same week. Implementation may use any combination of ERP workflow, RPA, document capture, iPaaS, and human procedures. The architecture is ERP-agnostic: SAP, Oracle, Dynamics 365, NetSuite, and Workday all map to the same roles.

2. Design principles

1. **Observe before you act.** Every agent starts at L0 (Observe) or L1 (Recommend). Promotion is evidence-gated. See `AUTONOMY_PROGRESSION.md`.
 2. **Prepare is not execute.** L2 means a complete packet sits with a named human. Nothing posts, pays, or changes master data at L2.
 3. **Guardrails are written, not implied.** L3 is limited to pre-approved, low-risk actions with hard stops. L4 is exception-based oversight of that same bounded set — not a new class of work.
 4. **Payment authorisation is not an agent job.** The Payment Proposal Review Agent may annotate a proposal. It may not release, approve, or transmit payment.
 5. **Duplicate & Anomaly is a signal, not a verdict.** Flags are hypotheses. They do not constitute fraud detection, and they do not authorise a hold or a release on their own.
 6. **One human owner per agent.** Accountability does not sit with “the SSC” or “Finance.” It sits with a named role.
 7. **Exclusions are as important as duties.** If an action is not listed, the agent does not do it.
 8. **Cost is a control.** Token, licence, and exception-handling cost are monitored. An agent that is expensive and wrong is retired or demoted, not scaled.
-

3. The five-stage autonomy model

Level	Name	Agent may	Agent may not	Human role
L0	Observe	Read, classify, log, produce shadow reports	Recommend a specific action as the system of record	Reviews reports; no reliance
L1	Recommend	Propose an action with rationale and evidence	Change ERP state; contact counterparties as the company	Accepts, edits, or rejects each recommendation
L2	Prepare	Assemble a complete, ready-to-act packet; draft communications	Execute the action; send external mail without approval	Approves the packet, then a human or downstream control executes
L3	Execute within guardrails	Perform a named, pre-approved, low-risk action inside written limits	Act outside the limit table; touch payment release; change bank details	Samples, exception review, limit maintenance
L4	Managed autonomy	Same bounded actions as L3, with exception-only human review	Expand its own scope; authorise payment; change policy	Oversight by exception; periodic recertification

Default for a new agent is **L0 or L1**. No agent is commissioned at L3 or L4. Payment-related actions have a hard ceiling: recommendation and preparation only.

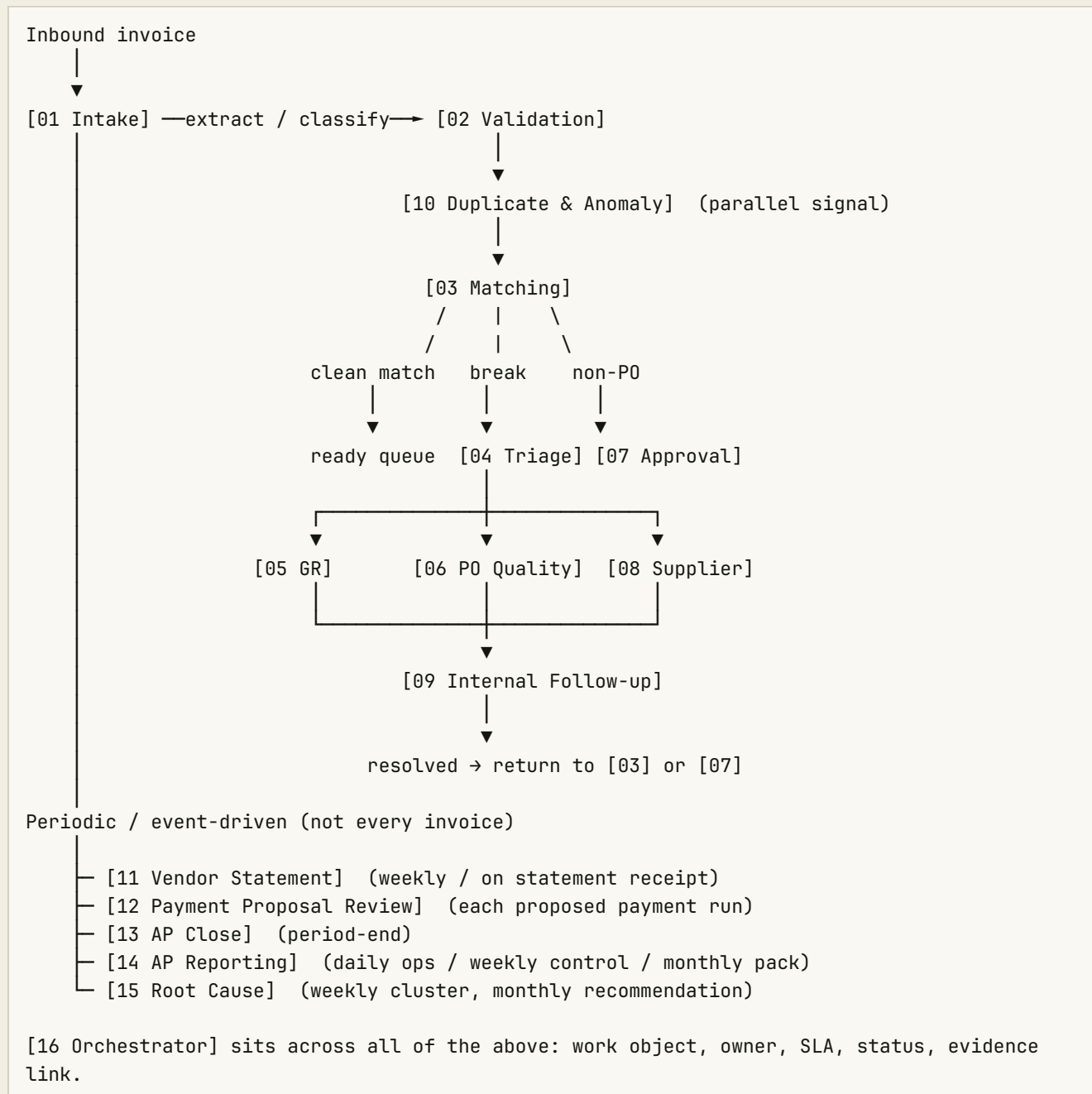
4. The sixteen agents

ID	Agent	Primary job	Default autonomy	Typical human owner
01	Invoice Intake	Capture, classify, extract, and register inbound invoices	L1	AP Operations Lead
02	Invoice Validation	Test completeness, supplier identity, tax fields, and posting readiness	L1	AP Quality Lead
03	Matching	Apply 2-way / 3-way / GR-IR match logic and tolerance	L1	AP Match Lead
04	Exception Triage	Classify breaks, assign owner, set next action and SLA	L1	Exception Desk Lead
05	Goods Receipt	Detect missing or aged receipts and assemble receiver packets	L1	Plant / Warehouse Finance Liaison
06	PO Quality	Detect PO defects that will become AP exceptions	L0	Procurement Operations Lead
07	Approval	Route non-PO and over-tolerance items per DOA; chase packets	L1	AP Approvals Coordinator
08	Supplier Resolution	Draft supplier queries; track replies; never commit the company	L1	Vendor Master / AP Supplier Desk
09	Internal Follow-up	Chase unresponsive internal owners with a complete fact pack	L1	Exception Desk Lead
10	Duplicate & Anomaly	Flag possible duplicates and unusual patterns for human review	L0	AP Controls Lead
11	Vendor Statement	Reconcile supplier statements to the AP subledger	L1	AP Reconciliations Lead
12	Payment Proposal Review	Annotate payment proposals; never authorise payment	L1	Treasury / AP Payments Lead
13	AP Close	Drive the period-end checklist and accrual candidate list	L1	Assistant Controller — Payables
14	AP Reporting	Produce operational and control reports from defined sources	L1	AP Analytics / Controller's office
15	Root Cause	Cluster exceptions and propose process fixes	L0	Finance Transformation Lead
16	Orchestrator	Sequence work, enforce SLAs, and record handoffs	L1	AP Process Owner

Each agent has a full specification in this folder. Do not charter an agent from this table alone.

5. How work moves through the stack

5.1 Invoice lifecycle (operational path)



5.2 Interaction rules

- **Intake does not decide match.** It produces a validated capture record or a capture exception.
- **Validation does not post.** It produces a pass / fail / query against a written checklist.
- **Matching does not invent receipts or POs.** Missing GR or defective PO is handed to Agents 05 and 06.
- **Triage does not resolve.** It names the break type, the owner, the next action, and the clock.
- **Supplier Resolution drafts; humans send or approve sending.** No commercial commitment in an agent draft.
- **Duplicate & Anomaly never releases or pays.** A flag adds a review task. Clearing the flag is human.
- **Payment Proposal Review annotates.** Dual human approvers release the run.

- **Orchestrator routes and records.** It does not post invoices, change vendors, or release cash.
- **Root Cause does not implement process change.** It proposes. Process owners accept or reject.

5.3 Work object

Every invoice, exception, statement line, and payment-proposal line is a **work object** with:

Field	Purpose
Object ID	Stable identifier (invoice, exception, statement item, proposal line)
Source system	ERP company, legal entity, source mailbox, or portal
Current agent	Which agent is accountable for the next action
Human owner	Named role or named person
Status	new / in_review / waiting_internal / waiting_supplier / ready / held / closed
Autonomy context	Level in force for this object type
Evidence pack URI	Link to extraction, match worksheet, correspondence, approval
SLA clock	Start, due, breach
Escalation state	None / L1 / L2 / L3 as defined in the owning agent spec
Cost token	Capture, match, exception-minutes, model-cost estimate

The Orchestrator is the system of record for work-object state. ERP remains the system of record for accounting.

6. RACI across the stack

R = does the work · A = owns the outcome · C = consulted · I = informed

“Agent” in the R column means the agent prepares or recommends; a human still A-owns.

6.1 Invoice-to-match

Activity	Intake	Validation	Matching	Triage	Dup/ Anom	AP Ops Lead	Controller	Internal Audit
Capture & extract	R	C	I	I	I	A	I	I
Completeness / tax field check	C	R	I	I	C	A	C	I
Duplicate / anomaly flag	I	C	C	I	R	A	I	I
2-way / 3-way match	I	C	R	I	C	A	I	I
Tolerance application	I	I	R	I	I	A	C	I
Exception classification	I	C	C	R	C	A	I	I
Post invoice (ERP)	—	—	C*	—	—	R/A	I	I

*Matching may recommend post at L1/L2. Posting at L3 is only for a pre-approved clean-match class after evidence gates. Controller remains A for the posting policy.

6.2 Exception resolution

Activity	GR	PO Quality	Approval	Supplier Res	Internal FU	Exception Lead	Buyer / Receiver	Plant Controller
Missing GR chase pack	R	I	I	I	C	A	C	I
PO defect diagnosis	I	R	I	I	C	A	C	I
DOA routing packet	I	I	R	I	C	A	C	C
Supplier query draft	I	C	I	R	I	A	C	I
Internal chase	C	C	C	I	R	A	R (response)	I
Write-off / concession	—	—	—	C	I	C	C	A

6.3 Periodic control

Activity	Vendor Stmt	Pymt Review	Close	Reporting	Root Cause	Orchestrator	AP Mgr	Treasury	Controller
Statement reconcile	R	I	C	I	C	I	A	I	I
Annotate payment proposal	I	R	I	I	I	I	C	A*	I
Authorise / release payment	—	—	—	—	—	—	C	Human R/A	C
Close checklist & accruals	C	I	R	C	C	I	C	I	A
Ops / control pack	I	I	C	R	C	I	C	I	A
Process-fix proposal	C	C	C	C	R	I	C	I	A
Work routing & SLA	I	I	I	I	I	R	A	I	I

*Treasury (or the named Payments Lead) is accountable for the integrity of the *proposal review*. Authorisation of the run is a separate human control and is never an agent R or A.

7. What stays human — non-negotiable

These decisions are outside every agent's authority, at every autonomy level:

1. **Payment authorisation and release.** Dual (or policy-specified) human approval. Bank file transmission. Positive Pay exception decisions.
2. **Vendor bank-detail create or change.** Any payment-method or beneficiary-account change.
3. **New vendor creation (or its approval).** Agents may assemble a request pack. They do not create the vendor.

4. **Delegation of authority changes.** Limit tables, substitute approvers, and emergency DOA are human policy.
5. **Tolerance and match-policy changes.** Agents apply the table; they do not edit it.
6. **Period-close attestation.** The Controller (or delegate) signs the close. Agent 13 prepares the file.
7. **Write-off, credit, and commercial concession above the written threshold.**
8. **Tax position on ambiguous or cross-border invoices.** Agent 02 flags; Tax decides.
9. **Legal dispute, formal collection, or settlement language.**
10. **Clearing a high-value duplicate or anomaly flag.** Agent 10 raises; a named control owner clears.
11. **Override of a payment hold placed by policy** (audit hold, stop-pay, sanctioned-party review).
12. **Expansion of an agent's own scope or autonomy level.** Promotion follows
AUTONOMY_PROGRESSION.md.

If a vendor, integrator, or internal team proposes that an agent “just release the run if everything looks fine,” the answer is no.

8. Shared services operating picture — Northline Industrials

Northline Industrials is a fictional mid-market manufacturer used throughout this library. Figures are illustrative, not a benchmark and not a promise.

Item	Value
Business	Precision fasteners, fabricated assemblies, industrial MRO components
Legal entities	US (Dayton OH HQ + plant), US (Birmingham AL), Canada (Hamilton ON), Mexico (Monterrey)
ERP	Dynamics 365 Finance; shared services on a single AP instance with four company codes
Volume	15,000 invoices / month (~180,000 / year)
Active vendors	~2,400
Invoice mix	72% PO; 18% non-PO services; 7% utilities / rent / recurring; 3% other (T&E out of scope)
Match policy	3-way for inventory and direct materials; 2-way for contracted services ≤ \$10,000; GR-based consignment
SSC	Cleveland; 18 AP specialists, 3 team leads, AP Manager; plant finance liaisons at four sites
Payment runs	Tuesday and Thursday; dual human approval; Positive Pay
Currencies	USD, CAD, MXN
Tax regimes	US sales/use, Canadian GST/HST, Mexican IVA
Starting baseline (observational)	First-time match 61%; exception rate 28%; average cycle 9.4 days; duplicate-suspect rate 0.8% of invoices

The stack is commissioned in waves, not as sixteen concurrent L3 agents.

Wave 1 (observe and recommend): Agents 01, 02, 03, 04, 10, 16 — L0/L1.

Wave 2 (prepare): Agents 05, 07, 08, 09, 12 — L1, selected object types to L2.

Wave 3 (periodic): Agents 11, 13, 14, 15, 06 — L0/L1.

Wave 4 (earned execute): Only clean-match posting and low-risk reminder send, and only after evidence gates in AUTONOMY_PROGRESSION.md.

Northline does not set a savings target as a success criterion for commissioning. Success is defined as evidence quality, cycle-time visibility, exception-mix clarity, and control completeness. Any later cost discussion is a management judgement, not an agent KPI that implies guaranteed savings.

9. Control environment the stack must respect

Control	How the stack treats it
Delegation of authority	Agent 07 applies the table; does not edit it
Segregation of duties	Intake/match vs. payment release vs. vendor master vs. bank file remain separate human roles
Three-way match policy	Agent 03 applies; Agents 05/06 explain breaks; humans resolve policy exceptions
Vendor master dual control	Agent 08 never writes bank details; change packs go to Vendor Master
Duplicate review	Agent 10 flags; AP Controls Lead clears above threshold
Payment dual approval	Agent 12 annotates; two humans release
Period-end cut-off	Agent 13 lists candidates; Controller attests
Audit trail	Every agent writes an evidence record (see each spec)
Access	Agents use service accounts with least privilege, logged, and reviewed quarterly
Model / prompt change	Treated as a change-controlled configuration, not an informal tweak

This toolkit does not assert that use of the stack constitutes compliance with any standard. It produces evidence that a control owner can present. The control owner decides sufficiency.

10. Data domains the stack consumes

Domain	Typical sources	Consumed by
Invoice image / XML / EDI / PDF	Email, vendor portal, OCR/IDR, cXML, EDI 810	01, 02, 10
Capture extract	IDR / human keying	01, 02, 03, 10
Vendor master	ERP	02, 08, 10, 11, 12
PO / contract / outline agreement	ERP / CLM	03, 06, 07
Goods receipt / service entry	ERP / WMS	03, 05
AP open items / parked documents	ERP	03, 04, 11, 12, 13
Approval log / DOA table	ERP / workflow	07, 12
Payment proposal / payment batch	ERP / treasury	12
Vendor statements	Email / portal	11
Correspondence	Shared mailbox / ticket	08, 09
GL / close calendar	ERP / close manager	13, 14
Exception history	Orchestrator store	04, 15, 16

Comma-separated or multi-ID queries against ERP APIs must follow that ERP's own parameter rules. Do not assume one client wrapper fits every host.

11. What “good” looks like at 90 days (Northline, Wave 1)

These are operating-discipline measures, not financial promises.

Measure	Intent
Every inbound invoice has a work object within the Intake SLA	Completeness of capture
Validation fail reasons are coded, not free-text only	Exception mix is analysable
Match worksheets exist for a defined sample of PO invoices	Matching is inspectable
Duplicate flags have a clear / confirm / escalate outcome recorded	No silent ignore
Orchestrator can show aging by owner, not only by queue	Management visibility
No payment released without two human approvers	Control held
Autonomy register shows every agent at L0 or L1	No silent promotion
Cost per 1,000 invoices (licence + model + exception minutes) is recorded	Cost is visible

If first-time match or cycle time moves, that is an observed outcome. It is not a commissioning target that implies the stack “delivers ROI.”

12. How to use this library on Monday

1. Appoint the AP Process Owner (Orchestrator A) and the AP Controls Lead (Duplicate & Anomaly A).
2. Complete a charter from `AGENT_CHARTER_TEMPLATE.md` for Agents 01, 02, 03, 04, 10, and 16 only.
3. Apply `HUMAN_VS_AGENT_DECISION_FRAMEWORK.md` to ten recent exceptions and one payment run. Record where the human must remain.
4. Stand the agents at L0 for two weeks: shadow reports, no recommendations in the system of record.
5. Promote individually to L1 only when the evidence gates in `AUTONOMY_PROGRESSION.md` are met.
6. Do not open Agents 12 or 13 at anything above L1. Do not discuss L3 for payment-adjacent work.
7. Schedule a 30-day control review with Internal Audit as an observer, not as the operator.

13. File map

File	Contents
<code>00_AGENT_STACK_OVERVIEW.md</code>	This document
<code>AGENT_01_INVOICE_INTAKE.md ... AGENT_16_ORCHESTRATOR.md</code>	Working specifications
<code>AGENT_CHARTER_TEMPLATE.md</code>	Blank charter + completed Matching Agent example
<code>HUMAN_VS_AGENT_DECISION_FRAMEWORK.md</code>	Decision rights, examples, refusal rules
<code>AUTONOMY_PROGRESSION.md</code>	Evidence gates from L0 to L4

14. Language discipline

Do not write, present, or allow vendors to write:

- guaranteed savings, guaranteed cycle-time reduction, or ROI
- fraud detection, fraud prevention, or “the agent will catch fraud”

- compliance as an outcome of using the stack
- accounting accuracy as a property of the agent
- autonomous payment safety
- “touchless AP” as a committed operating state

Write instead: observed rates, sampled accuracy, evidence completeness, hold rates awaiting human decision, and control exceptions.

15. Document control

Field	Value
Toolkit	Evidence Room — AP Agent OS
Object	Agent Library — stack overview
Owner	AP Process Owner (named at commissioning)
Review cadence	Quarterly, or when an agent is promoted
Related	All files in this folder
Fictional operating example	Northline Industrials
Status	Working specification

Autonomy Progression — How an Agent Earns L0 → L4

Product: Evidence Room — AP Agent OS

Audience: AP Process Owner, agent human owners, Controller, Internal Audit, IT

Use: Promote, hold, or demote an agent with written evidence. Informal “it looks good, turn it up” is a control breach.

Principle: Agents earn responsibility. Default is never full autonomy. Payment authorisation has no promotion path.

1. The only legal levels

Level	Name	Live behaviour	Typical time in level before a promotion case may be opened
L0	Observe	Shadow only. Output is not the system of record.	Minimum 10 operating days or 500 in-scope objects, whichever is later
L1	Recommend	Recommendation is visible to the operator. Human accept / edit / reject is mandatory.	Minimum 20 operating days and 1,000 in-scope objects <i>after</i> L1 start
L2	Prepare	Complete packet; human approval required before any execution	Minimum 20 operating days at L2 on the <i>same object class</i>
L3	Execute within guardrails	Named low-risk actions inside a limit table	Recertify every 90 days
L4	Managed autonomy	Same actions as L3; human review is exception-based	Recertify every 60 days; sample remains mandatory

Times are floors, not targets. An agent can remain at L1 indefinitely. That is a successful outcome.

Northline’s Matching Agent (see charter example) commissioned at L1 on 7 April 2026. A promotion case to L2 prepare-for-post was not eligible before 5 May 2026 *and* 1,000 worksheets. L3 was a separate case, US-OH inventory 3-way only, header ≤ \$5,000, no flags.

2. What “earn” means

Promotion is a change-controlled decision. It requires:

1. A signed charter at the current level.
2. An evidence pack that meets the gate table for the *from* → *to* move.
3. A limit table for the new level (object types, amounts, entities, hours of operation).
4. A named detection path for failure (who notices, how, how fast).
5. A rollback plan that can be executed in one business hour.
6. Sign-off: human owner, control reviewer, AP Process Owner. Controller signs any class-C posting promotion. Treasury signs any payment-*adjacent* L2 (annotation remains the ceiling).
7. An autonomy-register entry with version, date, and evidence URI.

Missing any one item means the case is declined. The agent stays at the current level.

3. Hard ceilings — not promotable

These do not appear in any L3/L4 limit table:

Action	Ceiling
Payment authorisation, release, bank-file transmission	Human. Agent 12 ≤ L2 annotate
Vendor bank / payment-method change	Human dual control
New vendor create	Human (pack ≤ L2)
DOA, tolerance, tax-position, match-policy edit	Human
Close attestation	Human (Agent 13 ≤ L2 prepare)
Clear high-value Agent 10 flag	Human
External commercial commitment	Human (Agent 08 ≤ L2 draft)
Write-off / concession above threshold	Human
Scope self-expansion / prompt change that alters behaviour	Change control

If a vendor roadmap shows “L4 payments,” discard that slide.

4. Gate tables

Each gate is binary. Partial credit is not a pass.

4.1 L0 → L1 (Recommend)

Gate	Pass rule	Evidence
G0.1 Coverage	Shadow output exists for ≥ 95% of in-scope objects in the window	Orchestrator coverage report
G0.2 Completeness	Required fields present on ≥ 99% of shadow outputs	Schema validation log
G0.3 Shadow vs human	On a sample of 100 (or 10% if smaller), material disagreement ≤ 8% <i>or</i> disagreements are coded and explained	Dual-review worksheet
G0.4 No silent scope	Zero objects processed that were out of charter	Exception list (must be empty)
G0.5 Failure handling	Every listed failure mode was either observed and handled, or tabletop-tested	Test record
G0.6 Cost meter	Cost ledger is running (even if numbers are small)	Cost extract
G0.7 Language	Outputs contain none of the forbidden claims (fraud, guaranteed savings, compliance, accuracy-as-property)	Sample review
G0.8 Owner	Human owner and backup confirmed active	RACI + out-of-office test

Fail any gate: remain L0. Fix. Restart the clock.

4.2 L1 → L2 (Prepare)

Gate	Pass rule	Evidence
G1.1 Volume	≥ 1,000 in-scope objects <i>or</i> 20 operating days, whichever later	Register
G1.2 Accept quality	Accept rate ≥ 80% and accept-after-edit is coded; reject reasons classified	Orchestrator
G1.3 Material error	Material error rate ≤ 1.0% of accepted items; no cluster ≥ 5 in 5 days	Desk + ERP
G1.4 Packet quality	On sample of 50, a second reviewer could execute from the packet without asking the agent owner a question, ≥ 90%	Independent review
G1.5 Escalation	Escalations fired per charter; no missed mandatory escalate	Escalation log
G1.6 SoD / access	Access review clean; no write privilege beyond charter	IT cert
G1.7 Duplicate interaction	100% of Agent 10 open flags blocked proceed-type recommendations	Join report
G1.8 Rollback drill	Pause-to-L0 drill completed in ≤ 60 minutes	Drill record

4.3 L2 → L3 (Execute within guardrails)

Open this case only for classes C, D, or E in the Decision Framework, and only for a **named subclass** (example: “US-OH 3-way inventory, single PO, header ≤ \$5,000, no Agent 10 flag, tolerance version AP-MAT-004.3”).

Gate	Pass rule	Evidence
G2.1 Stability	G1.2–G1.5 still true for the <i>subclass</i> over the last 20 days	Subclass extract
G2.2 Limit table	Written, versioned, hashed; includes amount, entity, hours, document type, exclusion list	Controlled doc
G2.3 Detection path	Independent report shows each L3 action within T+1 business day	Detection job
G2.4 Sample plan	≥ 10% or 25 items/week (greater) reviewed after the fact for first 90 days	Sample log
G2.5 Reversibility	Documented reversal for every L3 action type; tested	Reversal test
G2.6 No cash path	Written confirmation that L3 cannot reach payment release	Architecture review
G2.7 Controller (class C) or Process Owner (class D/E) sign-off	Signed	Charter vNext
G2.8 Kill switch	Named humans who can disable L3; test passed	Drill

4.4 L3 → L4 (Managed autonomy)

L4 is not new work. It is exception-based oversight of the **same** L3 limit table.

Gate	Pass rule	Evidence
G3.1 Tenure	≥ 90 days at L3 on an unchanged limit table	Register
G3.2 Material error	≤ 0.3% of L3 actions; zero uncontained incidents	Incident log
G3.3 Sample remaining	Sample may reduce to ≥ 5% or 15 items/week (greater), never zero	Plan
G3.4 Exception design	Written list of exceptions that still require human-before (not after)	Limit table vN
G3.5 Recertification calendar	60-day recert booked	Calendar
G3.6 Cost	Cost not rising while quality is falling (3-week paired trend)	Cost + quality
G3.7 Recert of access and prompts	Prompt/version hash unchanged or change-controlled	Hash log

If the limit table needs to grow, that is a **new L3 case**, not an L4 privilege.

5. Demotion and pause (automatic)

Promotion is earned. Demotion is triggered.

Trigger	Immediate level	Who is notified	Restart
Material error cluster ≥ 5 in 5 days	L0	Owner, Process Owner, Controller if class C	New L0 clock
Any cash movement or bank-file action by an agent	Full stop all agents in that path	Treasury, Controller, Process Owner, IT Security	Incident + recert
Agent 10 flag overwritten or auto-cleared	L0 for Agents 03, 10, 12	Controls Lead	Incident
Access found with excess write privilege	Pause	IT Security	Recert access
Accept rate < 60% for 2 weeks (L1)	L0	Owner	Root-cause (Agent 15 may assist)
Packet fail rate > 20% (L2)	L1	Owner	Packet redesign
L3 reversal rate > 1% in a week	L2	Controller	Limit table review
Cost > 2× baseline <i>and</i> quality falling, 3 weeks	Pause	Process Owner	Cost review — this is a brake, not a savings KPI
Owner absent with no backup > 2 business days	Pause new recommendations	Process Owner	Backup named
Prompt / model change without change control	L0	IT + Owner	Treat as new agent

Automatic triggers fire even if someone is “sure it was a one-off.”

6. Evidence pack — standard contents

Folder: /evidence-room/agents/{id}/autonomy/{yyyyMMdd}_{from}_to_{to}/

#	File	Notes
01	charter_current.pdf	Signed
02	charter_proposed.pdf	Redline + proposed level
03	limit_table.md	Object class, amounts, hours, exclusions
04	kpi_extract.csv	Gates Gx.x columns
05	sample_review.xlsx	Dual review, named reviewers
06	failure_mode_test.md	Tabletop or live
07	access_cert.pdf	IT
08	agent10_join.csv	Flag blocking proof (if proceed-type agent)
09	rollback_drill.md	Time to pause
10	detection_path.md	How wrong execution is seen
11	language_review.md	Forbidden-claim scan
12	cost_ledger.csv	Not a savings case
13	decision_memo.md	Approve / decline / defer, named
14	register_entry.json	Written after approval only

Northline Matching L1 → L2 pack (illustrative) was 41 pages plus extracts. The decline reason on the first attempt (28 April 2026) was G1.4: packets omitted tolerance hash on 8 of 50 samples. They resubmitted 19 May 2026 and passed. That delay is correct behaviour.

7. Limit table pattern (required from L2 upward)

```

Limit table: AGENT_03 / NL / US-OH / 3-way inventory
Version: 2026-05-19.1
Hash: (sha256)
Autonomy: L2 prepare-for-post [or L3 post]
In:
  - Company 1000 only
  - Document type: vend invoice PO
  - Item type: inventory / direct material
  - Single PO, single vendor
  - Header amount ≤ 5000 USD
  - Tolerance policy AP-MAT-004.3 (hash ...)
  - Agent 10 flag = none
  - Hours: Mon-Fri 07:00-18:00 ET
  - Currency: USD
Out (never):
  - Payment
  - CA, MX, US-AL
  - Services, freight, non-PO
  - Multi-PO
  - Tax variance
  - Any override
Detection: daily report 'NL.M3.L3.actions' to Match Lead + Controls
Sample: 10% or 25/week
Kill switch: Priya Shah, Elena Ruiz, James Okoye
Rollback: disable job NL.M3.L3; queue falls to L1 recommend

```

If a field is missing, the table is invalid.

8. Recertification

Level	Cadence	What is re-tested	Possible outcomes
L1	Quarterly	Sample disagreement, language, access, owner	Stay / pause
L2	Quarterly	Packet quality, escalations, cost	Stay / L1
L3	90 days	Limit table still true; material error; detection path; sample	Stay / L2 / L0
L4	60 days	All L3 tests + exception list still complete	Stay / L3

Recert is a scheduled gate, not a celebration. Northline books it on the close calendar so it cannot be “forgotten in a busy month.”

9. Multi-entity and multi-ERP promotion

Promotion does not travel.

Dimension	Rule
Legal entity	New charter, new clock
ERP	New access cert, new detection path (D365 ≠ SAP ≠ NetSuite)
Document type	New subclass, new L3 case
Amount band	Raising \$5,000 to \$25,000 is a new L3 case
Model / prompt / vendor tool	New L0 if behaviour can change
SSC location	If owners change, recert L1

Northline’s US-OH L3 clean-match, if earned, does not apply to Hamilton ON. GST/HST and a different plant receipt process make it a different agent instance.

10. Roles in a promotion case

Role	Duty	May not
Human owner	Assemble pack, propose limit table	Approve their own promotion alone
Control reviewer	Challenge gates, sample	Be the owner
AP Process Owner	Approve register entry	Skip Controller on class C
Controller	Sign class C posting promotions	Be asked to “just endorse” without the pack
Treasury	Sign any payment-adjacent L2	Approve L3 payments (does not exist)
IT / ERP owner	Access and kill switch	Grant write “temporarily”
Internal Audit	Observe on request	Operate the agent
Agent 15 Root Cause	Explain clusters that block a gate	Lobby for promotion

11. Worked progression — Northline Matching Agent

Illustrative. Not a promise that any reader's agent will move at this pace.

Date	Event
10 Mar 2026	L0 shadow on US-OH + US-AL PO invoices. No recommendations in D365.
24 Mar 2026	G0.1–G0.8 reviewed. Disagreement 6% on 100-sample, mostly UOM. Remain L0; PO Quality (Agent 06) opened at L0 to study UOM.
7 Apr 2026	L1 charter signed. Specialists see worksheets.
28 Apr 2026	Premature L2 case. Declined on G1.4 (tolerance hash missing).
19 May 2026	L2 prepare-for-post passed for US-OH inventory ≤ \$5,000. Human still posts.
18 Aug 2026	Earliest L3 eligibility <i>if</i> G2.* pass. Limit table frozen 90 days prior.
—	Payment proposal review remains L1. No joint promotion with Agent 12.

What stayed human the entire time: payment release, vendor bank changes, tolerance edits, Agent 10 flag clearance, CAMX invoices, anything > \$5,000 at L2/L3.

12. Cost as a brake, not a business case

Cost monitoring exists so an expensive, low-quality agent is paused. It does not exist to justify the programme.

A promotion memo that leads with “this will save X FTEs / Y% ROI” is out of standard. Rewrite: coverage, accept rate, material error, reversibility, detection path, limit table.

If leadership wants a financial discussion, that discussion happens *outside* the autonomy register, without putting a savings number in the agent KPI pack.

13. Monday checklist — opening a promotion case

1. Confirm the from-level charter is signed and the clock has cleared the floor.
2. Extract KPIs for the *subclass*, not the whole SSC.
3. Run the Agent 10 join (if the agent can recommend proceed).
4. Complete a fresh 50-item (or 100-item at L0 → L1) dual review.
5. Tabletop the kill switch.
6. Hash the limit table.
7. Write the decision memo with a decline option filled in (forces honesty).
8. Book the recert date before you sign.
9. Update Orchestrator so work objects carry the new level.
10. Tell specialists what changed in one page. If you cannot explain the new ceiling in one page, the ceiling is wrong.

14. Document control

Field	Value
Toolkit	Evidence Room — AP Agent OS
Object	Autonomy progression and evidence gates
Review	Quarterly, or after any automatic demotion
Related	Charters, Decision Framework, Agent 16 (register of levels)

Human vs Agent Decision Framework

Product: Evidence Room — AP Agent OS

Audience: AP Manager, Controller, Exception Desk, Treasury, Internal Audit, Finance Transformation

Use: Decide, for a given work object, whether an agent may observe, recommend, prepare, or (rarely) execute — and what must remain human.

Principle: Agents earn responsibility. Default is never full autonomy. Payment authorisation always remains human.

1. Why this document exists

Most AP “automation” failures are decision-rights failures. A tool posts a match it should have held. A draft email becomes a commercial commitment. A duplicate flag is treated as proof of fraud — or is ignored because last week’s flags were noisy. A payment proposal looks clean, so someone lets the bot release it.

This framework is the refusal rule and the assignment rule. Use it when chartering an agent, when promoting autonomy, when an exception feels urgent, and when a vendor demonstrates a feature.

If the framework and a feature conflict, the framework wins.

2. The question to ask of every work object

Work through these in order. Stop at the first “human-only” answer.

1. Does this action move cash, change a beneficiary, or release a payment file?

Human only. Agent may annotate. Stop.

2. Does this action create or change vendor master data that affects payability?

Human only (or human dual control). Agent may assemble a pack. Stop.

3. Does this action change policy, tolerance, DOA, or tax position?

Human only. Stop.

4. Does this action attest a control (close, SOX narrative, audit representation)?

Human only. Agent may prepare the file. Stop.

5. Is the object above the written value, complexity, or jurisdiction threshold for this agent?

Human decides. Agent may prepare. Stop.

6. Is there an open Duplicate & Anomaly flag, sanctions review, stop-pay, or audit hold?

Human clears or maintains the hold. Agent does not recommend proceed. Stop.

7. Is the recommendation reversible without cash movement and without legal commitment?

If no — human only. If yes — continue.

8. Do we have sampled evidence that this agent, on this object type, at this entity, meets the gate for the proposed level?

If no — level is at most L1 recommend, or L0. Stop.

9. Is a named human owner available to accept, edit, or reject within the SLA?

If no — do not raise autonomy to hide the staffing gap. Queue waits. Stop.

10. If the agent is wrong, who notices, how, and how fast?

If the detection path is not written — do not execute. Recommend or prepare only.

3. Decision classes

Class	Examples	Agent ceiling	Human must
A. Cash movement	Payment release, bank file, Positive Pay exception, wire, card, offset	Annotate only (Agent 12 ≤ L2)	Authorise, release, transmit
B. Payability master data	Vendor bank, payment method, withholding, block/unblock for pay	Pack only	Dual-control change
C. Accounting state (reversible, no cash)	Park, hold, match-accept, invoice post on clean PO match	L1 default; L3 only after gates on a named class	Policy ownership; sample; reversal rights
D. Documentary / capture	Extract, classify, register, attach image	L1 default; L3 only for low-risk register after gates	Review low-confidence; own legal archive
E. Routing and chasing	Assign owner, SLA clock, reminder to internal user	L1/L2; L3 only for internal reminders on a whitelist	Own the assignment policy
F. External commitment	Supplier email that agrees price, qty, due date, concession	Draft only (Agent 08 ≤ L2)	Approve and send; no implied authority
G. Judgement / policy exception	Override tolerance, accept short-ship as complete, waive tax variance	Recommend facts only	Decide
H. Investigation signal	Possible duplicate, unusual amount, new billing address	Flag only (Agent 10 default L0)	Investigate, clear, or hold
I. Attestation	Close sign-off, accrual completeness, audit PBC	Prepare only	Sign
J. Design change	New match rule, new agent scope, prompt change that alters behaviour	Propose (Agent 15)	Change control

4. Autonomy mapped to decision classes

Level	Permitted classes (typical)	Forbidden regardless of level
L0 Observe	Shadow reports on C–J	A, B; any live recommendation as system of record
L1 Recommend	C–J as recommendations	A, B execution; sending external mail; clearing H flags
L2 Prepare	Complete packets for C–G, I	Execution; A/B; expanding scope
L3 Execute within guardrails	Named subset of C, D, E only, limit table in the charter	A, B, F send, G decide, H clear, I sign, J implement
L4 Managed autonomy	Same subset as L3, exception-only review	Anything not already in the L3 limit table

There is no path from L4 to payment authorisation. L4 is not a promotion out of the framework. It is a thinning of routine review on a bounded, already-safe class.

5. Value, complexity, and jurisdiction modifiers

Even inside class C or D, raise the human requirement when any modifier is true.

Modifier	Rule at Northline (illustrative)	Effect
Header amount	> \$50,000	Second human on accept; no L3
Header amount	> \$100,000	Plant Controller or VP Finance; Agent 07 packet
Multi-PO or multi-entity	Any	Out of Wave-1 match scope; human / Triage
Cross-border tax	CA/MX invoices, US use-tax judgement	Tax desk; Agent 02 flags only
New vendor < 90 days	Any payment or master change	Vendor Master + Controls
First invoice from a vendor	Extract + validation human review	No L3 intake
Price change vs PO	Any	Buyer decision; Agent 06/08 facts only
Quantity over-receipt	Any	Warehouse + buyer
Duplicate or anomaly flag	Any open flag	Controls Lead; no proceed recommend
Related-party / intercompany	Any	Human accounting
Customs, duty, freight allocation	Any	Trade / cost accounting
Employee or T&E	Any	Out of AP Agent OS
Weekend / emergency pay	Any	Treasury human path; agent off

Write your own modifier table in the charter. Do not copy Northline's dollars if your DOA differs. Do copy the idea: modifiers promote *human* involvement, not agent autonomy.

6. Worked decisions — Northline Industrials

Fifteen thousand invoices a month. Cleveland SSC. Dynamics 365. Use these as calibration, not as folklore.

6.1 Clean 3-way match, \$4,200, Dayton inventory, no flags

- Class C, no modifiers.
- Wave 1: Agent 03 L1 recommends match; specialist accepts; specialist or junior poster posts.
- After gates: this *class* may be proposed for L3 post. Payment of the item still waits for the Tuesday/Thursday human-authorized run.

6.2 Same invoice, Agent 10 open flag “possible duplicate of inv 88321”

- Class H blocks class C.
- Agent 03 must emit RECOMMEND_NO_ACTION_ANOMALY_HOLD.
- Elena Ruiz (Controls Lead) investigates. She may clear or hold.
- The flag is not a fraud finding. Language in the ticket: “possible duplicate — human review.”

6.3 Missing GR, \$18,500, Birmingham, PO complete, warehouse silent 6 days

- Agent 03 codes MISSING_GR. Agent 05 prepares a receiver packet (L2 if chartered).
- Receiver or plant liaison confirms or disputes.
- Agent 09 chases internally if the packet ages.
- No one in the stack creates a GR to “make it match.” Creating a GR is a warehouse/finance human action.

6.4 Service invoice \$8,400, contract on file, 2-way in policy, cost centre owner on leave

- Agent 07 prepares the DOA packet and routes to the deputy in the DOA table.
- If no deputy: wait. Do not invent an approver. Do not let Orchestrator “find someone senior.”
- Agent does not approve.

6.5 Supplier emails “price is \$12.10 not \$11.40, please update the PO”

- Agent 08 may draft a *question*, not an agreement.
- Buyer decides. Agent 06 may record a PO quality defect if the PO was incomplete.
- Agent 03 does not override price.

6.6 Payment proposal Tuesday: 612 items, \$4.1m, two Agent 10 uncleared flags, one vendor with bank change yesterday

- Agent 12 annotates: hold the two flagged items; hold the bank-change vendor pending dual-control evidence.
- Payments Lead + Treasury approver release the rest.
- Agent 12 does not deselect-and-release. Humans deselect. Humans release.

6.7 Vendor statement shows \$27,000 “overdue” that AP does not recognise

- Agent 11 proposes: on-statement-not-in-AP list, with possible matched invoices and unmatched.
- Supplier Resolution drafts a query.
- No payment is added to the proposal because a statement line exists. Statements are not invoices.

6.8 Month-end: GR/IR \$2.3m aged > 60 days

- Agent 13 lists accrual candidates and aged GR/IR.
- Assistant Controller proposes; Controller attests close.
- The agent does not book the accrual.

6.9 Root Cause cluster: 22% of Dayton exceptions are “UOM mismatch EA vs BOX”

- Agent 15 proposes a PO template change and a vendor onboarding checklist item.
- Procurement Operations accepts or rejects.
- No agent writes the template.

6.10 Vendor demo: “our agent auto-releases payments under \$5k if match is clean”

- Framework class A. Refusal.
- Record the refusal in the vendor evaluation file.
- Northline's \$5k is a *DOA* limit for humans, not an agent payment limit.

7. Language that means “human only”

Treat these phrases in tickets, prompts, or vendor copy as stop words. They require a human decision, not more model confidence.

- “just this once”
- “the CFO said to pay”
- “we always override this vendor”

- “it’s probably not a duplicate”
- “update the PO to match the invoice”
- “post a dummy GR”
- “release so we keep the discount”
- “the statement is the invoice”
- “change the bank so it goes through”
- “approve on their behalf, they’re travelling”
- “catch the fraud”
- “we’ll be out of compliance if we don’t”
- “guaranteed savings if we turn on auto-post”

The correct operational response is: hold, prepare a packet, name the human, state the class.

8. Staffing is not an autonomy argument

A missing specialist is a queue problem. It is not evidence that the agent should post, send, or pay.

Pressure	Allowed response	Disallowed response
Month-end volume	Overtime, temporary specialists, narrower in-scope set	Silent L3 on a new document type
Approver on leave	DOA deputy only	Orchestrator picks a convenient manager
Supplier threat to stop ship	Human commercial / procurement decision	Agent agrees price or promises pay date
Discount expiry	Human expedite on a prepared packet	Agent releases payment
Audit week	Complete evidence packs	Hide rejects; backfill worksheets

If the business wants faster cycle time, it funds owners, cleans PO quality, and earns promotion on a *stable* class. It does not skip gates.

9. Duplicate & Anomaly — special rule

Agent 10 produces **signals**. Signals have four human outcomes only:

Outcome	Meaning	Who
Clear — not a duplicate / not anomalous	Proceed under normal match/pay path	Controls Lead or delegate
Confirm — duplicate or anomaly substantiated	Hold, reverse, or recover per procedure	Controls Lead + Match/ Payments
Escalate — needs investigation beyond AP	Fraud examination, legal, or security <i>if</i> those functions are invoked by policy	Named control owner — not the agent
Defer — more data required	Clock recorded; object stays held	Controls Lead

The agent does not choose among these four. It does not use the word “fraud” in user-facing output. It does not claim a detection rate. A quiet month of flags is not evidence that there is no issue. A loud month is not evidence that there is.

10. Payment authorisation — special rule

The payment path has three distinct jobs. Do not collapse them.

Job	Who	Agent?
Build proposal (ERP standard)	Payments specialist / ERP job	No (ERP function)
Review / annotate proposal	Agent 12 + Payments Lead	Agent annotates; human accepts annotations
Authorise and release	Two humans per Northline policy (Payments Lead + Treasury)	Never

Discounts, due-date logic, and “pay this to save a relationship” are inputs to the human authorisers. They are not agent release criteria.

11. RACI for the decision itself

When a work object is at a decision point:

Step	Responsible	Accountable	Consulted	Informed
Classify the decision (A–J)	Owning agent's human owner	AP Process Owner	Controls Lead	Orchestrator (log)
Apply modifiers	Owning agent (recommend)	Human owner	Tax / Treasury / Buyer as relevant	Plant Controller if \$
Choose autonomy action	Charter (already written)	Human owner	Process Owner if charter is silent	Internal Audit if class A/B/I
Execute if L3 class C/ D/E	Agent within limit table	Human owner	Controls (sample)	Process Owner
Record evidence	Agent + human	Human owner	—	Audit on request

If the charter is silent, the action is human-only until the charter is revised under change control.

12. One-page card (print for the SSC wall)

1. Cash, bank, vendor payability → human.
2. Policy, DOA, tax position, close sign-off → human.
3. Open anomaly / stop-pay / sanctions → human.
4. External promise → draft only.
5. No owner, no go.
6. No evidence gate, no promotion.
7. Urgent is not a class.
8. Agent 10 flags are hypotheses.
9. Agent 12 annotates; humans pay.
10. When in doubt, prepare a packet and name a person.

13. How Transformation uses this on Monday

Take the last ten exceptions and the last payment run. For each line, write: class, modifiers, who decided, what the agent would have been allowed to do at the current charter level, and what a vendor demo would have done. The gaps are your control conversation.

Northline’s April 2026 exercise (illustrative): of ten exceptions, six were class G (judgement), two class C (reversible accounting), one class F (supplier price), one class H (possible duplicate). Zero were lawful L3 candidates. The payment run contained two class A decisions that a demo script would have auto-released. The exercise took 70 minutes. It prevented a bad Wave-2 design.

14. Document control

Field	Value
Toolkit	Evidence Room — AP Agent OS
Object	Human vs Agent Decision Framework
Review	Quarterly, or after any attempted payment-adjacent promotion
Related	All agent specs; autonomy progression; charters

Agent Charter Template

Product: Evidence Room — AP Agent OS

Use: One charter per agent, per legal-entity scope (or per SSC if the SSC is the control environment). Complete before the agent is allowed to produce recommendations in the system of record.

Related: 00_AGENT_STACK_OVERVIEW.md, AUTONOMY_PROGRESSION.md, HUMAN_VS_AGENT_DECISION_FRAMEWORK.md, and the agent specification.

This file contains (A) a blank charter and (B) a completed example for the Matching Agent at Northline Industrials. Copy the blank. Do not edit the example in place — it is a teaching artefact.

A. Blank charter

A1. Identification

Field	Value
Agent ID / name	
Specification file	
Legal entities in scope	
ERP company codes / ledgers	
Document types in scope	
Document types out of scope	
Commissioning date	
Charter owner (human, named)	
Deputy	
Control reviewer (not the owner)	
Orchestrator work-queue name	

A2. Purpose (four sentences maximum)

State the job in operational language. Do not state a savings, ROI, fraud, compliance, or accuracy guarantee.

- 1.
- 2.
- 3.
- 4.

A3. Autonomy

Field	Value
Commissioning level	L0 / L1 (circle one; L2+ requires a promotion record)
Allowed object types at this level	
Hard ceiling (level this agent will never exceed without a new charter)	
Payment-adjacent? (Y/N)	If Y, ceiling is L2 prepare / annotate only
Promotion evidence pack location	

A4. Job description (attach or paste)

Paste the “Job description” section from the agent specification and strike any sentence that does not apply to this entity. Add entity-specific sentences below.

Entity-specific additions:

-

A5. Inputs

Input	Source system	Freshness required	Owner of source	Mandatory?
-------	---------------	--------------------	-----------------	------------

A6. Tools / data required

Tool or data set	Access pattern	Privilege	Secret / key owner	Residual risk
------------------	----------------	-----------	--------------------	---------------

A7. Responsibilities (in-scope actions)

Numbered. Each line is an action the agent is allowed to attempt at the commissioning level.

- 1.
- 2.
- 3.

A8. Explicit exclusions

Numbered. If an action is not listed in A7, it is excluded. Still write the dangerous ones explicitly.

1. Payment authorisation or release
2. Vendor bank-detail create or change
- 3.
- 4.

A9. Human owner

Field	Value
Role	
Named person (at commissioning)	
Time-to-cover if absent	
Backup	
Escalation manager	
Decision rights (see framework)	

A10. Approval requirements

Action the agent may propose	Approver	Threshold / condition	Evidence required before approval	SLA for approval
------------------------------	----------	-----------------------	-----------------------------------	------------------

A11. Escalation criteria

Condition	Escalate to	Time	Required packet
-----------	-------------	------	-----------------

A12. Output standard

Output	Format	Destination	Freshness	Naming / ID rule
--------	--------	-------------	-----------	------------------

Minimum fields every output must carry: object ID, agent ID, autonomy level in force, evidence URI, timestamp, human owner, recommendation (if L1+), confidence / exception code.

A13. Control requirements

Control	How this agent supports it	Test / sample
---------	----------------------------	---------------

A14. Audit evidence

Evidence type	Retention	Location	Who can retrieve
---------------	-----------	----------	------------------

A15. KPIs (operating discipline — not financial promises)

KPI	Definition	Source	Cadence	Use
-----	------------	--------	---------	-----

Forbidden as KPIs: guaranteed savings, ROI, “fraud caught,” “compliance rate,” “accounting accuracy.”

A16. Performance history fields

These fields are written every operating period (weekly at L0/L1; daily if volume requires).

Field	Description
Period	
Volume presented	
Volume acted (by type)	
Human accept / edit / reject counts	
Material error count (definition attached)	
Escalations	
SLA breaches	
Cost (licence + model + exception minutes)	
Autonomy level in force	
Incidents	
Demotion / pause events	

A17. Failure handling

Failure mode	Detect	Immediate action	Human notification	Resume rule
Source unavailable				
Extract / model timeout				
Confidence below floor				
Conflicting recommendations				
Suspected duplicate of agent output				
Privilege / access error				

A18. Cost monitoring

Cost element	Meter	Owner who sees it	Pause threshold	Notes
Capture / IDR				
Model / inference				
Connector / iPaaS				
Exception minutes (human)				
Rework after reject				

Pause threshold is an operating brake, not a savings target.

A19. Handoffs

From	To	Trigger	Artefact
------	----	---------	----------

A20. Sign-off

Role	Name	Date	Statement
Charter owner			I own this agent's outcomes and exclusions.
Control reviewer			I have read exclusions, payment ceiling, and evidence plan.
AP Process Owner			This agent is on the stack register at the stated level.
IT / ERP owner			Access listed in A6 is provisioned as least privilege.
Internal Audit (optional observer)			Noted. Observation is not approval of design sufficiency.

Promotion, demotion, or scope change requires a new signed charter version. Informal prompt edits that change behaviour are a change-control breach.

B. Completed example — Matching Agent, Northline Industrials

Teaching artefact. Fictional company. Not a client file. Not a performance promise.

B1. Identification

Field	Value
Agent ID / name	AGENT_03 Matching Agent
Specification file	AGENT_03_MATCHING.md
Legal entities in scope	Northline Industrials US-OH (Dayton), US-AL (Birmingham). CA and MX remain L0-observe only under a separate charter.
ERP company codes / ledgers	US-OH 1000, US-AL 1100; Dynamics 365 Finance, ledger USMF
Document types in scope	PO invoices (3-way inventory / direct; 2-way contracted services ≤ \$10,000)
Document types out of scope	Non-PO, utilities, rent, intercompany, employee expenses, customs entries, consignment until Wave 3
Commissioning date	7 April 2026
Charter owner (human, named)	Priya Shah, AP Match Lead, Cleveland SSC
Deputy	Marcus Bell, AP Operations Lead
Control reviewer (not the owner)	Elena Ruiz, AP Controls Lead
Orchestrator work-queue name	NL.MATCH.US.PO

B2. Purpose

1. Apply Northline's written 2-way and 3-way match rules to in-scope PO invoices after Validation has passed.
2. Produce a match worksheet that a specialist can accept, edit, or reject in one sitting.
3. Hand breaks to Exception Triage with a coded reason; do not invent receipts or change POs.
4. Remain at L1 until the promotion record in the autonomy register says otherwise.

B3. Autonomy

Field	Value
Commissioning level	L1
Allowed object types at this level	PO invoices, companies 1000 and 1100, header amount ≤ \$50,000, single PO, single receipt or open-receipt set already in D365
Hard ceiling	L3 for <i>clean in-tolerance match recommend-to-post only</i> , after gates; never payment; never tolerance edit
Payment-adjacent?	N for match itself; downstream payment remains human
Promotion evidence pack location	//northline/evidence-room/agents/03/autonomy/

B4. Job description (entity-specific)

The Matching Agent compares invoice header and lines to PO and goods-receipt (or service-entry) data in Dynamics 365 Finance using Northline procedure AP-MAT-004. It writes a worksheet: matched / quantity-price-tax break / missing GR / PO defect / routed-out-of-scope. It does not post. It does not create GR. It does not change PO price, quantity, tax, or vendor.

Entity-specific additions:

- Dayton plant uses warehouse-managed receipts; Birmingham uses finance-posted GR. The agent must read both receipt tables.
- Tolerance: quantity $\pm 2\%$ or 1 unit (greater); price $\pm 1\%$ or \$25 (greater); tax variance referred to Tax, not auto-cleared.
- Multi-PO invoices are out of scope at commissioning and route to Triage code MULTI_P0.

B5. Inputs

Input	Source system	Freshness required	Owner of source	Mandatory?
Validated invoice extract	Agent 01/02 work object	Same day as validation pass	AP Quality Lead	Y
PO header and lines	D365 PurchTable / PurchLine	Near-real-time	Procurement Ops	Y
Product receipt / GR	D365 VendPackingSlip / WMS	Near-real-time	Plant warehouse	Y for 3-way
Service entry / timesheet accept	D365 / project	T+0 business day	Project / plant	Y for 2-way services
Tolerance table AP-MAT-004	SharePoint controlled doc + ERP matching setup	Version-pinned	Controller	Y
Duplicate/anomaly flag	Agent 10	Same object	AP Controls Lead	N (if present, hold match recommend)
Vendor master (tax, currency, block)	D365 VendTable	Daily	Vendor Master	Y

B6. Tools / data required

Tool or data set	Access pattern	Privilege	Secret / key owner	Residual risk
D365 OData / custom inquiry	Read PO, GR, invoice pending	Read-only service account svc-ap-match	IT Security	Over-privilege if write is added later
Orchestrator queue API	Read/write work object	Queue write	AP Process Owner	Status spoofing — signed events required
Invoice image store	Read	Read	AP Ops	PII / pricing on image
Tolerance table	Read, version hash	Read	Controller	Stale version — pin hash in every worksheet
IDR confidence payload	Read	Read	AP Quality	Over-trust of extract — Validation must have passed

No write to D365 vendor, PO, or payment tables.

B7. Responsibilities (L1)

1. Retrieve the validated extract and the related PO / GR set.
2. Determine match path: 3-way inventory/direct, 2-way services ≤ \$10,000 with contract reference, or out-of-scope.
3. Apply AP-MAT-004 tolerances. Record the version hash.
4. Produce a worksheet with line-level compare, break codes, and a single recommendation:
RECOMMEND_MATCH, RECOMMEND_HOLD_MISSING_GR, RECOMMEND_ROUTE_PO_QUALITY, RECOMMEND_ROUTE_TRIAGE, RECOMMEND_NO_ACTION_ANOMALY_HOLD.
5. If Agent 10 has an open flag on the object, force RECOMMEND_NO_ACTION_ANOMALY_HOLD.
6. Hand the worksheet to the human match specialist via queue NL.MATCH.US.PO.
7. Write audit evidence (B14).
8. Update Orchestrator status and SLA clock.

B8. Explicit exclusions

1. Payment authorisation or release.
2. Vendor bank-detail create or change.
3. Posting the vendor invoice in D365.
4. Creating or editing a product receipt.
5. Changing PO price, quantity, tax group, vendor, or account assignment.
6. Editing the tolerance table.
7. Clearing an Agent 10 flag.
8. Contacting the supplier.
9. Approving a non-PO invoice.
10. Applying a quantity or price override, even if “the buyer said so” in email — that is a human worksheet edit.
11. Mexican IVA or Canadian GST determination.
12. Intercompany or customs invoices.

B9. Human owner

Field	Value
Role	AP Match Lead
Named person	Priya Shah
Time-to-cover if absent	Next business day
Backup	Marcus Bell
Escalation manager	AP Manager, Cleveland SSC
Decision rights	Accept / edit / reject every L1 worksheet; request promotion evidence; pause the agent

B10. Approval requirements

Action the agent may propose	Approver	Threshold / condition	Evidence required	SLA
RECOMMEND_MATCH	Match specialist (human)	All in-scope; header ≤ \$50,000	Worksheet + image + PO/GR + tolerance hash	Same business day if received by 14:00 ET
RECOMMEND_MATCH	Plant Controller + Match Lead	Header > \$50,000	Same, plus buyer confirmation if price break was manually edited	2 business days
Route to GR / PO Quality / Triage	Exception Desk Lead (auto-route OK at L1)	Any coded break	Break code + facts	Immediate
Any posting (not in force)	n/a	Not authorised at L1	—	—

B11. Escalation criteria

Condition	Escalate to	Time	Required packet
Same invoice fails match 3 times with changing extracts	AP Quality Lead	On third fail	Extract versions
PO and GR exist but tax variance > \$100	Tax desk	Same day	Tax worksheet
Header > \$50,000 with quantity <i>and</i> price break	Plant Controller	Same day	Full worksheet
Agent 10 flag + match would otherwise be clean	AP Controls Lead	Before any human match accept	Flag + worksheet
Service account cannot read GR for > 2 hours	IT + Match Lead	2 hours	Job log
Specialist reject rate > 15% over 200 invoices	AP Process Owner	Weekly review	Reject codes

B12. Output standard

Output	Format	Destination	Freshness	Naming
Match worksheet	JSON + PDF snapshot	Orchestrator + evidence store	On completion	NL-M3-{company}-{invoice}-{yyyyMMddHHmm}
Break handoff	Work object update	Agents 04/05/06	Immediate	Same object ID
Daily shadow register	CSV	Controls Lead	18:00 ET	NL-M3-shadow-yyyyMMdd

Minimum fields: object ID, agent 03, level L1, evidence URI, timestamp, owner Priya Shah or on-duty specialist, recommendation, confidence, tolerance hash, company, invoice, PO, GR IDs, line compares, Agent 10 flag state.

B13. Control requirements

Control	How this agent supports it	Test / sample
3-way match policy	Applies AP-MAT-004; cannot post	Weekly sample 25 worksheets vs ERP
SoD: match vs pay	No payment privilege	Quarterly access review
Tolerance integrity	Version hash on every worksheet	Monthly hash reconcile to controlled doc
Duplicate hold	Cannot recommend match over open flag	100% test of flagged objects
Audit trail	Worksheet + image + user accept log	Retrievable by invoice in < 15 minutes

B14. Audit evidence

Evidence type	Retention	Location	Who can retrieve
Worksheet JSON/PDF	7 years	Evidence store agents/03	Match Lead, Controls, IA
Tolerance hash + policy PDF	7 years	Controlled docs	Controller
Human accept/edit/reject log	7 years	Orchestrator	AP Process Owner
Service-account access log	1 year	SIEM	IT Security
Promotion / pause records	Life of programme	Autonomy register	Process Owner

B15. KPIs

KPI	Definition	Source	Cadence	Use
Worksheet completeness	% of in-scope invoices with a worksheet same day as validation pass	Orchestrator	Daily	Coverage
Human accept rate	Accepts / (accept + edit + reject), by break code	Orchestrator	Weekly	Quality of recommendation
Material error	Human-accepted worksheet later reversed for match error	ERP + desk log	Monthly	Demotion input
Misroute rate	Breaks sent to wrong agent, confirmed by Triage	Orchestrator	Weekly	Handoff quality
SLA on \$50k+ packets	% escalated packets complete on time	Orchestrator	Weekly	Control timeliness
Cost per 1,000 in-scope invoices	Model + connector + specialist minutes	Cost ledger	Monthly	Pause threshold

No savings, ROI, or “accuracy guarantee” KPI.

B16. Performance history fields

Period; volume presented; worksheets produced; accept / edit / reject; material errors; escalations; SLA breaches; cost; level in force (L1); incidents; demotion/pause; Agent 10 holds encountered; multi-PO routed-out count; companies 1000/1100 split.

B17. Failure handling

Failure mode	Detect	Immediate action	Notify	Resume
D365 read fail	Connector health	Queue freeze; no stale match	IT + Match Lead	After successful probe + one sample worksheet
Timeout	Job SLA 120s	Retry once; then fail object to human	On-duty specialist	Object-level only
Confidence / incomplete extract	Validation not pass	Refuse to match; return to 02	Quality Lead if repeat	When 02 passes
Conflicting GR set	>1 plausible GR set	RECOMMEND_ROUTE_TRIAGE code GR_AMBIGUOUS	Exception Desk	n/a
Open Agent 10 flag	Flag state	Force anomaly hold	Controls Lead	After human clear
Access error	401/403	Stop all jobs	IT Security	After recertified privilege

B18. Cost monitoring

Cost element	Meter	Owner who sees it	Pause threshold	Notes
Model / inference	Tokens / invoice	Match Lead + Process Owner	> 2× first-month baseline for 3 weeks <i>and</i> accept rate falling	Pause is a brake
D365 connector	Calls	IT	Sustained error rate > 5%	
Specialist minutes	Time on reject/edit	AP Manager	Edit+reject > 40% for 2 weeks	Agent is costing more than it saves in attention — demote to L0
Rework	Reversals	Controller	Any material error cluster ≥ 5 in a week	Immediate L0

B19. Handoffs

From	To	Trigger	Artefact
Agent 02	Agent 03	Validation pass	Validated extract
Agent 10	Agent 03	Open flag	Flag record (blocks recommend-match)
Agent 03	Human specialist	Worksheet ready	Worksheet
Agent 03	Agent 04	Coded break	Break object
Agent 03	Agent 05	Missing GR	GR chase pack seed
Agent 03	Agent 06	PO defect code	PO defect seed
Human specialist	ERP (human)	Accepted match	Human posts

B20. Sign-off (illustrative)

Role	Name	Date	Statement
Charter owner	Priya Shah	7 Apr 2026	I own this agent's outcomes and exclusions.
Control reviewer	Elena Ruiz	7 Apr 2026	Exclusions, payment ceiling, and evidence plan read.
AP Process Owner	James Okoye	7 Apr 2026	On the stack register at L1.
IT / ERP owner	Chris Nguyen	7 Apr 2026	svc-ap-match is read-only as listed.
Internal Audit (observer)	Samira Haddad	8 Apr 2026	Noted.

C. How to fill this in 90 minutes

1. Open the agent specification. Copy Job description, Exclusions, KPIs, Failure handling.
2. Strike anything not true for this entity (country, document type, ERP table).
3. Name the owner and the control reviewer. They cannot be the same person.
4. Set commissioning level to L0 unless you already have two weeks of shadow evidence.
5. Write the payment sentence even if the agent is not payment-adjacent: "This agent does not authorise payment."
6. Pin the policy document versions (match, DOA, duplicate).
7. List every system access. If write access is requested, stop and get the Process Owner.

8. Sign. File under the autonomy register. Tell Orchestrator the queue name.

A charter that says “the agent will reduce exceptions and catch fraud” is rejected. Rewrite in operating language.

AGENT 01 — Invoice Intake

Stack position: First operational agent. Upstream of Validation, Duplicate & Anomaly, and Matching.
Default autonomy: L1 Recommend (commission at L0).
Human owner (typical): AP Operations Lead
Payment authority: None.
Northline instance: Cleveland SSC, Dynamics 365 Finance, 15,000 invoices/month inbound.

1. Position in the stack

Intake is the front door. It turns inbound artefacts — PDF, image, EDI 810, cXML, portal XML, email body — into a **capture record** attached to an Orchestrator work object. It does not decide whether the invoice is payable, matched, or duplicated. It does not register the accounting document in the ERP unless a later, separate L3 subclass is earned (and even then, only a named low-risk register action).

If Intake is noisy, every downstream agent inherits the noise. If Intake is over-trusted, Validation becomes decorative. Treat capture confidence as a field, not as a posting licence.

2. Job description

The Invoice Intake Agent monitors agreed inbound channels, classifies each artefact (invoice, credit, statement, reminder, junk, duplicate-of-file), extracts header and line fields against a written schema, attaches the original, and opens or updates a work object for Agent 02.

It routes obvious non-invoices to a human miscellaneous queue. It does not invent a vendor if the extract does not resolve. It does not key a document into the ERP at L0/L1. It does not reply to suppliers.

The job is complete when every in-scope artefact from the day’s channels is either (a) a capture record with a work object or (b) a coded intake exception with an owner. Silence — an email that vanished — is a failure.

3. Operating intent and cadence

Mode	Cadence	Output
Channel poll	Every 15 minutes during SSC hours; hourly off-hours	New artefacts listed
Extract	Near-real-time after poll	Capture record
Human low-confidence review	Continuous during SSC hours	Accept / correct / reject extract
Daily completeness	16:30 ET	Channel vs work-object reconcile
Weekly	Friday	Channel mix, exception codes, cost

Northline SSC hours: 07:00–18:00 ET, Monday–Friday. Off-hours poll continues; human review waits for the next shift. Do not auto-register overnight at L1.

4. Inputs

Input	Description	Source	Mandatory
Inbound artefact	File or message	Email, portal, EDI, scan	Y
Channel metadata	From, received time, message ID, envelope	Mail gateway / EDI VAN / portal	Y
Vendor identity hints	VAT/GST/RFC, name, email domain, vendor ID on face	Artefact	N
Vendor master (read)	For <i>suggestion</i> of vendor ID only	ERP	N at extract; Y before Validation pass
Capture schema	Field list, types, legal-entity variants	Controlled doc AP-CAP-001	Y
Prior artefact hash	Detect file-level duplicate of the same PDF	Evidence store	Y
Work object store	Open or create	Orchestrator	Y
Exclusion list	Employee expense, customs broker packs, bank advices	Charter	Y

Header fields typically extracted: vendor name, vendor identifier on face, invoice number, invoice date, due date, currency, header net, tax, gross, PO number(s), ship-to, bill-to, tax registration numbers, payment terms on face, credit-note indicator.

Line fields: line number, description, quantity, UOM, unit price, line net, tax code on face, PO line (if printed), item/SKU (if printed), account assignment hints (do not post from hints).

5. Tools / data required

Tool / data	Purpose	Privilege	Notes
Shared mailbox / graph API	Read invoice@ / country boxes	Read + move-to-folder	No send at L0/L1
Vendor portal retrieve	Pull available invoices	Read	Portal T&Cs are human
EDI 810 / cXML listener	Structured intake	Read	Map version-pinned
OCR / IDR engine	Unstructured extract	Inference	Confidence per field
Hash service	SHA-256 of payload	—	File-level dup of artefact
ERP vendor search	Suggest vendor	Read	Do not create vendor
Evidence store	Image + extract	Write	Retention 7 years typical
Orchestrator	Work object	Write	Status <code>intake_new</code>
Channel inventory	Which pipes are in scope	Read	Controlled list

ERP-agnostic mapping:

ERP	Typical register target (later wave, human or earned L3 only)	Intake still reads
D365 F&O	Pending vendor invoice / document attachment	VendTable, document types
SAP S/4	Parked / incoming invoice (MIRO park) / VIM	LFA1, BKPF attachments
Oracle Fusion	Payables invoice in incomplete	Suppliers, attachments
NetSuite	Vendor bill pending	Vendor, files
Workday	Supplier invoice draft	Supplier, attachments

Intake's job is the capture record. The ERP document is a downstream, chartered action.

6. Responsibilities

1. Poll each in-scope channel on the cadence. Record poll success/fail.
2. Classify artefact: invoice, credit, statement, dunning, remittance, junk, unreadable, file_duplicate.
3. Compute payload hash. If hash exists, attach to the existing object as file_duplicate; do not open a second financial object.
4. Extract fields to schema AP-CAP-001. Store per-field confidence.
5. Suggest a vendor ID only when a deterministic key hits (VAT/GST/RFC + name match, or printed vendor number that exists). Otherwise vendor_unresolved.
6. Open work object: channel, hash, extract, image URI, classification, confidence vector, suggested vendor, legal-entity guess.
7. Route statement to Agent 11 queue seed; dunning to Agent 08 seed; junk/unreadable to human miscellaneous.
8. Produce the daily completeness reconcile: artefacts received vs objects opened vs leftover in channel.
9. Write audit evidence.
10. Stop and fail closed if the IDR vendor, mailbox, or store is unavailable (see failure handling).

7. Explicit exclusions

1. Payment authorisation or release.
2. Vendor create or vendor bank change.
3. Posting or parking in ERP at L0/L1 (parking is a separate earned subclass, not implied by “intake”).
4. Matching, tax determination, or duplicate *financial* decision (file hash ≠ financial duplicate).
5. Replying to the supplier (“we have your invoice”).
6. Changing PO, creating GR, or assigning GL accounts from a header guess.
7. Treating a statement or dunning letter as an invoice.
8. Splitting a multi-invoice PDF into posted documents without a human confirm of split boundaries (at L1, split *proposals* are allowed).
9. Processing employee T&E, payroll, or customs entries.
10. Claiming the extract is “accurate” or “compliant with e-invoicing law.” Jurisdiction e-invoice mandates are a Tax/IT control, not this agent’s outcome.
11. Deleting the original artefact.
12. Raising its own autonomy.

8. Human owner

Field	Northline (illustrative)	Generic
Role	AP Operations Lead	Same
Named	Marcus Bell, Cleveland	Named in charter
Backup	Shift lead, invoices	Named
Escalation	AP Manager	AP Manager
Time-to-cover	Same day	Same day — Intake cannot sit ownerless
Owns	Channel inventory, extract schema, low-confidence desk, completeness reconcile	Same
Does not own	Vendor master, tax position, payment	Same

The low-confidence desk (typically 2–3 specialists at Northline volume) corrects extracts. They are not “the bot’s assistants”; they are the control.

9. Approval requirements

Agent output	Human action required	Who	When
High-confidence extract, known vendor, single invoice PDF	Review not required at L1 <i>for use by Agent 02</i> ; still subject to sample	Ops Lead sample plan	Sample daily
Any field confidence < floor (Northline: 0.85 header ID fields, 0.80 lines)	Correct or reject	Low-confidence desk	Before Agent 02 starts
vendor_unresolved	Assign vendor or return to sender procedure	Vendor Master + Ops	Same day
Multi-invoice PDF split proposal	Confirm page boundaries	Low-confidence desk	Before split objects go live
Legal-entity ambiguous (bill-to matches two companies)	Assign company	Ops Lead	Before Validation
First invoice from vendor < 90 days	Mandatory review	Ops + Vendor Master	Before Validation
Proposed L3 auto-register (not in force at commissioning)	Separate promotion case	Process Owner + Controller	After gates

No approval in Intake authorises payment.

10. Escalation criteria

Condition	To	Timing	Packet
Channel poll fail > 45 minutes in SSC hours	IT + Ops Lead	45 min	Job log
Daily completeness gap (artefacts > objects)	Ops Lead	16:45 ET	Reconcile
Unreadable rate > 8% of a day's volume	IDR vendor owner + Ops	Next morning	Sample images
Sudden volume > 2× 20-day average on one channel	Ops Lead	Same day	Channel dump — possible flood or misdirected mailbox
Suspected mailbox compromise / unexpected sender pattern	IT Security + Ops	Immediate	Headers
Schema cannot represent a new legal e-invoice format	Tax + IT + Ops	Before forcing a mapping	Sample + regulation note
Low-confidence queue > 1 business day aging	AP Manager	Daily	Aging

11. Output standard

11.1 Capture record (minimum)

Field	Rule
object_id	Orchestrator ID
agent_id	01
autonomy_level	L0 or L1
channel	email / edi / cxml / portal / scan
received_at	Channel timestamp
payload_hash	SHA-256
classification	Enum in §6
legal_entity_guess	Company code or unresolved
vendor_suggest	ERP ID or unresolved
header{}	Schema fields + per-field confidence
lines[]	Schema fields + per-field confidence
image_uri	Immutable store
extract_uri	JSON
file_duplicate_of	Object ID or null
human_review_state	not_required / pending / corrected / rejected
owner	Role + name
timestamp	UTC

11.2 Daily completeness report

Channels, artefacts in, objects out, leftovers, unreadable count, file-duplicates, cost estimate, poll failures.

11.3 Naming

NL-I1-{channel}-{yyyyMMdd}-{seq} for artefacts; work object ID is system-assigned and never reused.

12. Control requirements

Control	Agent behaviour	Test
Completeness of inbound	Daily reconcile	Sample 3 days/month: mailbox vs objects
Original retained	No delete	Quarterly store test
File-level duplicate	Hash attach, no second object	Inject known PDF
Least privilege	Read channels; no ERP write at L1	Access review
SoD	Intake staff ≠ payment releasers	HR/role review
Change control	Schema and IDR prompt/version hashed	Version log
PII / pricing	Store access logged	SIEM sample

Intake does not satisfy e-invoicing legal archival on its own. Tax and Legal specify the archive. Intake feeds it.

13. Audit evidence

Evidence	Retention	Location
Original artefact	Per legal archive + min 7 years	Evidence store
Extract JSON + confidence	7 years	Evidence store
Human corrections (before/after)	7 years	Orchestrator
Channel poll logs	1 year	SIEM / job
Completeness reports	7 years	Close-adjacent folder
Schema versions	Life of programme	Controlled docs
File-duplicate joins	7 years	Orchestrator

Retrieval target: from invoice number or message ID to image + extract in 15 minutes.

14. KPIs

Operating discipline. Not savings. Not “straight-through.”

KPI	Definition	Cadence	Use
Channel completeness	Objects / artefacts (ex junk)	Daily	Coverage
Time to work object	Receipt → object	Daily	Freshness
Low-confidence rate	% below floor	Daily	IDR health
Correction rate	Fields changed by human / fields presented	Weekly	Extract quality
Misclassification rate	Human flips class	Weekly	Classifier health
Unreadable rate	%	Weekly	Image / vendor issue
File-duplicate rate	%	Weekly	Sender behaviour
Leftover artefacts	Count at 16:30	Daily	Silent loss
Cost per 1,000 artefacts	IDR + connector + desk minutes	Monthly	Pause brake

Forbidden: “capture accuracy guaranteed,” “touchless rate,” ROI.

15. Performance history fields

Period; artefacts by channel; objects opened; file-duplicates; unreadable; leftover; low-confidence volume; correction rate; misclass; poll failures; IDR version; schema version; autonomy level; incidents; pause events; cost; companies processed; first-invoice-from-new-vendor count.

16. Autonomy level

Level	Allowed	Northline commissioning
L0	Shadow extracts, no work objects as SOR (parallel folder)	First 10 days
L1	Work objects + recommendations; human on low-confidence, unresolved vendor, splits, new vendors	Default live
L2	Prepare ERP register packet (attachments, suggested fields) for a human clerk	After gates, optional
L3	Auto-register <i>named subclass</i> (e.g. structured EDI 810 from 12 listed vendors, hash-unique, vendor resolved)	Separate case; no payment
L4	Exception-only review of that same EDI subclass	Recert

Default: **L1**. Ceiling without a new charter: **L1**. File-hash duplicate handling is not L3 financial matching.

17. Failure handling

Failure	Detect	Immediate	Notify	Resume
Mailbox / portal down	Poll fail	Stop claiming completeness; mark channel red	IT + Ops 45 min	After poll success + leftover sweep
IDR timeout	Job SLA	Retry once; then unreadable	Ops	Object-level
Store write fail	5xx	Freeze intake (do not extract without archive)	IT	After write test
Schema mismatch	Validation of JSON	Fail object to human	Ops	After schema fix (change control)
Clock skew / missing received_at	Parse	Use poll time; flag time_inferred	Ops weekly	n/a
Flood	2× volume	Do not drop; enlarge queue; extra desk	AP Manager	After source confirmed
Suspected malicious payload	AV / sandbox	Quarantine; do not open in ERP	Security	After clearance

Never “skip archive and key it anyway.”

18. Cost monitoring

Element	Meter	Pause threshold (illustrative)
IDR / model	Pages × unit cost	2× baseline for 3 weeks <i>and</i> correction rate worsening
Portal / EDI	Connector fees	Error rate > 5% sustained
Desk minutes	Low-confidence + leftover	Desk > 40% of artefacts for 2 weeks → IDR is not earning L1
Rework	Downstream Validation rejects for capture error	Cluster ≥ 20/week → L0

Cost is a brake. Do not write a savings case into this table.

19. Handoffs

From	To	Trigger	Artefact
Channel	Agent 01	New artefact	Payload
Agent 01	Agent 02	Capture record ready (and human review if required)	Capture record
Agent 01	Agent 10	Object exists	Image + extract (parallel)
Agent 01	Agent 11	Class statement	Seed
Agent 01	Agent 08	Class dunning	Seed
Agent 01	Human misc	Junk / unreadable	Artefact
Agent 01	Agent 16	All status changes	Events

20. Configuration parameters (charter these)

Parameter	Northline starting value	Change control
Header confidence floor	0.85 on invoice #, date, gross, vendor name	Yes
Line confidence floor	0.80	Yes
Poll interval	15 min / 60 min off hours	IT
Channels in scope	apinvoices@northline.example, EDI 810 VAN, two OEM portals	Yes
Max PDF pages auto-split proposal	40	Yes
File-hash algorithm	SHA-256	Yes
New-vendor review window	90 days	Yes

21. First 90 days

Days 1–14 (L0): Shadow extract on US-OH + US-AL email + EDI only. Completeness reconcile built. No specialist relies on the extract.

Days 15–45 (L1): Work objects live. Low-confidence desk staffed. CA/MX email still L0 until GST/IVA fields are in schema.

Days 46–90: Open L2 register-packet case only if G1.* gates pass. Do not discuss L3 except for a short EDI vendor list. Agent 15 may look at unreadable clusters (bad fax from two foundries).

Success: leftover ≈ 0 at 16:30; correction reasons coded; no ERP write privilege on the service account.

22. Worked example — Northline Industrials

Tuesday 09:12 ET. Mailbox receives a 4-page PDF from ar@lakeshoresteel.example, subject “Inv 459102.”

1. Poll picks up message ID AAMk-771. Hash 9f3c... is new.
2. Classifier: invoice (0.96). Not a statement.
3. Extract: vendor name Lakeshore Steel Co.; invoice 459102; date 2026-04-06; PO 45007821; USD; net 18,400; tax 1,196; gross 19,596; 12 lines, EA, item numbers printed.
4. Vendor suggest: VAT/EIN + name hits V-10442 in D365. Legal entity: bill-to Dayton plant → company 1000.
5. Confidences: invoice # 0.97, gross 0.94, lines 0.88–0.93. All above floor.
6. Not a first invoice (vendor age 6 years). Work object W0-1048821 opened. Image stored.
7. Agent 02 and Agent 10 subscribed.
8. Completeness: this artefact is one of 612 before 16:30; leftover 0.

Same morning, second mail: identical PDF forwarded by a buyer. Hash matches. Classification file_duplicate. Attached to W0-1048821. No second invoice will be validated. (Financial duplicate of a *different* number is Agent 10’s problem, not hash.)

Same morning, third mail: 18-page PDF, three invoices, low split confidence. Desk confirms pages 1–6 / 7–12 / 13–18. Three objects. Agent does not guess.

Birmingham portal: 40 OEM invoices retrieved. Structured XML. Extract floors met. Company 1100. Same path.

Hamilton box (out of Wave 1 live): still L0 shadow. GST fields not forced into US schema.

No payment, no post, no supplier reply occurred.

23. Sample output artefact (abridged)

```
object_id: W0-1048821
agent_id: 01
autonomy_level: L1
channel: email
received_at: 2026-04-07T13:12:04Z
payload_hash: 9f3c...
classification: invoice
legal_entity_guess: 1000
vendor_suggest: V-10442
header.invoice_number: 459102 (0.97)
header.gross: 19596.00 USD (0.94)
header.po: 45007821 (0.91)
lines: 12
image_uri: evidence://nl/01/2026/04/W0-1048821.pdf
human_review_state: not_required
owner: Marcus Bell / low-confidence desk on duty
file_duplicate_of: null
downstream: [02, 10]
```

24. What this agent does not replace

Mailroom policy, legal e-invoice archive, vendor onboarding, EDI mapping project, mailbox security, or the specialist who corrects a bad extract. It does not replace Validation.

25. Document control

Field	Value
Spec	AGENT_01 Invoice Intake
Default autonomy	L1 (start L0)
Review	Quarterly or on channel/schema change
Related	Agents 02, 10, 11, 08, 16; charter template

AGENT 02 — Invoice Validation

Stack position: After Intake, before Matching. Runs in parallel awareness with Duplicate & Anomaly (does not wait for Agent 10 except where a flag already exists).

Default autonomy: L1 Recommend (commission at L0).

Human owner (typical): AP Quality Lead

Payment authority: None.

Northline instance: Procedure AP-VAL-002; US entities live first; CA/MX L0 until tax field packs are signed.

1. Position in the stack

Validation answers one question: **is this capture record complete and internally consistent enough to enter match, approval, or exception?** It is a gate, not a posting engine.

A pass is a licence to *continue in the stack*. It is not a licence to pay, and it is not a representation that the invoice is legally perfect in every jurisdiction.

2. Job description

The Invoice Validation Agent applies a written checklist to each Intake capture record: required fields present, types valid, supplier identity resolvable to a single unblocked vendor, currency allowed for the company, invoice number format usable, dates plausible, arithmetic (net + tax ≈ gross), PO present when policy requires, credit-note sign, and tax *fields present* (not tax *position decided*).

It emits PASS, FAIL (coded), or QUERY (human must interpret). It does not calculate use tax, does not change vendor master, and does not “fix” the invoice to make it pass.

3. Operating intent and cadence

Mode	Cadence	Output
Object-driven	On Intake ready	Validation worksheet
Human QUERY desk	SSC hours	Interpret / fail / return to Intake
Daily	17:00 ET	Fail-code mix, QUERY aging
Weekly	Friday	Checklist version vs fail mix (Agent 15 input)

Do not batch Validation to “overnight” if Matching needs same-day worksheets. Northline runs Validation continuously 07:00–18:00 ET.

4. Inputs

Input	Source	Mandatory
Capture record + image	Agent 01	Y
Validation checklist AP-VAL-002 (version-hashed)	Controlled docs	Y
Vendor master: block, currency, tax IDs, payment hold, one-time flag, age	ERP	Y
Company allowed currencies and bill-to identifiers	ERP / policy	Y
PO existence check (number on face vs ERP, not a match)	ERP	If PO policy applies
Open Agent 10 flag (if already raised)	Agent 10	If present, note on worksheet; do not auto-fail solely because a <i>new</i> flag might later appear
Prior invoices for number-collision hint	ERP / Orchestrator	N (hint only; Agent 10 owns duplicate decision)
Legal-entity assignment	Agent 01 or human	Y

5. Tools / data required

Tool / data	Privilege	ERP notes
Capture JSON + image	Read	—
Vendor master	Read	D365 VendTable; SAP LFA1/LFB1; Oracle Suppliers; NetSuite Vendor; Workday Supplier
PO header existence	Read	Existence and vendor-on-PO vs invoice vendor — not quantity match
Block / hold codes	Read	Pay hold ≠ validation fail automatically; record it
Checklist engine (rules, not a black-box score)	Read rules	Every fail needs a code
Orchestrator	Write status	val_pass / val_fail / val_query
Tax field dictionary	Read	Which IDs must be <i>present</i> by entity — Tax owns the dictionary

No write to vendor, PO, or invoice posting tables.

6. Responsibilities

1. Confirm Intake human_review_state is not pending or rejected.
2. Apply checklist in published order. Stop-on-fatal vs continue-to-collect is defined in AP-VAL-002 (Northline: continue-to-collect so specialists see the full fail set).
3. Resolve vendor: exactly one candidate above the deterministic threshold. Zero or many → FAIL_VENDOR or QUERY_VENDOR.
4. Test vendor not blocked for *invoicing* (distinguish invoice-block from payment-block; payment-block is recorded as NOTE_PAY_HOLD and still may PASS for match).

5. Test arithmetic: $|\text{net} + \text{tax} - \text{gross}| \leq \0.02 or currency equivalent; else FAIL_ARITH.
6. Test dates: invoice date not > 8 days future; not older than policy (Northline: 18 months without Controls waiver).
7. Test currency $\in$ company allowed set.
8. Test PO presence when document type requires it (Northline PO-required categories).
9. Test tax *fields present* per entity dictionary (US: none extra at header beyond amounts; CA: GST/HST registration on vendor or face; MX: RFC/UUID as specified by Tax). Missing field → FAIL_TAX_FIELD. Do not compute tax due.
10. Emit worksheet with codes, PASS/FAIL/QUERY, and evidence pointers.
11. Hand PASS to Matching (PO) or Approval (non-PO) per document type. Hand FAIL/QUERY to human Quality desk and, if coded as capture error, back to Intake.

7. Explicit exclusions

1. Payment authorisation or release.
2. Tax determination, use-tax accrual, withholding judgement, or e-invoice legal validity opinion.
3. Financial duplicate decision (Agent 10).
4. 2-way / 3-way match (Agent 03).
5. Vendor create, unblock, or bank edit.
6. Changing extract fields silently — corrections go through Intake human-review so the audit shows who changed what.
7. Treating NOTE_PAY_HOLD as a reason to hide the invoice from the stack.
8. Approving non-PO spend.
9. Declaring the invoice “compliant.”
10. Auto-passing first invoices from vendors < 90 days (mandatory QUERY).
11. CA/MX live PASS if those dictionaries are not signed (Northline Wave 1).

8. Human owner

Field	Northline	Generic
Role	AP Quality Lead	AP Quality Lead
Named	Keisha Ward	Named
Backup	Ops Lead (Marcus Bell)	Named
Escalation	AP Manager; Tax desk on tax-field disputes	Same pattern
Owns	Checklist content with Controller; QUERY desk; fail taxonomy	Same
Does not own	Match tolerances, payments, vendor master writes	Same

9. Approval requirements

Output	Human	Condition
PASS	Sample only at L1	Sample plan 25/day or 5%, greater
FAIL	Specialist confirms code or recodes	Before supplier/internal chase on the back of a fail
QUERY_VENDOR	Quality + Vendor Master	Same day
QUERY_TAX_FIELD	Tax desk	Before any match recommend
First invoice / vendor < 90 days	Quality + Vendor Master	Mandatory
Arithmetic fail that “looks like tax rounding”	Tax or Quality, not the agent	Do not loosen \$0.02 in the job
Waiver of age > 18 months	Controls Lead	Written

10. Escalation criteria

Condition	To	Timing
QUERY aging > 1 business day	Quality Lead	Daily
Fail code FAIL_TAX_FIELD volume jump	Tax	2 days of jump
Vendor master systematically missing tax IDs	Vendor Master	Weekly pack
Intake correction loop > 2 on same object	Ops Lead	On third return
Checklist vs ERP setup conflict	Controller	Before changing either
Service account cannot read vendor	IT	Immediate

11. Output standard

Field	Rule
object_id	Same as Intake
agent_id	02
result	PASS / FAIL / QUERY
codes[]	Controlled taxonomy
notes[]	Including NOTE_PAY_HOLD
vendor_id	Resolved or null
document_path	PO_MATCH / NON_PO_APPROVAL / CREDIT / STOP
checklist_version + hash	Mandatory
tax_dictionary_version	Mandatory
evidence_uri	Worksheet
owner	Quality role
autonomy_level	Recorded

Fail taxonomy (minimum — extend, do not freestyle):

FAIL_VENDOR, FAIL_ARITH, FAIL_DATE, FAIL_CURRENCY, FAIL_PO_REQUIRED, FAIL_TAX_FIELD, FAIL_CREDIT_SIGN, FAIL_SCHEMA, QUERY_VENDOR, QUERY_ENTITY, QUERY_TAX_FIELD, QUERY_NEW_VENDOR, NOTE_PAY_HOLD, NOTE_ANOMALY_OPEN.

12. Control requirements

Control	Support	Test
Completeness before match	No Agent 03 start without PASS	Join report
Tax field presence	Dictionary, not determination	Tax samples CA/MX when live
Vendor identity	Single resolved ID	Sample 25 PASSes
Checklist integrity	Hash on worksheet	Monthly
SoD	Quality ≠ payment releaser	Role review
No silent extract edit	Edits via Intake	Before/after log

13. Audit evidence

Worksheets 7 years; checklist versions life-of-programme; QUERY decisions 7 years; sample logs 7 years; access 1 year. Retrieval by object ID in 15 minutes.

14. KPIs

KPI	Definition	Use
Same-day validation	% of ready Intake objects with a result same SSC day	Flow
PASS rate	PASSes / results (context, not a target to maximise)	Mix
Fail mix	Count by code	Process signal
QUERY aging	>1 day count	Staffing / master data
Recode rate	Human changes agent code	Checklist quality
False PASS (later found missing required field)	Count	Demotion input
Cost per 1,000 objects	Model + desk	Brake

Do not KPI “tax accuracy” or “compliance rate.”

15. Performance history fields

Period; objects presented; PASS/FAIL/QUERY; codes; recodes; false PASSes; new-vendor QUERYs; pay-hold notes; checklist hash; dictionary hash; autonomy; incidents; cost; entity split.

16. Autonomy level

Level	Behaviour
L0	Shadow checklist
L1	Live result; humans on QUERY/FAIL confirm; sample PASS
L2	Prepare return-to-vendor pack for FAIL codes on a whitelist (Agent 08 sends only after approval)
L3	Auto-PASS only structured EDI that already passed Intake L3 subclass <i>and</i> deterministic vendor <i>and</i> arithmetic — still not tax determination
L4	Exception review of that EDI subclass

Hard ceiling without new charter: L1. Tax QUERY never auto-resolves.

17. Failure handling

Failure	Immediate	Resume
Vendor read fail	QUERY not PASS	After ERP probe
Checklist hash missing	Stop agent	After pin
Intake still pending review	Refuse	When Intake ready
Arithmetic on tax-inclusive face formats	QUERY not creative math	Tax writes a rule if needed
Conflicting vendor keys	QUERY_VENDOR	Human
Clock/date timezone issues	Use company local date; flag	n/a

18. Cost monitoring

Desk minutes on QUERY, recode clusters, model cost. Pause if recode > 20% for 2 weeks or false PASS cluster ≥ 10/week → L0.

19. Handoffs

From	To	Trigger
01	02	Capture ready
02	03	PASS + PO path
02	07	PASS + non-PO path
02	Quality desk	FAIL / QUERY
02	01	Capture error
02	08	L2 return pack only if chartered
02	10	Worksheet available (10 may already have flagged)
02	16	Status

20. Configuration parameters

Parameter	Northline start
Arithmetic tolerance	\$0.02
Future date slack	8 days
Max age	18 months
New-vendor window	90 days
Continue-to-collect fails	Yes
Live entities	1000, 1100

21. First 90 days

L0 two weeks on US companies. L1 when G0.* pass. Keep CA/MX on L0 until Tax signs dictionaries. Do not maximise PASS rate as a goal — that trains the agent to wave things through.

22. Worked example — Northline Industrials

Object W0-1048821 (Lakeshore Steel, \$19,596, PO 45007821, company 1000).

Checklist AP-VAL-002.6 hash a11e...:

- Vendor V-10442 unique, not invoice-blocked. Payment block: none. Age 6 years.
- Arithmetic: $18,400 + 1,196 = 19,596$. Pass.
- Date 6 Apr 2026, received 7 Apr. Pass.
- Currency USD allowed. Pass.
- PO required and present in ERP, vendor on PO = V-10442. Existence only. Pass.
- Tax fields for US-OH: amounts present. Pass (no determination).
- Agent 10: no open flag at validation time.
- Path: PO_MATCH. Result: PASS.

Same day W0-1049010: Canadian vendor, Hamilton, GST number blank on vendor and face. Entity still L0-live? Charter says CA is L0 — shadow only; human validates. If someone had turned on L1 early, agent must FAIL_TAX_FIELD / QUERY_TAX_FIELD, not PASS.

W0-1049104: Gross \$10,000, net \$10,000, tax \$0, services, US. PASS on fields. Path NON_PO_APPROVAL if no PO. Use-tax might be due — **Tax/accrual later, not this agent.**

W0-1049220: Two vendor candidates share a similar name. QUERY_VENDOR. Keisha's desk picks V-08811. Agent does not pick the higher name-score.

23. Sample output artefact (abridged)

```
object_id: W0-1048821
agent_id: 02
result: PASS
document_path: PO_MATCH
vendor_id: V-10442
codes: []
notes: []
checklist_version: AP-VAL-002.6
checklist_hash: a11e...
tax_dictionary_version: TAX-US-2026.1
autonomy_level: L1
owner: Keisha Ward
downstream: 03
```

24. What this agent does not replace

Tax department, vendor master quality, e-invoicing legal review, Matching, or Agent 10. A PASS is not an accounting opinion.

25. Document control

Field	Value
Spec	AGENT_02 Invoice Validation
Default autonomy	L1 (start L0)
Review	Quarterly or checklist/tax-dictionary change

AGENT 03 — Matching

Stack position: After Validation PASS on the PO path. Hands clean recommendations to a human poster; hands breaks to Triage, Goods Receipt, and PO Quality. Blocked by open Agent 10 flags.

Default autonomy: L1 Recommend (commission at L0).

Human owner (typical): AP Match Lead

Payment authority: None. Posting, if later earned at L3, is still not payment.

Northline instance: AP-MAT-004; companies 1000/1100 in Wave 1; see completed charter in AGENT_CHARTER_TEMPLATE.md.

1. Position in the stack

Matching applies the organisation’s written 2-way, 3-way, and GR/IR rules to a validated PO invoice. It is the most tempting place to “just post it.” Resist. A match worksheet is an argument. A human (or a later earned L3 subclass) posts. Cash still waits for Agent 12 and two human authorisers.

2. Job description

The Matching Agent loads the validated extract, the PO, and the receipt or service-entry set. It applies the pinned tolerance table. It writes a line-level worksheet and a single recommendation code. It does not create receipts, change POs, override tolerances, or post. If Agent 10 has an open flag, the only recommendation allowed is hold.

3. Operating intent and cadence

Mode	Cadence	Output
Object-driven	On Validation PASS + PO path	Worksheet
Specialist accept/edit/reject	SSC hours	Human decision
Daily	17:15 ET	Accept mix, break codes, \$50k+ aging
Weekly	Friday	UOM / price / missing-GR mix for Agents 06 and 15

Northline: specialists work NL.MATCH.US.PO. Target: worksheet same day if PASS arrived by 14:00 ET.

4. Inputs

Input	Source	Mandatory
Validated extract + image	Agents 01/02	Y
PO header/lines, remaining qty, price, tax group, vendor, account assignment	ERP	Y
Product receipts / service entries / timesheet accepts	ERP / WMS / project	Y for the chosen path
Tolerance table AP-MAT-004 + hash	Controlled + ERP matching setup	Y
Agent 10 flag state	Agent 10	Y (must read)
Contract reference (2-way services)	ERP / CLM	Y for 2-way path
Prior match worksheets (rework)	Orchestrator	If rework

5. Tools / data required

Tool	Privilege	ERP mapping (read)
PO	Read	D365 PurchTable/Line; SAP EKKO/EKPO; Oracle PO; NetSuite Purchase Order; Workday Purchase Order
GR / SES	Read	D365 product receipt / WMS; SAP EKBE/MIGO; Oracle Receipts; NetSuite Item Receipt; Workday Receipt
Pending invoice	Read	Do not write at L1
Tolerance	Read, hash	Pin version on every worksheet
Orchestrator	Write worksheet + status	—

Service account example: svc-ap-match **read-only**. Write on invoice post is a different account, different charter, L3-only, still not payment.

6. Responsibilities

1. Refuse if Validation is not PASS or path is not PO_MATCH.
2. Read Agent 10. If open flag → RECOMMEND_NO_ACTION_ANOMALY_HOLD only.
3. Choose path: 3-way inventory/direct; 2-way contracted services ≤ policy cap; else OUT_OF_SCOPE to Triage.
4. Identify candidate GR/SES set. If ambiguous → GR_AMBIGUOUS to Triage. Do not pick the set that “makes it work.”
5. Compare quantity, price, UOM, currency, tax *amounts* vs *PO/receipt*, account assignment presence (not redesign).
6. Apply tolerances. Record each test and the hash.
7. Emit one recommendation (see §11).
8. Seed Agent 05 if missing GR; seed Agent 06 if PO defect codes fire.
9. Present worksheet to the specialist queue.

10. Record accept/edit/reject. Edits are human; the agent does not learn an override as a new secret tolerance.

7. Explicit exclusions

1. Payment authorisation or release.
2. Posting the vendor invoice at L0/L1.
3. Creating or reversing GR/SES.
4. Changing PO price, qty, tax, vendor, or account.
5. Editing tolerances.
6. Clearing Agent 10 flags.
7. Multi-PO invoices (Northline Wave 1) — route out.
8. Price override because “email from buyer.”
9. Dummy GR to clear GR/IR.
10. CA/MX live match before those charters exist.
11. Intercompany, customs, consignment (until chartered).
12. Claiming the match is “accurate accounting.”

8. Human owner

Field	Northline
Role / name	AP Match Lead / Priya Shah
Backup	Marcus Bell
Escalation	AP Manager; Plant Controller > \$50k with dual breaks
Owns	Worksheet quality, specialist standard, promotion cases
Does not own	Warehouse GR create, PO change, payment, tax position

9. Approval requirements

Recommendation	Approver	Extra
RECOMMEND_MATCH ≤ \$50k	Match specialist	Worksheet + hash + image
RECOMMEND_MATCH > \$50k	Specialist + Plant Controller	Buyer if price was human-edited
Hold / route codes	Auto-route OK	Triage owns next
Post in ERP	Human poster at L1/L2; L3 only on signed subclass	Never payment
Any tolerance exception	Human only	Recorded as edit, not agent success

10. Escalation criteria

Condition	To	Timing
3 extract versions fail match	Quality (02/01)	Third fail
Tax variance > \$100	Tax	Same day
Qty and price break, header > \$50k	Plant Controller	Same day
Open flag + otherwise clean	Controls Lead	Before accept
GR read fail > 2 hours	IT + Match Lead	2 hours
Reject rate > 15% / 200 invoices	Process Owner	Weekly
Ambiguous GR set on consignment	Plant Finance	Same day

11. Output standard

Recommendation enum (closed):

Code	Meaning	Next
RECOMMEND_MATCH	In tolerance, no flag	Human post / later L3 subclass
RECOMMEND_HOLD_MISSING_GR	PO ok, receipt missing	Agent 05
RECOMMEND_ROUTE_PO_QUALITY	PO defect	Agent 06
RECOMMEND_ROUTE_TRIAGE	Other break	Agent 04
RECOMMEND_NO_ACTION_ANOMALY_HOLD	Agent 10 open	Controls
OUT_OF_SCOPE	Charter exclusion	Triage

Worksheet minimum: object ID; agent 03; level; company; invoice; PO; GR/SES IDs; line compare table (inv qty/price/UOM vs PO vs GR); tolerance tests; hash; flag state; recommendation; owner; timestamp; evidence URI.

12. Control requirements

Control	Support	Test
3-way policy	Applied, not posted by agent at L1	25 worksheets/week vs ERP
SoD match vs pay	No payment privilege	Quarterly access
Tolerance integrity	Hash	Monthly
Duplicate hold	Cannot recommend match	100% flagged objects
Audit trail	Accept log	15-minute retrieve

13. Audit evidence

Worksheets 7 years; hashes + policy PDFs 7 years; accept/edit/reject 7 years; access 1 year; promotion records life-of-programme.

14. KPIs

KPI	Definition	Use
Worksheet coverage	In-scope PASSES with worksheet same day	Flow
Accept / edit / reject	Rates by code	Quality
Material error	Accepted then reversed for match error	Demotion
Misroute	Wrong next agent	Handoff
Flag-block integrity	100%	Control
\$50k SLA	Escalations on time	Control
Cost / 1,000	Model + specialist minutes	Brake

No “first-time-match %” as a promised outcome. Northline may *observe* FTM as a process measure on the reporting pack (Agent 14), not as this agent's success bonus.

15. Performance history fields

Period; presented; worksheets; recommendations by code; accept/edit/reject; material errors; flag holds; multi-PO out; entity split; hash; level; incidents; cost; GR-ambiguous count.

16. Autonomy level

Level	Behaviour
L0	Shadow worksheets
L1	Live recommend; human post
L2	Prepare-for-post packet (journal-ready fields)
L3	Post <i>only</i> named subclass (e.g. US-OH 3-way inventory, ≤ \$5,000, no flag, hours-bound)
L4	Exception oversight of that subclass

Payment never enters this ladder. Raising the \$5,000 cap is a new L3 case.

17. Failure handling

Failure	Action
ERP read fail	Freeze queue; no stale match
Timeout	Retry once; fail to human
Validation not PASS	Refuse
Ambiguous GR	Triage GR_AMBIGUOUS
Open flag	Anomaly hold
401/403	Stop all jobs; Security

18. Cost monitoring

Inference tokens, connector errors, specialist edit/reject minutes, reversal rework. Pause: 2× cost + falling accept rate for 3 weeks; or material-error cluster ≥ 5/week → L0.

19. Handoffs

02 → 03 on PASS PO; 10 → 03 blocks; 03 → human specialist; 03 → 04/05/06; human → ERP post; 03 → 16 events.

20. Configuration parameters

Parameter	Northline start (AP-MAT-004)
Qty tolerance	±2% or 1 unit (greater)
Price tolerance	±1% or \$25 (greater)
Tax variance	Not auto-cleared; > \$100 escalate Tax
2-way services cap	\$10,000 + contract ref
Multi-PO	Out of scope
Header second-approver	\$50,000
L3 subclass (future)	Not in force

21. First 90 days

Follow the worked progression in AUTONOMY_PROGRESSION.md. Do not open L3 in the first 90 days. Spend the period coding breaks honestly — missing GR vs bad PO vs UOM — so Agents 05, 06, and 15 have material.

22. Worked example — Northline Industrials

W0-1048821 Lakeshore Steel \$19,596, PO 45007821, Dayton inventory.

PO: 12 lines, item fasteners, UOM EA, prices match face. WMS shows product receipt PR-88311 two days earlier, qty exact on 11 lines; line 7 invoice 1,000 EA vs receipt 980 EA (2.04% over). Tolerance is 2% or 1 unit — **fails** (20 units and 2.04% > 2%).

Recommendation: RECOMMEND_ROUTE_TRIAGE with break QTY_OVER on line 7. Not RECOMMEND_MATCH. Not a dummy GR for 20 EA. Agent 04 assigns: warehouse to confirm short-ship vs invoice error. Agent 08 may later draft a query if warehouse confirms 980 received.

Specialist accepts the *route*, not a match. Cycle time on this object will look “slow.” That is correct.

Second object W0-1049400: \$3,110, 3-way, all lines in tolerance, no flag. RECOMMEND_MATCH. Specialist accepts. Junior poster posts in D365. Item sits open until Thursday’s human-authorized payment run. Agent 03’s job ended at the worksheet.

Third W0-1049555: Clean numbers, but Agent 10 flag possible_duplicate vs invoice 88321. RECOMMEND_NO_ACTION_ANOMALY_HOLD. Priya cannot accept match until Elena clears or confirms.

23. Sample output artefact (abridged)

```
object_id: W0-1048821
agent_id: 03
recommendation: RECOMMEND_ROUTE_TRIAGE
break_codes: [QTY_OVER]
line_7: inv_qty 1000 vs gr 980 vs po 1000; tol 2%|1u FAIL
tolerance_version: AP-MAT-004.3
tolerance_hash: c90a...
agent10_flag: none
autonomy_level: L1
owner: Priya Shah
next: 04
```

24. What this agent does not replace

ERP matching configuration, warehouse discipline, buyer PO hygiene, the poster, or the payment authoriser. It does not replace judgement on over-receipts.

25. Document control

Field	Value
Spec	AGENT_03 Matching
Default autonomy	L1 (start L0)
Charter example	AGENT_CHARTER_TEMPLATE.md section B

AGENT 04 — Exception Triage

Stack position: Receives coded breaks from Matching, Validation fails that need assignment, and out-of-scope objects. Assigns owner, next action, and SLA. Does not resolve.

Default autonomy: L1 Recommend (commission at L0).

Human owner (typical): Exception Desk Lead

Payment authority: None.

Northline instance: Queue NL.XCP.US; taxonomy AP-XCP-001.

1. Position in the stack

Triage is air-traffic control for broken invoices. Most SSC pain is not “too many exceptions”; it is exceptions without an owner, a code, or a clock. This agent exists to make every break *assignable*. It does not become a second matcher, a buyer, or a warehouse clerk.

2. Job description

The Exception Triage Agent reads the inbound break (codes, worksheet, image, vendor, amount, age) and emits a triage record: exception class, human owner role, recommended next agent or human action, SLA due, and escalation ladder. A human Exception Desk specialist accepts, recodes, or reassigns. The agent does not close exceptions by “accepting the variance.”

3. Operating intent and cadence

Mode	Cadence	Output
Object-driven	On break handoff	Triage record
Desk accept	SSC hours	Owner live
Sweep	10:00 and 15:00 ET	Unowned / breached
Weekly	Friday	Mix by class / plant / vendor for Agent 15

4. Inputs

Input	Source	Mandatory
Break payload	Agents 02, 03, 07, 11, 12 (holds)	Y
Taxonomy AP-XCP-001	Controlled	Y
Owner matrix (role by class, plant, \$)	Controlled	Y
SLA table	Controlled	Y
Work object history	Orchestrator	Y
Open related exceptions (same PO/vendor)	Orchestrator	N
DOA / plant calendar	HR / finance	For assignment only

5. Tools / data required

Tool	Privilege
Orchestrator read/write	Status, owner, SLA
Taxonomy + owner matrix	Read, version-hashed
ERP inquiry (open PO, GR, vendor block)	Read — to <i>confirm facts</i> , not to fix
Mail / ticket (internal)	Draft at L2 only
People directory	Read deputies in the matrix only

No ERP write. No supplier send.

6. Responsibilities

1. Accept only objects with a coded break or OUT_OF_SCOPE. If codes are blank, QUERY_UNCODED to the sending agent's owner — do not invent a class.
2. Map codes → class (examples: MISSING_GR, QTY_OVER, PRICE, UOM, PO_DEFECT, TAX_FIELD, MULTI_PO, GR_AMBIGUOUS, ANOMALY_HOLD, NON_PO_UNROUTED, STATEMENT_UNMATCHED).
3. Apply owner matrix. If matrix is silent, assign Exception Desk Lead — never “Finance.”
4. Set next action: Agent 05/06/07/08/09, Controls (10), human specialist, Tax, Vendor Master.
5. Set SLA from table (Northline examples: missing GR 5 business days to first receiver response; price break 3 days to buyer; anomaly hold 2 days to Controls).
6. Detect clusters (same PO, same vendor + class ≥ 5 open) and tag CLUSTER for Agent 15.
7. Emit triage record; wait for desk accept at L1.
8. Start SLA clock only after accept (or at L2 if chartered for auto-assign on a whitelist of classes).
9. Re-triage when new facts arrive (GR posted, PO changed by human). Do not leave stale owners.

7. Explicit exclusions

1. Payment authorisation or release.
2. Resolving the break (no qty write-off, no price accept, no GR create).
3. Overriding match tolerances.
4. Clearing Agent 10 flags (can route to Controls; cannot clear).
5. Contacting suppliers (Agent 08) or internals (Agent 09) except as a *handoff*.
6. Assigning a person not in the matrix because they “usually help.”
7. Closing an exception for inactivity.
8. Changing the taxonomy ad hoc in a prompt.
9. Promising a supplier a resolution time.

8. Human owner

Field	Northline
Role / name	Exception Desk Lead / Tomasz Wójcik
Backup	Priya Shah (Match Lead) — backup only, SoD watch
Escalation	AP Manager
Owns	Taxonomy with Process Owner; owner matrix; SLA table; desk quality
Does not own	Plant warehouses, buying, cash

At 15,000 invoices/month and a 28% observational exception rate, the desk is a real team, not a side duty. Staffing is a management decision. The agent does not compensate for an empty desk by auto-closing.

9. Approval requirements

Action	Human	Notes
Class + owner at L1	Desk accept	Recode allowed
Auto-assign (L2 whitelist)	Charter list only (e.g. MISSING_GR → plant liaison)	Sample 10%
SLA extension	Desk Lead	Reason coded
Close exception	Resolver + confirm in ERP/facts	Agent never closes alone
Matrix change	Process Owner + Controller if \$ owners change	Change control

10. Escalation criteria

Condition	To	Timing
Unaccepted triage > 4 SSC hours	Desk Lead	Same day
SLA breach	Owner's manager + Desk Lead	On breach
Second breach	AP Manager + Plant Controller if plant-owned	+2 days
Cluster ≥ 10 same vendor/class	Agent 15 + Procurement Ops	Weekly or ad hoc
Anomaly hold > 2 days	Controls Lead	Calendar
Matrix miss (no owner)	Process Owner	Immediate

11. Output standard

Triage record minimum:

Field	Rule
object_id	Same
agent_id	04
class	Taxonomy
owner_role / owner_name	Named
next_action	Enum
next_agent	05/06/07/08/09/10/human
sla_due	Timestamp
cluster_id	If any
taxonomy_hash	Mandatory
matrix_hash	Mandatory
human_state	pending_accept / accepted / recoded
autonomy_level	Recorded

12. Control requirements

Control	Support	Test
No ownerless exceptions	Matrix + sweep	Daily unowned = 0
SoD	Desk does not release payments	Role review
Taxonomy stability	Hash	Monthly
No silent close	Close only with evidence	Sample 20 closes
Anomaly path	Class ANOMALY_HOLD always to Controls	100%

13. Audit evidence

Triage records 7 years; accept/recode 7 years; SLA extensions 7 years; taxonomy/matrix versions life-of-programme; sweep reports 1 year.

14. KPIs

KPI	Definition	Use
Time to accepted owner	Handoff → accept	Flow
Recode rate	Human class ≠ agent class	Taxonomy quality
Unowned count	Snapshot	Control
SLA breach %	By class	Management
Ping-pong	Reassigns > 2	Matrix quality
Cluster tags	Count	Agent 15
Cost / 1,000 exceptions	Desk + model	Brake

Do not KPI “exceptions eliminated” as an agent success — that encourages improper closes.

15. Performance history fields

Period; inbound by sending agent; classes; recodes; unowned hours; breaches; extensions; ping-pong; clusters; closes (human); level; incidents; cost.

16. Autonomy level

Level	Behaviour
L0	Shadow assignment
L1	Recommend; desk accepts
L2	Auto-assign whitelist classes; still no resolve
L3	Auto-assign + start Agent 09 reminder on whitelist (internal only)
L4	Exception oversight of that whitelist

Never L3 resolve, write-off, or supplier commit.

17. Failure handling

Failure	Action
Blank codes	Return to sender agent
Matrix miss	Assign Desk Lead + escalate
Orchestrator write fail	Freeze; do not assign in side spreadsheets as SOR
Owner on leave, no deputy	Desk Lead holds; do not invent
Duplicate triage records	Merge; keep audit

18. Cost monitoring

Desk accept minutes, ping-pong minutes, model cost. Pause if recode > 25% for 2 weeks or ping-pong > 15% — the matrix is wrong; demote to L0 and fix the matrix.

19. Handoffs

03/02/07/11/12 → 04; 04 → 05/06/07/08/09/10/human; 04 → 15 cluster; 04 → 16 events; resolvers → 04 re-triage or human close.

20. Configuration parameters

Parameter	Northline start
Accept SLA (desk)	4 SSC hours
Missing GR owner	Plant finance liaison
Price owner	Buyer on PO
UOM owner	Buyer + Master Data
Anomaly owner	Controls Lead
Cluster threshold	5 open same vendor+class

21. First 90 days

Spend L0 aligning taxonomy with how specialists *actually* talk. A taxonomy nobody uses will be recoded to death. Do not auto-assign until recode < 15% for two weeks.

22. Worked example — Northline Industrials

W0-1048821 QTY_OVER line 7, \$19,596, Dayton.

Triage: class QTY_OVER; owner role Warehouse Supervisor Dayton (from matrix, not “someone in shipping”); next Agent 05 for receipt confirmation *and* Agent 09 if no response in 2 days; SLA first response 5 business days; not a cluster yet (2 open Lakeshore qty issues — below 5).

Tomasz accepts. He does not change the qty. He does not ask Matching to “use 2.1% just this once.”

Friday sweep: 14 MISSING_GR Birmingham > SLA. Escalate to Birmingham Plant Controller. Agent 09 packets go out. Agent 15 tagged because Birmingham missing GR is 19% of US-AL exceptions this month.

23. Sample output artefact (abridged)

```
object_id: W0-1048821
agent_id: 04
class: QTY_OVER
owner_role: Warehouse Supervisor Dayton
owner_name: (from matrix — L. Grant)
next_action: CONFIRM_RECEIPT_QTY
next_agent: 05
sla_due: 2026-04-14T21:00:00Z
taxonomy_hash: e2bb...
human_state: accepted
autonomy_level: L1
```

24. What this agent does not replace

The Exception Desk, plant ownership, or root-cause work. A well-triaged exception is still an exception.

25. Document control

Field	Value
Spec	AGENT_04 Exception Triage
Default autonomy	L1 (start L0)

AGENT 05 — Goods Receipt

Stack position: Downstream of Matching breaks MISSING_GR, GR_AMBIGUOUS (facts only), and QTY classes that need warehouse confirmation. Upstream of Internal Follow-up if the receiver is silent.

Default autonomy: L1 Recommend (commission at L0).

Human owner (typical): Plant / Warehouse Finance Liaison (per site), with AP Exception Desk as process owner of the packets.

Payment authority: None.

Northline instance: Dayton WMS receipts; Birmingham finance-posted GR; Hamilton and Monterrey L0.

1. Position in the stack

Many AP “exceptions” are warehouse facts that never reached Finance. This agent assembles a **receiver packet**: PO, invoice lines waiting, what ERP/WMS already shows, what is missing, and a single question. It does not create a goods receipt to clear GR/IR. Creating a receipt is a warehouse or plant-finance human action under their procedure.

2. Job description

The Goods Receipt Agent detects invoices (or match breaks) waiting on a receipt or quantity confirmation, builds a packet for the named receiver or liaison, and records the response. It monitors aging of unmatched receipts and of invoices waiting for receipts. It does not post GR, reverse GR, or tell the warehouse to “just receive it so we can pay.”

3. Operating intent and cadence

Mode	Cadence	Output
Object-driven	On Triage next=05	Packet
Plant digest	08:00 local, SSC days	List of open packets by receiver
Aging	Daily 16:00 ET	GR/IR and waiting-invoice ages (signal, not accrual)
Response capture	Continuous	Coded warehouse reply

Accrual booking is Agent 13 + Controller. This agent supplies a cleaner list; it does not book.

4. Inputs

Input	Source	Mandatory
Triage record + match worksheet	Agents 03/04	Y
PO remaining qty, delivery schedule, plant/warehouse	ERP	Y
Existing receipts / inbound deliveries / ASN	ERP / WMS	Y
Receiver / buyer names on PO	ERP	Y
Plant calendar / shutdowns	Plant ops	N
Prior packets on same PO	Orchestrator	N
Invoice image (line qty)	Evidence store	Y

5. Tools / data required

Tool	Privilege	ERP / WMS
PO + inbound	Read	D365 Purch / WHS; SAP INB/EKBE; Oracle Receiving; NetSuite IR; Workday Receipt
WMS inventory moves	Read	May explain "it's on the dock"
Orchestrator	Write packets	—
Internal directory	Read	Liaison + supervisor
Plant digest channel	Write digest (L1 recommend; send at L2 if chartered)	Teams/email

No GR post privilege on the agent account. Ever. A human uses a personal or warehouse account.

6. Responsibilities

1. Confirm class is receipt-related. If the break is price-only, refuse and return to Triage.
2. Reconstruct facts: ordered, previously received, invoiced, open.
3. Ask one primary question: NOT_RECEIVED confirm? QTY_DISPUTE count? GR_AMBIGUOUS which receipt? DOCK_NOT_POSTED is it physical?
4. Name the receiver from PO / WMS task, not from memory.
5. Produce packet + plant digest line.
6. Capture response codes: WILL_POST_GR, NOT_OURS, SHORT, OVER, DAMAGED, WRONG_ITEM, ALREADY_POSTED (point to ID), NEED_BUYER.
7. If WILL_POST_GR, wait for ERP evidence; do not take a verbal as a receipt.
8. If silent past SLA, hand to Agent 09 with the same packet (no new narrative).
9. Feed aged unmatched GR and aged waiting invoices to Agent 13 as *candidates*, labelled as such.

7. Explicit exclusions

1. Payment authorisation or release.
2. Creating, reversing, or dummy-posting GR/SES.

3. Instructing staff to receive goods they have not received.
4. Changing PO quantity.
5. Booking GR/IR accruals.
6. Supplier communication (Agent 08) unless Triage redirects.
7. Clearing quantity breaks by applying a hidden tolerance.
8. WMS configuration changes.
9. Speaking for “the plant” without a named human.

8. Human owner

Site	Northline liaison (illustrative)	Escalation
Dayton (1000)	Plant Finance Liaison R. Patel	Warehouse Manager → Plant Controller
Birmingham (1100)	Plant Finance Liaison A. Cole	Same
Hamilton / Monterrey	Not live	—
Process owner of agent	Exception Desk Lead (Tomasz)	AP Manager

Liaisons own fact quality. The Desk owns packet standard. Split is deliberate.

9. Approval requirements

Action	Human
Packet content at L1	Liaison or receiver accepts facts / answers
Sending plant digest	L1: digest is a recommend in Orchestrator; L2: auto-send to named lists
GR post	Warehouse / appointed clerk only
Treat verbal as done	Never — wait for ERP
Close missing-GR exception	Match Lead after ERP GR + rematch

10. Escalation criteria

Condition	To	Timing
No response 2 business days	Supervisor + Agent 09	On day 2
No response 5 business days	Plant Controller	On day 5
WILL_POST_GR but no ERP GR in 2 days	Liaison	Day 2 after promise
Physical on dock > 3 days, no post	Warehouse Manager	Day 3
Repeated NOT_OURS on same vendor	Buyer + Agent 06	Weekly
Safety / damaged goods	Plant quality — human path	Immediate

11. Output standard

Packet minimum: object ID; PO; plant; warehouse; receiver name; ordered/received/invoiced table; invoice image link; question enum; requested action; SLA; liaison; agent 05; level; evidence URI.

Response record: code; responder identity; timestamp; ERP document ID if claimed; “verbal only” flag (always insufficient).

12. Control requirements

Control	Support	Test
No dummy GR	Agent cannot post	Access review + sample new GRs vs packets
Evidence before rematch	ERP GR ID required	Join 03 rematch
Named receiver	Matrix	Sample 20 packets
GR/IR listing ≠ booking	Labelled candidates	Close review

13. Audit evidence

Packets + responses 7 years; plant digests 1 year; access 1 year; candidate lists supplied to Close 7 years.

14. KPIs

KPI	Definition	Use
Packet completeness	Required fields present	Quality
Time to first coded response		Flow
Verbal-only rate	Should trend down	Control
Promise-break (WILL_POST without GR)		Plant discipline
Mis-question rate	Recode by liaison	Agent quality
Cost / 1,000 packets		Brake

Do not KPI “GR created by agent” — that would be a control failure.

15. Performance history fields

Period; packets; classes; response codes; verbal-only; promise-breaks; Agent 09 handoffs; plants; level; incidents; cost.

16. Autonomy level

Level	Behaviour
L0	Shadow packets
L1	Packets in Orchestrator; humans send/answer
L2	Auto-send digest to named internal lists
L3	Auto-send individual packet to named receiver whitelist
L4	Exception oversight of whitelist send

No L3 GR post. Dummy GR is a dismissal offence for a human; it is impossible for a correctly privileged agent.

17. Failure handling

Failure	Action
WMS/ERP read fail	No packet from stale stock; mark DATA_UNAVAILABLE
Receiver not on PO	Assign liaison; do not guess
Duplicate packets	One packet per object; digest can summarise
Plant shutdown	SLA pause per calendar; do not escalate as neglect

18. Cost monitoring

Model cost, liaison minutes, chase minutes (09). Pause if mis-question > 20% for 2 weeks.

19. Handoffs

04 → 05; 05 → receiver/liaison; 05 → 09; 05 → 13 candidates; 05 → 03 rematch seed when ERP GR exists; 05 → 08 if warehouse confirms invoice error; 05 → 16.

20. Configuration parameters

Parameter	Northline start
First chase	2 business days
Plant Controller escalate	5 business days
Promise window	2 days
Live plants	Dayton, Birmingham

21. First 90 days

Map who actually receives at each plant — PO “receiver” fields are often wrong. Fix the matrix before L2 auto-send. Measure verbal-only; that is the behaviour to change, not “automation.”

22. Worked example — Northline Industrials

W0-1048821 qty 1,000 invoiced vs 980 received, Dayton WMS.

Packet to L. Grant (Warehouse Supervisor) and R. Patel (liaison): line 7 item FS-440, PO 45007821, receipt PR-88311 980 EA, invoice 1,000 EA, question QTY_DISPUTE. Asks: recount, confirm short-ship, or confirm overbill. Does not ask “can you receive 20 more.”

L. Grant responds SHORT — two boxes damaged, claim with carrier. Agent 08 later drafts a debit-note query to Lakeshore. Agent 03 will rematch at 980 if a credit or a reduced invoice arrives. Nobody posts 20 EA.

Parallel missing GR W0-1050022 Birmingham \$6,400. Digest 08:00 CT to A. Cole. Response DOCK_NOT_POSTED. Human clerk posts GR that afternoon. Agent 05 records ERP ID. Object returns to Agent 03.

23. Sample output artefact (abridged)

```
object_id: W0-1048821
agent_id: 05
plant: Dayton
question: QTY_DISPUTE
ordered: 1000
received: 980
invoiced: 1000
receiver: L. Grant
liaison: R. Patel
ask: recount or confirm short / overbill
forbidden_ask: post 20 EA
autonomy_level: L1
```

24. What this agent does not replace

WMS, receiving SOP, cycle count, or plant leadership. It replaces the AP email that says “please receive this PO” with a fact packet.

25. Document control

Field	Value
Spec	AGENT_05 Goods Receipt
Default autonomy	L1 (start L0)

AGENT 06 — PO Quality

Stack position: Observes POs that generate AP exceptions and, where chartered, inspects new POs for defect patterns *before* invoice. Feeds Root Cause and Procurement Operations. Does not rewrite POs.

Default autonomy: L0 Observe (promotion to L1 recommend-only).

Human owner (typical): Procurement Operations Lead

Payment authority: None.

Northline instance: Wave 3 in the stack overview; UOM and missing-tax-on-PO are the first study objects.

1. Position in the stack

AP cannot match what Procurement did not specify. This agent exists so Finance Transformation and Procurement look at the **same defect list**, not at each other. It is not a purchasing cop and not a second buyer.

Default L0 is intentional. A recommend-level PO agent without buyer trust becomes ignored mail.

2. Job description

The PO Quality Agent identifies purchase orders (and change orders) whose attributes are known to cause AP breaks: missing or conflicting UOM, price unit vs order unit, missing tax/jurisdiction fields, vendor on PO ≠ intended payee, incomplete lines, missing account assignment on non-stock, outline agreement not referenced, receipt-required flag wrong for the item type, and service POs without a contract reference when policy requires one.

It produces defect records and, at L1, recommended buyer actions. Buyers decide. The agent does not change the PO.

3. Operating intent and cadence

Mode	Cadence	Output
Retrospective	Daily	POs linked to new Agent 03/04 breaks
Prospective (optional, later)	Near-real-time on PO approve	Defect hints <i>after</i> buyer submit, before invoice
Weekly	Friday	Top defects by buyer group / plant / vendor
Monthly	With Agent 15	Proposed template / catalogue fixes

Northline Wave 1: retrospective only, L0. Prospective L1 is a separate charter.

4. Inputs

Input	Source	Mandatory
Exception objects with PO_DEFECT, UOM, PRICE, TAX on PO	03/04	Y for retrospective
PO header/lines / change history	ERP	Y
Item master UOM and order unit	ERP / PIM	Y for UOM class
Vendor master payee vs PO vendor	ERP	Y
Contract / outline agreement	CLM / ERP	For services policy
PO template catalogue	Procurement	N
Policy AP-P0Q-001 defect list	Controlled	Y

5. Tools / data required

Tool	Privilege
PO and change log	Read
Item / UOM	Read
Contract header	Read
Exception store	Read
Orchestrator	Write defect records (L1); shadow folder at L0
Buyer directory	Read

No PO change privilege. No invoice privilege.

6. Responsibilities

1. On each tagged exception, pull the PO version at *invoice date*, not only current (changes after invoice are a different story).
2. Test defect list. Code each hit. A PO may have multiple defects.
3. Distinguish DEFECT_ON_PO vs INVOICE_DISAGREES_WITH_GOOD_PO — the latter is not PO quality.
4. Attribute to buyer group / plant / vendor / item category for clustering.
5. At L0, publish a weekly shadow register. At L1, open a defect record for the buyer with a recommended action (CONFIRM_UOM, ADD_CONTRACT_REF, CORRECT_TAX_JURISDICTION, SPLIT_FREIGHT_LINE, etc.).
6. Never recommend “change the PO to match the invoice” as a default. That is a commercial decision. The recommendation is “resolve the specification.”
7. Feed Agent 15 with defect frequencies. Feed Agent 08 only when the defect requires a supplier master or supplier process change — still as a draft path.
8. Track whether a defect was fixed on a *change order* by a human, and whether subsequent invoices on that PO still broke.

7. Explicit exclusions

1. Payment authorisation or release.

2. Creating or changing POs, contracts, or item masters.
3. Directing a buyer to raise price to match an invoice.
4. Vendor selection or sourcing.
5. Approving spend.
6. Blocking PO release in the ERP at L0/L1 (a future hard-stop is a Procurement system control, separately chartered, not this agent “taking over release”).
7. Scoring buyers for performance management without HR/Procurement design (the weekly pack is diagnostic, not a bonus metric unless Procurement says so — AP does not).
8. Tax determination.
9. Claiming catalogue “compliance.”

8. Human owner

Field	Northline
Role / name	Procurement Operations Lead / Hannah Cho
AP counterpart	Priya Shah (match codes must stay aligned)
Escalation	CPO + AP Process Owner (James Okoye)
Owns	Defect list with Controller/AP; buyer communication standard
Does not own	AP posting, cash, tax position

9. Approval requirements

Action	Human
Defect list change	Procurement Ops + AP Match Lead + Controller if match policy affected
L1 defect record to buyer	Buyer accept/reject/defer
PO change	Buyer / authorised change process
Prospective scan on	Process Owner + Procurement
Any block of PO release	Out of this agent unless a separate control project

10. Escalation criteria

Condition	To	Timing
Defect class $\geq$ 15% of a plant's AP exceptions for 4 weeks	Plant Controller + Buyer manager	Monthly pack ad hoc
Same buyer group, same defect, $\geq$ 20 POs	Buyer manager	Weekly
Vendor systematically invoices different UOM	Vendor Master + Agent 08 path	Weekly
Price-on-PO systematically blank	Procurement Ops	Immediate (policy)
Agent attributes defect to invoice when PO was good	Match Lead (recode)	On complaint

11. Output standard

Defect record: PO, version/date, object IDs of related invoices, defect codes, ON_PO vs INVOICE_SIDE, recommended action (L1), buyer group, hashes of policy, evidence (PO snapshot + exception codes), agent 06, level.

Weekly register: top 10 codes, counts, example POs, whether a template fix is proposed (pointer to Agent 15).

12. Control requirements

Control	Support	Test
Snapshot at invoice date	Versioned PO read	Sample 15
No silent PO change	Access	Quarterly
Diagnostic vs punitive	Pack labelled diagnostic	Language review
Alignment with match codes	Joint taxonomy	Monthly with Agent 03

13. Audit evidence

Defect records 7 years; weekly registers 7 years; policy versions life-of-programme; PO snapshots stored with defects 7 years.

14. KPIs

KPI	Definition	Use
Attribution precision	% defects confirmed ON_PO by buyer/AP	Quality
Repeat-break after change order		Whether fixes work
Coverage	% of PO-tagged exceptions reviewed	L0/L1
Buyer response rate (L1)		Engagement, not obedience
Cost		Brake

No “PO compliance %” as a guaranteed control outcome.

15. Performance history fields

Period; exceptions reviewed; defects by code; attribution flips; buyer accept/reject; repeat-breaks; plants; level; incidents; cost.

16. Autonomy level

Level	Behaviour
L0	Shadow register
L1	Defect records to buyers
L2	Prepare change-order packet (fields suggested, not submitted)
L3	Not expected for PO write. Do not plan L3 PO change
L4	n/a for writes

Hard ceiling: L2 prepare. PO change remains human.

17. Failure handling

Failure	Action
Cannot read historical PO version	Mark VERSION_UNAVAILABLE; do not use current only without a flag
Item master UOM missing	Code MASTER_UOM_MISSING to Master Data, not the buyer alone
CLM down	Skip contract tests; do not PASS them

18. Cost monitoring

Model + analyst minutes. Pause L1 if attribution precision < 70% for 3 weeks — you are training buyers to ignore the pack.

19. Handoffs

03/04 → 06; 06 → buyers; 06 → 15; 06 → 08 (supplier process); 06 → Master Data; 06 → 16.

20. Configuration parameters

Parameter	Northline start
Mode	Retrospective L0
First defect list	UOM, blank price, missing contract ref on services, receipt flag vs item type
Cluster for escalate	20 POs or 15% of plant exceptions

21. First 90 days

Only UOM and blank price. Prove attribution precision. Hannah and Priya meet weekly for 30 minutes on 10 examples. If they cannot agree, do not go L1.

22. Worked example — Northline Industrials

Agent 15 notes 22% of Dayton exceptions are UOM EA vs BOX. Agent 06 L0 review of 40 POs: 31 have order unit BOX and price unit EA with no conversion on the PO; item master has conversion 50 EA/BOX. Invoices from two fastener vendors bill EA.

Attribution: DEFECT_ON_PO (conversion not carried) **and** a supplier billing-practice issue. Not “warehouse error.”

L0 weekly register lists the 31 POs, the two vendors, and a proposed template: always explode conversion on fastener category. Hannah rejects auto-changing 31 live POs. She accepts a catalogue change *going forward* and a buyer aid. Agent 06 does not submit the catalogue change.

A bad recommendation would have been: “raise PO qty to EA to match invoices.” That hides the conversion problem.

23. Sample output artefact (abridged)

```
po: 45007821
po_version_date: 2026-03-12
defect_codes: [UOM_CONVERSION_ABSENT]
attribution: ON_PO
related_objects: [W0-1048821]
recommend_L1: CONFIRM_CONVERSION_ON_TEMPLATE (not live)
buyer_group: Dayton MRO
agent_id: 06
autonomy_level: L0
```

24. What this agent does not replace

Catalogue management, buying policy, or supplier contracting. It makes PO-caused AP breaks visible in procurement language.

25. Document control

Field	Value
Spec	AGENT_06 PO Quality
Default autonomy	L0

AGENT 07 — Approval

Stack position: Non-PO path after Validation PASS; also over-tolerance and policy exceptions that require DOA. Chases packets. Does not approve.

Default autonomy: L1 Recommend.

Human owner (typical): AP Approvals Coordinator

Payment authority: None. Approval of an invoice is not payment authorisation.

Northline instance: DOA table FIN-DOA-2026.2; cost centre owner to \$5k; Plant Controller to \$25k; VP Finance to \$100k; CFO above.

1. Position in the stack

Approval is a **routing and completeness** job. The agent builds a packet a human approver can decide in one sitting: what it is, why it is non-PO or over-tolerance, who the DOA says must sign, what has already been received, and what is still missing. The agent never clicks Approve. Substitute approvers come only from the DOA table. Orchestrator does not “find someone senior.”

2. Job description

The Approval Agent determines the required approver(s) from the written DOA, assembles the packet (image, extract, GL/cost centre suggestion *as already on the request or PO exception*, prior related invoices, policy citations), routes to the named approver or deputy, records the decision, and chases through Agent 09 if silent. It does not invent cost centres, does not approve on someone’s behalf, and does not treat a Slack “ok” as an approval unless that channel is in the controlled procedure (Northline: it is not).

3. Operating intent and cadence

Mode	Cadence	Output
Object-driven	On path NON_PO_APPROVAL or match over-tolerance requiring DOA	Packet + route
Reminder	Per SLA table	Agent 09 handoff
Digest	Daily to each approver with >0 open	List
Monthly	DOA stale-user review with HR	Exceptions

4. Inputs

Input	Source	Mandatory
Validated non-PO object or over-tolerance worksheet	02 / 03 / 04	Y
DOA table (amount, account, plant, commodity)	Controlled + ERP workflow	Y
Cost centre / project owner	ERP	Y if coded
Deputy / out-of-office	DOA / HR, not calendar scraping alone	Y for routing
Invoice image + extract	01	Y
Related spend (same vendor+CC, 90 days)	ERP	N — context
Policy: when non-PO is allowed	Procurement / Finance	Y

5. Tools / data required

Tool	Privilege
DOA table	Read, version-hashed
ERP workflow read	Read (do not approve)
Directory + deputies	Read
Orchestrator	Write route state
Evidence store	Read attachments

ERP mapping: D365 workflow; SAP release strategy / Fiori; Oracle Approval Management; NetSuite approval routing; Workday approval chains. The agent **mirrors** the official engine. It does not become a second unofficial approver.

6. Responsibilities

1. Confirm the object is in an approval class. PO in-tolerance match does not come here.
2. Compute required steps from DOA (amount *and* account *and* plant). Document the path.
3. If cost centre missing on a non-PO, QUERY_CODING to the requester — do not guess GL.
4. If requester is also the only approver and policy forbids self-approval, route to the next step in DOA.
5. Build packet. Include “why non-PO” if a PO should have existed (flag for Agent 06, still needs approval).
6. Route to named human. If OOO and deputy listed, route deputy. If no deputy, wait and escalate to Approvals Coordinator — do not climb the org chart.
7. Accept only decisions from the official workflow or the written alternate (Northline: D365 workflow). Record reject reasons.
8. On approve, hand to posting clerk (human) or parked-document path — not to payment.
9. On reject, hand to requester / Agent 08 as directed; do not shop for another approver.

7. Explicit exclusions

1. Payment authorisation or release (a different two humans on the payment run).
2. Clicking Approve or using a service account in the approval step.
3. Self-approval workarounds.
4. Inventing deputies or “acting” approvers.
5. Accepting informal channels (chat, verbal, SMS) unless the procedure lists them — Northline does not.
6. Changing DOA limits.
7. Coding the invoice to make it fall under a friendlier approver.
8. Treating invoice approval as vendor-master or bank-change approval.
9. Auto-approving recurring utilities unless a *separate* recurring-control procedure exists (Northline Wave 3, human-designed).

8. Human owner

Field	Northline
Role / name	AP Approvals Coordinator / Nina Alvarez
Backup	Exception Desk Lead
Escalation	AP Manager; Controller on DOA disputes
Owns	Packet standard, chase SLA, stale-approver list
Does not own	DOA policy (Controller / CFO), cash

9. Approval requirements

This section is the point of the agent.

Decision	Who	System
Invoice / exception approve or reject	DOA-named human or deputy	D365 workflow
Packet completeness at L1	Coordinator sample	Orchestrator
Route start	Agent recommends; Coordinator confirms if QUERY	L1
Recurring utility auto-route	Not in force	—
Post after approve	AP clerk	ERP
Pay after post	Agent 12 + two authorisers	Payment run

10. Escalation criteria

Condition	To	Timing
No deputy, approver OOO	Coordinator	Day 0
No decision 3 business days	Approver's manager as <i>information</i> , not as approver unless DOA says so	Day 3
No decision 7 business days	Controller (process), still not a substitute approve	Day 7
Self-approval attempt	Coordinator + Controls	Immediate
Informal "ok" only	Coordinator — reject as evidence	Immediate
DOA table vs ERP workflow mismatch	Controller + IT	Same day

11. Output standard

Packet: object ID; amount; currency; vendor; why-approval-required; DOA path (steps, names, hashes); coding as supplied; related spend; image; "not payment"; agent 07; level.

Route event: step, person, sent-at, due, decision, system transaction ID of the workflow.

12. Control requirements

Control	Support	Test
DOA	Applied, not edited	25 packets/month vs table
No agent approve	Access: no approve privilege	Quarterly + transaction sample
No informal approve	Channel rule	Sample rejects
Self-approval	Block + escalate	Inject test
Invoice approve ≠ pay	Separate run	Payment SoD

13. Audit evidence

Packets 7 years; workflow IDs 7 years; DOA versions life-of-programme; informal-attempt incidents 7 years; access 1 year.

14. KPIs

KPI	Definition	Use
Packet completeness	Approver asked zero clarifying questions (sample)	Quality
Time to route	PASS → correct inbox	Flow
Mis-route rate	Wrong person per DOA	Quality
Informal-attempt count		Control culture
Aging >7 days		Management
Cost		Brake

Do not KPI “approval rate” — that pressures approvers.

15. Performance history fields

Period; packets; DOA steps; mis-routes; informal attempts; self-approval blocks; aging; decisions approve/reject; level; incidents; cost.

16. Autonomy level

Level	Behaviour
L0	Shadow routes
L1	Recommend route + packet; Coordinator/workflow is SOR
L2	Auto-start official workflow for packets that pass completeness on a whitelist (e.g. utilities with standing coding)
L3	Auto-chase via Agent 09 only
L4	Exception oversight of chase

Approve action: never an agent privilege.

17. Failure handling

Failure	Action
DOA hash missing	Stop routing
Workflow API down	Queue packets; do not collect chat approvals
Approver left company	HR + Controller; freeze objects on that person
Amount in MXN/CAD	Convert per DOA rule (Northline: company currency); document rate source

18. Cost monitoring

Coordinator minutes, reminder volume, model. Pause L2 if mis-route > 5% — wrong people seeing invoices is a control issue.

19. Handoffs

02/03/04 → 07; 07 → DOA human; 07 → 09; 07 → 06 (should-have-been-PO); 07 → clerk post; 07 → 16; never 07 → bank.

20. Configuration parameters

Parameter	Northline start
DOA	\$5k / \$25k / \$100k / CFO
Reminder	Day 3 inform manager; day 7 Controller
Official channel	D365 workflow only
Self-approval	Forbidden

21. First 90 days

Reconcile DOA document to D365 workflow. Most “approval automation” failures are those two disagreeing. Stay L1 until mis-route < 2% on a 100-sample.

22. Worked example — Northline Industrials

W0-1049104 non-PO services \$10,000, Dayton cost centre 5310, owner J. Hale.

DOA: \$10,000 > \$5k, ≤ \$25k → Plant Controller (Dayton) M. Singh, after cost-centre owner. Packet includes: last three invoices same vendor+CC (\$4,200, \$4,200, \$4,200) — context, not a reason to auto-approve. Agent 06 flagged “should have been PO” — still needs approval; flag is visible.

J. Hale OOO, deputy in DOA is K. Adeyemi. Route: Adeyemi then Singh. A buyer IMs Nina “just approve it.” Nina records informal-attempt; no approval.

After both workflow approves, clerk parks/posts. Thursday payment run still needs two payment authorisers. Hale’s deputy did not authorise cash.

23. Sample output artefact (abridged)

```
object_id: W0-1049104
agent_id: 07
why: NON_PO
doa_path: [CC owner deputy K. Adeyemi, Plant Controller M. Singh]
doa_hash: FIN-DOA-2026.2
self_approval: blocked_n/a
channel: D365 workflow
not_payment: true
autonomy_level: L1
```

24. What this agent does not replace

DOA policy, workflow configuration, or the approver's judgement. It replaces incomplete "please approve attached" emails.

25. Document control

Field	Value
Spec	AGENT_07 Approval
Default autonomy	L1 (start L0)

AGENT 08 — Supplier Resolution

Stack position: Drafts external queries when Triage says the next fact sits with the supplier. Tracks replies. Never commits the company.

Default autonomy: L1 Recommend (draft only).

Human owner (typical): AP Supplier Desk / Vendor Master Lead

Payment authority: None.

Northline instance: Shared mailbox ap.suppliers@northline.example; send only after human approval; no legal or settlement language.

1. Position in the stack

Supplier Resolution is a **correspondence factory with a muzzle**. It writes the question AP meant to ask, attaches the facts, and files the reply. It does not negotiate, promise a pay date, agree a price, or accept a concession. If a sentence would bind Northline, it does not belong in the draft.

2. Job description

The Supplier Resolution Agent prepares supplier-facing drafts for coded issues: invoice error, missing credit, quantity overbill, price disagreement, missing tax fields, statement-only items, and dunning responses that are factual. A named human edits and sends (or, at earned L2, approves send of a whitelist template). The agent files inbound replies to the work object and extracts *claimed* facts for humans to verify in ERP.

3. Operating intent and cadence

Mode	Cadence	Output
Object-driven	On Triage next=08 or Intake class dunning	Draft
Inbound file	Mail poll	Reply attached to object
Aging	Daily	Sent-waiting-reply
Weekly	Friday	Vendor-level repeat issues → Agents 06/15

4. Inputs

Input	Source	Mandatory
Triage + worksheets + GR facts	03/04/05/11	Y
Allowed template book AP-SUP-001	Controlled	Y
Forbidden-phrase list	Controlled	Y
Vendor contact (AR email)	Vendor master / last letter	Y — else QUERY
Prior correspondence	Evidence store	N
Dunning / statement artefacts	01 / 11	If that path

5. Tools / data required

Tool	Privilege
Template book + forbidden list	Read, hashed
Mailbox	Read inbound; no send at L0/L1
Vendor contacts	Read
Orchestrator	Write drafts and reply files
ERP	Read to verify claimed invoice/credit IDs

L2 send, if ever chartered, uses a controlled send path with an approval token — still not a free mailbox.

6. Responsibilities

1. Choose a template from the book. If no template fits, `QUERY_TEMPLATE` — do not free-compose commercial language.
2. Fill facts only: invoice numbers, quantities, PO, what warehouse recorded, what AP shows open. No adjectives about the supplier.
3. Run forbidden-phrase scan (see §7 and §20). Fail the draft if any hit.
4. Present draft to Supplier Desk human. Record edits (they train the template book, not a hidden prompt).
5. After human send, start reply SLA (Northline: 7 calendar days monitor, not a contractual deadline promised to the vendor).
6. File replies. Extract claimed credit numbers, revised invoices — mark `UNVERIFIED` until ERP shows them.
7. Never add a statement line to a payment proposal because the supplier says it is overdue.
8. Dunning response template: facts of what is open, what is held and why (process, not argument), who at Northline owns it. No pay-date promise.

7. Explicit exclusions

1. Payment authorisation, promised pay date, or “we will include you on Thursday.”
2. Agreeing price, quantity, interest, or concession.
3. Legal, collection, or settlement language.
4. Vendor bank-detail change (“please use this account”) — route to Vendor Master dual control; do not confirm.
5. Sending at L0/L1.
6. Free-composing outside the template book.
7. Threatening or blaming language.
8. Discussing other vendors or volume commitments.
9. Calling a duplicate flag “fraud.”
10. Binding tax advice.

Forbidden phrases (scan list — extend in the charter):

will pay, will be paid, payment date, we agree, we accept, please update the P0, we authorise,

settlement, without prejudice (unless Legal template), fraud, guaranteed, as per our verbal, CFO approved, ignore the discrepancy.

8. Human owner

Field	Northline
Role / name	AP Supplier Desk Lead / Olivia Grant
Backup	Vendor Master Lead
Escalation	AP Manager; Legal if supplier sends legal notice
Owns	Template book with Legal/AP; send discipline; contact quality
Does not own	Buying, cash, disputes above threshold

9. Approval requirements

Action	Human
Every send at L1	Supplier Desk (or named deputy)
Template add/change	Desk + AP Manager + Legal if commitment-adjacent
L2 auto-send whitelist	Only <code>missing_copy</code> and <code>please_resubmit_tax_fields</code> if gates pass
Concession / debit / credit accept	Buyer or Controller per threshold
Bank change in inbound mail	Vendor Master dual control — stop this agent

10. Escalation criteria

Condition	To	Timing
Inbound legal notice / attorney	Legal + AP Manager	Immediate; agent stops drafting
Inbound new bank details	Vendor Master + Controls	Immediate
No reply 7 days	Desk; optional second template <code>chaser_facts_only</code>	Day 7
Stop-ship threat	Procurement commercial human	Immediate; agent does not reply
Repeat same issue ≥ 5 invoices	Agents 06/15 + buyer	Weekly
Forbidden phrase in <i>human</i> edit	Desk Lead coaching	Same day

11. Output standard

Draft: object ID; template ID + hash; filled facts; scan result `CLEAN`; language `no_commitment`; recipient; owner; agent 08; level; “not sent.”

Send record (human): message ID, time, person.

Reply record: attachment URI; extracted claims `UNVERIFIED`; flags `LEGAL`, `BANK_CHANGE`, `STOP_SHIP`.

12. Control requirements

Control	Support	Test
No unauthorised send	No send privilege L1	Mail gateway
No commitment language	Scan + sample	25 drafts/week
Bank change path	Divert	Inject test
Legal path	Divert	Tabletop
SoD	Desk ≠ payment releaser	Roles

13. Audit evidence

Drafts, edits, sends, replies 7 years; template versions life-of-programme; incident (legal/bank) 7 years; scan logs 1 year.

14. KPIs

KPI	Definition	Use
Scan fail rate	Drafts blocked	Template quality
Human edit distance	Material rewrites	Template fit
Mis-template rate	Desk changes template	Taxonomy
Unverified-claim leak (used as fact)	Should be ~0	Control
Time to human send		Flow
Cost		Brake

No “supplier satisfaction” or savings KPI.

15. Performance history fields

Period; drafts; templates; scan fails; sends; legal/bank/stop-ship flags; edit distance; unverified leaks; level; incidents; cost.

16. Autonomy level

Level	Behaviour
L0	Shadow drafts
L1	Drafts; human send
L2	Prepare + one-click send after human token; optional whitelist auto-send of non-commitment templates
L3	Auto-send <i>only</i> missing_copy / tax-field resubmit on whitelist vendors
L4	Exception oversight of that send list

No L3 commercial letters. No pay-date template at any level.

17. Failure handling

Failure	Action
No contact on vendor	QUERY Vendor Master; do not use a personal buyer email as if it were AR
Scan engine down	No draft release
Mailbox send accidentally granted	Incident; revoke; L0
Reply cannot attach to object	Human file; do not lose the letter

18. Cost monitoring

Model, desk edit minutes. Pause L1 if edit distance stays high — templates are wrong. Pause all send paths if any unauthorised send occurs.

19. Handoffs

04/05/11/01(dunning) → 08; 08 → human send; inbound → 08 file; 08 → 03 when verified credit/revised invoice in ERP; 08 → Vendor Master on bank; 08 → Legal on notice; 08 → 06/15; 08 → 16.

20. Configuration parameters

Parameter	Northline start
Template book	AP-SUP-001
Forbidden list	\$7
Reply monitor	7 days
Live send	Human only
Languages	EN drafts; ES/FR human only in Wave 1

21. First 90 days

Write five templates only: missing invoice copy, tax-field resubmit, qty overbill question, price question (no accept), dunning factual response. Legal reads them once. Measure edit distance. Do not invent a sixth template in a prompt.

22. Worked example — Northline Industrials

W0-1048821 warehouse confirmed SHORT 20 EA damaged.

Template qty_question_v3. Facts: PO 45007821, invoice 459102, received 980, invoiced 1,000, item FS-440. Question: credit 20 EA or revised invoice? No “we will pay the 980 on Thursday.” No “we agree the unit price.”

Olivia edits one sentence for tone, sends. Reply three days later: “Credit CM-992 will issue.” Agent files, mark UNVERIFIED. When D365 shows CM-992, Agent 03 rematch path opens. Payment proposal never included a promised date.

Same week: dunning from a tooling vendor, \$27,000. Intake classed dunning. Draft lists what is open vs held (QTY_OVER on another object) with object-safe language. Supplier Desk sends. Agent 12 is not told to add the dunning amount.

Inbound: “New wiring instructions.” Agent stops. Vendor Master + Controls. Dual-control procedure. Agent 08 does not thank them or confirm.

23. Sample output artefact (abridged)

```
object_id: W0-1048821
agent_id: 08
template: qty_question_v3
scan: CLEAN
commitment: false
pay_date_mentioned: false
state: DRAFT_NOT_SENT
owner: Olivia Grant
autonomy_level: L1
```

24. What this agent does not replace

Vendor management, Legal, commercial negotiation, or Vendor Master. It replaces ad-hoc AP emails that accidentally promise cash.

25. Document control

Field	Value
Spec	AGENT_08 Supplier Resolution
Default autonomy	L1 (start L0)

AGENT 09 — Internal Follow-up

Stack position: Chases silent internal owners after Agents 05, 07, and other packets have a named human and an SLA. Does not re-triage from scratch and does not escalate to a convenient executive.

Default autonomy: L1 Recommend.

Human owner (typical): Exception Desk Lead

Payment authority: None.

Northline instance: Reminder copy book AP-IFU-001; official channels only (Orchestrator + email to corporate addresses).

1. Position in the stack

Internal Follow-up is the difference between a packet and a rumour. Triage names an owner. GR or Approval builds the packet. This agent notices silence and sends a complete fact pack again — not “any update?” — then escalates along the written ladder. It does not punish plants. It does not approve, receive, or pay to make the reminder go away.

2. Job description

The Internal Follow-up Agent monitors work objects in `waiting_internal` past the first-response SLA, emits a reminder that restates the question and the facts, records touches, and at the next rung informs the owner’s manager *as specified in the matrix*. It never adds a new technical judgement. If facts have changed, it returns the object to the owning agent (05/07/04) for a new packet.

3. Operating intent and cadence

Mode	Cadence	Output
Sweep	09:30 and 14:30 ET SSC days	Due / overdue list
Reminder propose	On first SLA breach	Draft reminder
Escalate propose	On second rung	Inform packet
Daily	16:45 ET	Aging by owner and plant

4. Inputs

Input	Source	Mandatory
Objects <code>waiting_internal</code>	Orchestrator	Y
Original packet (05/07/04)	Evidence store	Y
SLA + ladder	Taxonomy / matrix	Y
OOO / deputy	Same source the owning agent used	Y
Touch history	Orchestrator	Y
Copy book AP-IFU-001	Controlled	Y

5. Tools / data required

Tool	Privilege
Orchestrator	Read/write status and touches
Copy book	Read, hashed
Corporate mail / Teams (named lists)	Draft at L1; send at L2 if chartered
Directory	Read

No ERP write. No supplier mailbox. No personal (non-corporate) addresses.

6. Responsibilities

1. Select objects that are waiting on an *internal* party and are at or past SLA. Exclude waiting_supplier (Agent 08) and held anomaly (Agent 10).
2. Confirm the packet is still current (no new GR, no new approval). If stale, bounce to owning agent — do not chase a dead question.
3. Draft reminder: original question, facts table, link, due, who asked. Tone: neutral, specific, one ask.
4. At L1, Exception Desk releases the reminder. At L2, auto-send on a whitelist of rungs 1 only.
5. Record touch. Increment rung.
6. Rung 2: inform the manager listed in the matrix. The manager is not thereby the approver or the receiver unless DOA/matrix says so.
7. Rung 3: AP Manager + Plant Controller (plant-owned) or Controller (finance-owned). Still no substitute action.
8. Stop chasing if the object leaves waiting_internal.
9. Detect chase storms (same person > 15 open reminders) and notify Desk Lead — staffing/design issue, not more mail.

7. Explicit exclusions

1. Payment authorisation or “we will pay if you reply.”
2. Approving, posting GR, changing PO, or coding invoices.
3. Escalating to an executive not on the ladder (“copy the CFO”).
4. Chasing suppliers.
5. Chasing on anomaly holds (Controls owns that clock).
6. New technical analysis in the reminder body.
7. Public shaming lists (leaderboards of named individuals) without HR/Plant agreement — Northline packs are role-aged, names only on the owner’s own digest.
8. SMS or personal email.
9. Closing the exception because the owner is silent.

8. Human owner

Field	Northline
Role / name	Exception Desk Lead / Tomasz Wójcik
Backup	Nina Alvarez (Approvals) for 07-origin; R. Patel for 05-origin
Escalation	AP Manager
Owns	Copy book, rung timing, storm detection
Does not own	Plant discipline, DOA

9. Approval requirements

Action	Human
Rung 1 send at L1	Desk
Rung 2+	Desk Lead confirm first 90 days; later L2 if gates pass
Ladder change	Process Owner + Plant Controllers
Copy book change	Desk + Communications/HR if tone
Pause chases to a plant (shutdown)	Liaison

10. Escalation criteria

Condition	To	Timing
Rung 1 due	Owner	SLA
Rung 2	Manager (inform)	+2 business days
Rung 3	AP Manager + Plant Controller / Controller	+2 further
Chase storm	Desk Lead + owner's manager	Same day
Packet stale mid-chase	Owning agent	Immediate
Harassment complaint	HR + Process Owner	Immediate; pause that thread

11. Output standard

Reminder: object ID; rung; recipient role/name; facts hash of original packet; ask (unchanged); links; “this is not an approval request” if 05-origin; “this is not payment” always; agent 09; level.

Touch log: timestamp, channel, message ID, person who released (if L1).

12. Control requirements

Control	Support	Test
No substitute approve/receive	Language + access	Sample 25
Ladder integrity	Matrix hash	Monthly
No personal channels	Address validation	Inject
Pause on complaint / shutdown	Flag	Tabletop

13. Audit evidence

Reminders + touches 7 years; ladders life-of-programme; storm reports 1 year; complaint incidents 7 years.

14. KPIs

KPI	Definition	Use
Reply-after-rung-1 rate		Copy/SLA fit
Stale-chase rate	Chases on outdated packets	Quality
Storm count		Design
Rung-3 volume		Management load
Cost (mail + desk)		Brake

Do not KPI “exceptions closed after chase” as agent success — that can hide improper closes.

15. Performance history fields

Period; chases by rung and origin agent; replies; stale; storms; complaints; plants; level; incidents; cost.

16. Autonomy level

Level	Behaviour
L0	Shadow due lists
L1	Draft reminders; desk send
L2	Auto-send rung 1 whitelist
L3	Auto-send rung 1 + auto-inform rung 2 on whitelist
L4	Exception oversight of that send set

No L3 substitute action.

17. Failure handling

Failure	Action
Packet missing	Do not send “any update?”; rebuild via owning agent
Recipient left company	Freeze; HR/matrix
Mail bounce	Desk; do not retry personal address
Orchestrator clock wrong	Pause sweep

18. Cost monitoring

Volume of reminders is itself a cost (attention tax). If rung-1 volume grows while reply rate falls for 3 weeks, pause L2 and fix ownership — do not add rungs.

19. Handoffs

05/07/04 → 09; 09 → owner; 09 → manager (inform); 09 → owning agent if stale; 09 → 16.

20. Configuration parameters

Parameter	Northline start
Rung 2	+2 business days
Rung 3	+2 further
Storm	15 open reminders / person
Channels	Corporate email + Orchestrator digest

21. First 90 days

Human-send only. Watch storms at Birmingham (historically silent GR). Do not enable L2 until stale-chase < 5%.

22. Worked example — Northline Industrials

W0-1050022 Birmingham missing GR, packet sent Monday. No response Wednesday 09:30. Agent 09 drafts rung 1 to A. Cole with the same dock question. Tomasz releases. Cole replies Thursday DOCK_NOT_POSTED. Chase stops. Human posts GR.

W0-1049104 approval. Adeyemi silent 3 days. Rung 1. Still silent. Rung 2 informs Adeyemi’s manager — *not* as approver. Singh still must approve as Plant Controller when the first step completes. Agent does not skip Adeyemi.

Bad path refused: specialist asks to “copy James Okoye on everything aged 5 days.” Matrix does not list the Process Owner as rung 2. Refused.

23. Sample output artefact (abridged)

```
object_id: W0-1050022
agent_id: 09
rung: 1
recipient: A. Cole
ask: unchanged from packet 05 (DOCK_NOT_POSTED?)
stale: false
not_payment: true
autonomy_level: L1
state: DRAFT
```

24. What this agent does not replace

Plant management, DOA deputies, or staffing. It replaces empty chase mail.

25. Document control

Field	Value
Spec	AGENT_09 Internal Follow-up
Default autonomy	L1 (start L0)

AGENT 10 — Duplicate & Anomaly

Stack position: Parallel signal on every capture record. Can block Matching recommend-match and Payment Proposal proceed annotations. Does not detect fraud as a guarantee and does not clear its own flags.

Default autonomy: L0 Observe (promotion to L1 flag-as-recommendation).

Human owner (typical): AP Controls Lead

Payment authority: None. A flag is not a hold authority by itself until a human (or a written control procedure) places the hold.

Northline instance: Procedure AP-DUP-003. Language discipline: “possible duplicate / unusual pattern — human review.” Never “fraud.”

1. Position in the stack

This agent exists because humans miss collisions and odd patterns when volume is 15,000 a month. It also exists to be kept in its box. A noisy flagger trains people to ignore it. A confident flagger that uses the word “fraud” creates legal and cultural risk and still misses things.

Treat every output as a **hypothesis**. Four human outcomes only: Clear, Confirm, Escalate, Defer (see Decision Framework §9). The agent does not choose.

2. Job description

The Duplicate & Anomaly Agent compares a new capture record to open and recently paid invoices (and credits) using a written rule set: exact and near keys (vendor + invoice number + amount; vendor + amount + date window; image-hash near-duplicate; PO + amount collision; unusual first-invoice amount; unusual billing address vs master). It raises a flag with the rule IDs that fired and the candidate object IDs. It does not block payment by writing to the bank file. It does not investigate. It does not call law enforcement.

3. Operating intent and cadence

Mode	Cadence	Output
Object-driven	On Intake ready (parallel to Validation)	Shadow score (L0) or flag (L1)
Human disposition	SSC hours	Clear / Confirm / Escalate / Defer
Daily	17:30 ET	Open flags, aging, rule-fire mix
Monthly	With Internal Audit as observer if they wish	Rule performance, not “fraud caught”

4. Inputs

Input	Source	Mandatory
Capture record + image + hash	Agent 01	Y
Open AP items + paid history (Northline: 24 months)	ERP	Y
File-hash duplicates already known	Agent 01	Y — do not double-count as financial dup without rules
Vendor master (age, address, bank-change date)	ERP	Y for anomaly rules
Rule book AP-DUP-003	Controlled	Y
Prior dispositions (false-positive learning is <i>rule change</i> , not silent)	Orchestrator	N

5. Tools / data required

Tool	Privilege
Invoice history search	Read
Image perceptual hash (optional)	Read
Rule engine (deterministic first; model only as <i>secondary</i> hint, labelled)	Read rules
Orchestrator	Write flag records at L1
Evidence store	Write flag packet

Deterministic rules are the system of record for *why* a flag exists. A model score, if used, is HINT and cannot flag alone at commissioning.

ERP-agnostic: search by vendor + invoice number is universal; fuzzy number (459102 vs 4591O2) must be written as a rule, not hoped for.

6. Responsibilities

1. Run the rule book in published order. Record each fire with candidates.
2. Distinguish FILE_DUP (same artefact, Agent 01) from POSSIBLE_FINANCIAL_DUP (different artefacts, possible double pay).
3. Anomaly rules (examples): first invoice > \$25k; amount 10× vendor 12-month median; billing address ≠ master; invoice number reset pattern; weekend-dated high value. Each is a *pattern*, not a finding.
4. At L0, write shadow only. At L1, open flag on the work object and notify Controls queue.
5. Notify Agents 03 and 12: proceed-type recommendations forbidden while flag open.
6. Present disposition UI fields to the human. Require a reason code on Clear.
7. After Confirm: hand to Match/Payments for hold, reverse, or recover *per procedure* — this agent does not reverse.
8. After Escalate: notify the named control owner (could be Internal Audit or a fraud-examination function *if the company has one*). The agent still does not use “fraud detected.”
9. Weekly: false-positive rate by rule. Propose rule changes to Controls Lead (change control). Do not auto-tune thresholds.

7. Explicit exclusions

1. Payment authorisation, release, or silent drop from a payment proposal (Agent 12 humans deselect).
2. Auto-clear of flags.
3. Auto-hold in the bank / Positive Pay system.
4. Vendor bank change.
5. Any claim of fraud detection, fraud prevention, or completeness of detection.
6. Model-only flags at commissioning.
7. Scoring people or plants as “fraud risk.”
8. External reporting to banks or agencies.
9. Using Clear as “approved to pay” beyond “this flag is closed.”
10. Training on real “fraud cases” from other employers (no real client data).

8. Human owner

Field	Northline
Role / name	AP Controls Lead / Elena Ruiz
Backup	Assistant Controller — Payables
Escalation	Controller; IA as they request
Owns	Rule book, dispositions, language, join tests with 03/12
Does not own	Investigations beyond AP procedure unless escalated

Elena is not a fraud examiner unless Northline separately designates that role. The charter should not imply she is.

9. Approval requirements

Action	Human
Disposition of every L1 flag	Controls Lead or named delegate
Clear above \$10,000 or any bank-related anomaly	Controls Lead personally (not delegate)
Confirm (substantiated duplicate)	Controls + Match or Payments as relevant
Escalate	Controls + named function
Rule add/change/threshold	Controls + Process Owner; Controller if payment-adjacent
Model hint on	Separate charter; still not standalone

10. Escalation criteria

Condition	To	Timing
Flag aging > 2 business days	Controls Lead	Daily
High-value (> \$50k) open flag on a proposal day	Payments Lead + Controls	Before run
Confirmed duplicate already paid	Controller + Payments	Immediate
Bank-detail anomaly	Vendor Master + Controls	Immediate
Rule FP rate > 40% for a week	Pause that rule	Same week
Someone asks the agent to “catch fraud” in a demo	Process Owner refusal	n/a

11. Output standard

Flag record: object ID; candidate IDs; rule IDs that fired; FILE vs FINANCIAL vs ANOMALY; HINT model score if any (nullable); language field fixed: possible duplicate or unusual pattern – human review; state open/cleared/confirmed/escalated/deferred; disposition reason; owner; agent 10; level L0/L1; “not a fraud finding.”

Join signal to 03/12: agent10_open: true/false.

12. Control requirements

Control	Support	Test
No proceed over open flag	03/12 join 100%	Daily join
Human disposition	No auto-clear	Access + log
Language	Forbidden-word scan (fraud, guaranteed)	Weekly
Rule change control	Version hash	Monthly
High-value clear	Dual or named	Sample 100% > \$10k

This agent does **not** satisfy an anti-fraud programme. It produces review evidence.

13. Audit evidence

Flags + dispositions 7 years; rule versions life-of-programme; join reports 7 years; language reviews 1 year. Retrieval by invoice in 15 minutes.

14. KPIs

KPI	Definition	Use
Coverage	% of Intake objects scored	Completeness of signal
Precision by rule	Confirmed / (confirmed+cleared) <i>per rule</i>	Tune via change control
Time to disposition		Control SLA
Join integrity	Proceeds blocked	Control
Language incidents		Culture
Cost		Brake

Forbidden KPIs: “fraud caught,” “dollars saved from fraud,” “detection rate,” “assurance.”

Precision is not a promise of recall. A high precision rule can still miss collisions. Do not market precision as safety.

15. Performance history fields

Period; objects scored; flags by rule; dispositions; FP by rule; join breaks (should be 0); high-value clears; language incidents; level; pauses of rules; cost.

16. Autonomy level

Level	Behaviour
L0	Shadow scores, no live block
L1	Live flags; block proceed-recommend on 03/12
L2	Prepare hold packet for Payments Lead (still human places hold)
L3	Not for auto-confirm or auto-clear. Do not plan L3 disposition
L4	n/a for disposition

Ceiling: L2 prepare. Disposition remains human. Payment remains human.

17. Failure handling

Failure	Action
History search down	Do not PASS “no flag”; mark SCAN_UNAVAILABLE; 03/12 treat as open-hold equivalent until scan runs
Rule hash missing	Stop
Model vendor outage	Continue deterministic rules only
Flood of flags	Pause the noisy rule; do not auto-clear the flood

18. Cost monitoring

Search compute, review minutes. If review minutes explode and precision collapses, pause rules — attention is the control. Do not add a more expensive model to “fix” a bad rule book.

19. Handoffs

01 → 10; 10 → Controls human; 10 → 03/12 block signal; 10 → 04 class ANOMALY_HOLD; Confirm → Match/ Payments procedure; Escalate → named function; 10 → 16.

20. Configuration parameters

Parameter	Northline start
History window	24 months
Amount+date window	±7 days, same vendor, amount ±\$1
Invoice number exact + fuzzy	Written list of substitutions (O/O, spaces, prefixes)
First-invoice anomaly	> \$25,000
Median spike	10× 12-month median, min \$5,000
Model standalone	Off
Language	Fixed sentence

21. First 90 days

L0 only for at least 10 days / 500 objects. Measure shadow vs what specialists already caught. Promote L1 only if specialists agree the flags are intelligible. Do not turn on model hints in 90 days.

22. Worked example — Northline Industrials

W0-1049555 invoice 459210, V-10442, \$8,440, company 1000. Rule VENDOR_NUM_AMT hits paid invoice 88321 (\$8,440, same vendor, number different, date 11 months ago) — weak. Rule PO_AMT hits open invoice 459102 on same PO for \$19,596 — different amount, not a dup. Rule FUZZY_NUM does not fire. Anomaly MEDIAN_SPIKE does not fire.

Shadow L0: one weak candidate. At L1, Elena's book says VENDOR_NUM_AMT alone is HINT not a flag unless a second rule fires. No live flag. (If a vendor had insisted this is "AI fraud detection," they would have been wrong twice: no flag, and not fraud.)

Second case: invoice 459102 submitted via portal *and* a second PDF with number 459102-A, same amount \$19,596, same vendor, same PO. Rules AMT_DATE_VENDOR + PO_AMT + near image-hash fire. L1 flag open. Agent 03 must RECOMMEND_NO_ACTION_ANOMALY_HOLD. Thursday proposal: Agent 12 annotates hold. Elena confirms duplicate artefact-plus-suffix. Confirm. Payments deselects. Language on the ticket: "confirmed duplicate invoice submission — human review." Not "fraud."

Scan unavailable Thursday 07:00: Agent 10 marks SCAN_UNAVAILABLE. Agent 12 treats the whole proposal as needing a delayed run or a human extra sample — not "no flags so safe."

23. Sample output artefact (abridged)

```
object_id: W0-1049555
agent_id: 10
autonomy_level: L1
rules_fired: [VENDOR_NUM_AMT]
standalone_allowed: false
flag_state: not_opened
language: possible duplicate or unusual pattern – human review
fraud_claim: false
scan: complete
```

24. What this agent does not replace

Internal Audit, a fraud-examination function, vendor-master dual control, or a human looking at a weird invoice. It does not make payment safe.

25. Document control

Field	Value
Spec	AGENT_10 Duplicate & Anomaly
Default autonomy	L0
Language	No fraud / savings / guarantee claims

AGENT 11 — Vendor Statement

Stack position: Periodic / event-driven. Reconciles supplier statements to the AP subledger. Seeds Supplier Resolution and Triage. Never treats a statement line as an invoice or a payment instruction.

Default autonomy: L1 Recommend.

Human owner (typical): AP Reconciliations Lead

Payment authority: None.

Northline instance: Weekly for top 40 vendors by spend; on-receipt for others; Wave 3 in the stack plan.

1. Position in the stack

Statements are the supplier's picture of the world. They are useful because they show what AP does not have — and dangerous because AP teams sometimes pay from them. This agent produces a reconciliation worksheet: matched lines, in-AP-not-on-statement, on-statement-not-in-AP, amount disagreements, and credits parked on one side only. Humans decide what to query. Nobody adds a statement-only line to a payment proposal.

2. Job description

The Vendor Statement Agent intakes a statement (from Agent 01 classification or a portal pull), identifies the vendor and period, matches lines to open and recently paid items using a written key set, and emits a reconciling packet with coded differences. It does not create invoices from statement lines. It does not change due dates because the statement says “overdue.”

3. Operating intent and cadence

Mode	Cadence	Output
On statement artefact	Event	Worksheet
Top-40 pull	Weekly Monday	Request list + completed recons
Aging of unreconciled	Daily	Open packets
Month-end	Close calendar	Status to Agent 13 (completeness of <i>work</i> , not of AP)

4. Inputs

Input	Source	Mandatory
Statement artefact + extract	Agent 01 or portal	Y
AP open items + paid in statement period	ERP	Y
Unapplied credits / debit memos	ERP	Y
In-flight exceptions on that vendor	Orchestrator	Y — explain “overdue”
Match keys AP-STM-001	Controlled	Y
Vendor contact	Master	For later 08

5. Tools / data required

Tool	Privilege
Statement extract	Read
AP inquiry	Read
Orchestrator	Write worksheet
Portal pull (top 40)	Read

ERP: D365 vendor transactions; SAP FBL1N-equivalent; Oracle Payables activities; NetSuite vendor bill/ bill payments; Workday supplier invoices. Read only.

6. Responsibilities

1. Identify vendor uniquely. If not, QUERY — do not recon against the wrong vendor.
2. Pin statement date and currency. Multi-currency statements split or QUERY.
3. Apply keys in order: invoice number exact; invoice number fuzzy (same as Agent 10 list); amount+date; PO+amount. Record which key matched.
4. Classify each statement line: MATCHED_OPEN, MATCHED_PAID, ON_STMT_NOT_IN_AP, AMOUNT_BREAK, CREDIT_ONE_SIDED.
5. Classify each AP open item not on statement: IN_AP_NOT_ON_STMT (may be timing).
6. Link ON_STMT_NOT_IN_AP to possible Intake leftovers or Intake never-received. Seed Agent 01 completeness if the supplier claims they sent it.
7. Link “overdue” on statement to open exceptions (05/07/10). Annotate; do not treat as AP negligence by default.
8. Hand ON_STMT_NOT_IN_AP material items to Triage → Agent 08 draft “please send invoice artefact.”
9. Never create a parked invoice from the statement line.
10. Produce a vendor-level balance bridge: statement close vs AP close, with timing explanation where possible.

7. Explicit exclusions

1. Payment authorisation or adding statement-only lines to a proposal.
2. Creating invoices or credits from the statement.
3. Changing due dates or terms.
4. Promising payment of “overdue” on the statement.
5. Vendor bank details on a statement remittance advice — divert to Vendor Master.
6. Intercompany statements (separate charter).
7. Declaring the AP subledger “accurate.”
8. Auto-clearing differences within a quiet tolerance without a written rule (Northline: no quiet tolerance at commissioning).

8. Human owner

Field	Northline
Role / name	AP Reconciliations Lead / David Chen
Backup	Olivia Grant (Supplier Desk)
Escalation	AP Manager; Controller at month-end if top-40 incomplete
Owns	Key set, top-40 list, worksheet standard
Does not own	Cash, invoice create

9. Approval requirements

Action	Human
Accept worksheet	Reconciliations specialist
Query supplier	Via Agent 08 human send
Write off difference	Controller per threshold — not this agent
Change top-40 list	AP Manager
Any invoice create	Intake/Validation path only, with artefact

10. Escalation criteria

Condition	To	Timing
Vendor unresolved	Vendor Master	Same day
ON_STMT_NOT_IN_AP > \$25k single line	AP Manager	Same day
Top-40 weekly pack incomplete Friday	AP Manager	Friday
Statement shows paid item still open in AP	Payments + Match	Same day (possible unapplied)
Bank details on statement	Vendor Master + Controls	Immediate

11. Output standard

Worksheet: vendor; period; currency; statement URI; line table with class + key used; AP-only lines; bridge; open exception links; “not an invoice”; “not a payment instruction”; agent 11; level; hash of AP-STM-001.

12. Control requirements

Control	Support	Test
No pay-from-statement	Join with Agent 12 (no statement-only IDs on proposal)	Each run
No invoice-from-statement	Access	Quarterly
Top-40 coverage	Weekly	Close
Bank divert	Inject	Annual

13. Audit evidence

Worksheets 7 years; statements 7 years; key versions life-of-programme; join-to-payment-run 7 years.

14. KPIs

KPI	Definition	Use
Top-40 on-time		Coverage
Specialist accept rate		Quality
ON_STMT_NOT_IN_AP resolved via real invoice artefact		Process (not a savings claim)
Statement-only items appearing on a proposal	Must be 0	Control
Cost		Brake

15. Performance history fields

Period; statements; vendors; classes; accepts; \$ ON_STMT_NOT_IN_AP; proposal leaks (0); level; incidents; cost.

16. Autonomy level

Level	Behaviour
L0	Shadow recon
L1	Live worksheets
L2	Prepare Agent 08 packets for ON_STMT_NOT_IN_AP
L3	Not for invoice create or pay
L4	n/a for those actions

17. Failure handling

Failure	Action
Extract of statement poor	Human key lines; do not guess
ERP open-item fail	No worksheet from stale balances
Duplicate statements	Latest + delta

18. Cost monitoring

IDR on long PDFs, specialist minutes. If accept rate < 70%, stay L0 and fix extract.

19. Handoffs

01 → 11; 11 → human; 11 → 04/08; 11 → 13 status; 11 → 12 negative join; 11 → 16.

20. Configuration parameters

Parameter	Northline start
Top 40	Named list, quarterly
Quiet tolerance	None
History for MATCHED_PAID	180 days

21. First 90 days

Five vendors only, L0 then L1. Prove the proposal join is zero. Then expand toward 40.

22. Worked example — Northline Industrials

Lakeshore Steel April statement. Close \$214,400. AP open \$194,804. Bridge: \$19,596 invoice 459102 open and on statement (match); \$27,000 “overdue” line inv 44011 **not in AP** — no artefact in Intake. Class ON_STMT_NOT_IN_AP. David accepts. Agent 08 drafts “send invoice copy.” Agent 12 cannot add \$27,000. Agent 13 sees the item as a *possible* unrecorded liability *candidate* — Controller decides accrual, not this agent.

Dunning \$27,000 the same week is the same line, not a second fact.

23. Sample output artefact (abridged)

```

vendor: V-10442
period: 2026-04
bridge_gap: 27600
lines:
  - 459102 MATCHED_OPEN
  - 44011 ON_STMT_NOT_IN_AP 27000
not_invoice: true
not_payment_instruction: true
agent_id: 11
autonomy_level: L1

```

24. What this agent does not replace

Subledger recon to GL (different control), accrual judgement, or Intake. It replaces spreadsheet archaeology on a Monday.

25. Document control

Field	Value
Spec	AGENT_11 Vendor Statement
Default autonomy	L1 (start L0)

AGENT 12 — Payment Proposal Review

Stack position: Each proposed payment run. Annotates. Never authorises, releases, or transmits payment.

Default autonomy: L1 Recommend. Hard ceiling: L2 prepare/annotate.

Human owner (typical): AP Payments Lead, with Treasury accountable for run integrity.

Payment authority: None — and not earnable.

Northline instance: Tuesday and Thursday runs; dual human approval; Positive Pay; companies 1000/1100 live notes, CA/MX human-only review.

1. Position in the stack

This is the most dangerous agent to misunderstand. ERP already built a proposal. Humans already feel late. Vendors already shout. The agent’s job is to **put holds and questions on the page** so two human authorisers can see them. The agent’s job is never to make the run “go.”

If a vendor, integrator, or internal sponsor says “auto-release under \$5k,” refuse. \$5k is a human DOA number at Northline, not an agent payment limit.

2. Job description

The Payment Proposal Review Agent reads the proposal (or payment batch in draft), joins each line to work-object state, Agent 10 flags, vendor master change recency, invoice approval/match state, early-pay discount maths as *information*, stop-pay lists, and statement-only IDs (which must not appear). It emits an annotation pack: OK_FOR_HUMAN_REVIEW, HOLD_ANOMALY, HOLD_BANK_CHANGE, HOLD_OPEN_EXCEPTION, HOLD_NOT_APPROVED, HOLD_STOP_PAY, QUERY_DISCOUNT, LEAK_STATEMENT_ONLY. Humans deselect and then two humans authorise.

3. Operating intent and cadence

Mode	Cadence	Output
Pre-run	Tue/Thu 06:30 ET	Annotation pack
Intra-run	If proposal rebuilt	Re-annotate (full, not delta-only)
Post-run	After release	Compare released set vs annotations (detective)
Ad hoc	Emergency pay request	Same checklist; no shortcut

4. Inputs

Input	Source	Mandatory
Draft proposal / batch	ERP / treasury	Y
Work objects for each invoice	Orchestrator	Y
Agent 10 flag + scan-available bit	Agent 10	Y
Vendor bank-change log (90 days)	ERP / Vendor Master	Y
Stop-pay / sanctioned / audit hold lists	Controls / Legal / Treasury	Y
Match/approval state	Agents 03/07	Y
Discount terms vs dates	ERP	N — informational
Agent 11 leak check	Statement-only IDs	Y
Prior annotations this run	Orchestrator	If rebuild

5. Tools / data required

Tool	Privilege
Payment proposal read	Read only
Vendor master change	Read
Orchestrator	Write annotations
Hold lists	Read

The service account must not have payment release, bank-file download-to-transmit, or Positive Pay approve. If a role template “Payments” includes those, **do not use it**. Build a review-only role.

ERP: D365 payment journal draft; SAP F110 proposal; Oracle payment process request; NetSuite payment batch; Workday settlement run. Review-only.

6. Responsibilities

1. Freeze a copy of the proposal IDs and amounts (hash) so annotations attach to a version.
2. For each line, evaluate the hold tests in published order (see §20). First fatal hold wins; still record all fires.
3. If Agent 10 SCAN_UNAVAILABLE, mark the **run** SCAN_DEGRADED — authorisers must decide delay vs extra sample. The agent does not decide.
4. LEAK_STATEMENT_ONLY is a control incident, not a quiet deselect by the agent.
5. Discount QUERY is maths only (“discount expires today; invoice has open qty break”). No “pay now to save.”
6. Produce the pack before 07:30 ET on run days (Northline). If late, Payments Lead delays the run — the agent does not skip tests.
7. After human deselects, re-hash. After two authorisers release, detective-compare: any line that had a fatal hold but released → incident.
8. Emergency single payments get the same tests. Urgency is not a class.

7. Explicit exclusions

1. Authorising payment.
 2. Releasing a journal, batch, or bank file.
 3. Transmitting to bank or Positive Pay exception decide.
 4. Deselecting lines in the ERP (humans deselect).
 5. Adding lines (statement, dunning, “relationship”).
 6. Changing vendor bank to make a payment succeed.
 7. Promising suppliers they are on the run.
 8. Using discount, relationship, or “CFO said” as a release criterion.
 9. Auto-release under any dollar cap.
 10. Clearing Agent 10 flags.
 11. Declaring the run “safe.”
-

8. Human owner

Field	Northline
Role / name	AP Payments Lead / Grace Okonkwo
Treasury counterpart	Treasury Manager (second authoriser)
Backup	AP Manager (still needs a second human)
Escalation	Controller + Treasurer
Owns	Annotation standard, run clock, detective compare
Does not own	Bank contract, investment of cash, DOA for invoices

Two authorisers cannot be the agent owner plus the agent.

9. Approval requirements

Action	Who
Accept annotations	Payments Lead
Deselect lines	Payments Lead (or procedure)
Authorise / release	Payments Lead and Treasury (Northline)
Transmit	Treasury / bank platform — human
Delay run due to SCAN_DEGRADED	Payments + Treasury
Release a line against a fatal annotation	Forbidden without Controls + Controller written override — still human, still not the agent

10. Escalation criteria

Condition	To	Timing
Pack not ready 07:30	Payments Lead — delay	Clock
SCAN_DEGRADED	Controls + Treasury	Before run
LEAK_STATEMENT_ONLY	Process Owner + Controls	Immediate
Fatal hold released (detective)	Controller + Treasurer + Security	Immediate
Bank-change line on proposal	Vendor Master + Controls	Before run
Service account gained release privilege	IT Security — stop agent	Immediate

11. Output standard

Pack: run ID; proposal hash; company; line annotations (invoice, vendor, amount, codes, fatal Y/N); run-level SCAN_DEGRADED; “not an authorisation”; agent 12; level; owner.

Detective report: released IDs vs fatal holds; empty is expected.

12. Control requirements

Control	Support	Test
Dual human release	Agent has no release	Access + each run
Anomaly join	Fatal on open flag	100%
Bank-change hold	90-day window	Inject
No statement-only	Leak test	Each run
Detective compare	After release	Each run
SoD	Reviewer ≠ sole authoriser	Roles

13. Audit evidence

Packs 7 years; proposal hashes 7 years; authoriser IDs (from ERP) 7 years; detective reports 7 years; privilege reviews 1 year; incidents 7 years.

14. KPIs

KPI	Definition	Use
Pack on-time		Run discipline
Annotation accept / override	Overrides coded	Quality
Detective incidents	Must be 0	Control
Statement leaks	Must be 0	Control
SCAN_DEGRADED runs		Dependency
Cost		Brake

Forbidden: “payments automated,” “safe pay,” “discount captured \$,” ROI.

15. Performance history fields

Period; runs; lines; annotations by code; overrides; detective incidents; degraded scans; leaks; level; privilege incidents; cost.

16. Autonomy level

Level	Behaviour
L0	Shadow annotations
L1	Live pack; humans deselect and authorise
L2	Prepare deselect list in a side file for humans to apply
L3	Does not exist for release
L4	Does not exist for release

There is no promotion case that ends in agent-authorised payment.

17. Failure handling

Failure	Action
Cannot read proposal	No pack; no run on hope
Orchestrator down	No pack; delay run
Agent 10 scan down	SCAN_DEGRADED
Privilege creep	Stop
Rebuild mid-morning	Full re-annotate; old pack void

18. Cost monitoring

Compute is small; **incident cost is large**. The brake is: any detective incident → L0 and full recert. Do not “save time” by skipping the pack on a heavy Thursday.

19. Handoffs

ERP proposal → 12; 10/03/07/11/Vendor Master → 12 joins; 12 → Payments Lead + Treasury; 12 → detective; 12 → 16; never 12 → bank.

20. Configuration parameters (hold tests)

Order (Northline start):

1. HOLD_STOP_PAY
2. HOLD_ANOMALY (open flag)
3. HOLD_BANK_CHANGE (90 days, unless Controls has a written clearance ID)

4. HOLD_NOT_APPROVED / not matched when required
 5. HOLD_OPEN_EXCEPTION (qty/price/GR still open — policy: do not pay contested qty)
 6. LEAK_STATEMENT_ONLY
 7. QUERY_DISCOUNT (info)
 8. Else OK_FOR_HUMAN_REVIEW
-

21. First 90 days

L0 for four run days minimum. Then L1. Dual approval unchanged. No “pilot auto-release.” Tabletop a detective incident.

22. Worked example — Northline Industrials

Thursday 16 April 2026, 06:31 ET. Proposal 612 lines, \$4.1m, companies 1000/1100.

- Two lines, Agent 10 open flags → HOLD_ANOMALY (includes 459102-A).
- One vendor, bank change yesterday → HOLD_BANK_CHANGE.
- Invoice 459102 \$19,596, open QTY_OVER → HOLD_OPEN_EXCEPTION.
- Discount query on a clean \$2,200 invoice (info).
- Scan complete. No statement-only IDs.
- Pack ready 07:12.

Grace deselects the four fatal holds. Treasury second-approves the remainder. Bank file human-transmitted. Detective: empty. Agent 12 did not release \$4.1m. It annotated \$4.1m.

Vendor demo the same week: “we would have released everything under \$5k including the bank-change vendor.” Refusal logged.

23. Sample output artefact (abridged)

```
run_id: PAY-2026-04-16-THU
proposal_hash: 7aa1...
lines: 612
fatal_holds: 4
scan: complete
authorisation: HUMAN_ONLY
agent_id: 12
autonomy_level: L1
not_a_release: true
```

24. What this agent does not replace

Treasury, bank platforms, Positive Pay, dual approval, or cash forecasting. It replaces a proposal that nobody annotated.

25. Document control

Field	Value
Spec	AGENT_12 Payment Proposal Review
Default autonomy	L1 (start L0)
Ceiling	L2 annotate; never release

AGENT 13 — AP Close

Stack position: Period-end. Drives the close checklist and lists accrual *candidates*. Does not attest and does not book.

Default autonomy: L1 Recommend. Ceiling: L2 prepare.

Human owner (typical): Assistant Controller — Payables

Payment authority: None.

Northline instance: Calendar FIN-CLOSE-2026; US entities first; Controller attests.

1. Position in the stack

Close is an attestation event. The agent is a file clerk with a memory: what is open, what is aged, what statements are unfinished, what GR/IR looks like, what Intake leftovers existed at cut-off, what invoices arrived after cut-off with prior-period dates. The Controller signs. The agent does not.

2. Job description

The AP Close Agent executes a written checklist against evidence already in the stack and ERP: parked documents, unposted match recommends, open exceptions above a dollar list, GR/IR aging, unapplied credits, statement top-40 status, Agent 10 open flags, payment journals not yet released (informational), and a candidate accrual list built from rules the Controller owns. It produces a close file. Humans book and attest.

3. Operating intent and cadence

Mode	Cadence	Output
Soft close	Business day -2	Flash file
Hard close	Business day 0-1 per calendar	Close file
Post-close	+2	Late items list (detective)
Quarterly	Recert checklist vs policy	Version

4. Inputs

Input	Source	Mandatory
Close calendar + cut-off rules	Controlled	Y
Checklist AP-CLS-001	Controlled	Y
Orchestrator open objects	16	Y
GR/IR / uninvoiced receipts	ERP + Agent 05 candidates	Y
Parked / pending invoices	ERP	Y
Agent 11 top-40 status	11	Y
Agent 10 open flags	10	Y
Accrual policy (what <i>may</i> be proposed)	Controller	Y

5. Tools / data required

Tool	Privilege
ERP inquiry (subledger, GR/IR, parked)	Read
Orchestrator	Read + write close file
Calendar	Read

No GL post, no accrual journal write, no period-open/close toggle on the agent account.

6. Responsibilities

1. Open the period's close object on the calendar date. List checklist items as open.
2. Pull evidence for each item. Mark `evidence_present` or gap.
3. Build GR/IR aging and waiting-invoice aging from Agent 05 + ERP. Label **candidates**.
4. Apply accrual policy rules mechanically (e.g. "uninvoiced receipt > \$X and receipt date in period"). Emit *proposed* lines with source IDs. Do not book.
5. List invoices received after cut-off with invoice dates in the period — cut-off candidates for human judgement.
6. List open Agent 10 flags and open fatal payment holds — so close is not "clean" by ignoring them.
7. Hand the file to Assistant Controller. Record questions.
8. After Controller attestation, freeze the file. Post-close detective lists items that would have changed the file.
9. Never roll the ERP period.

7. Explicit exclusions

1. Payment authorisation.
2. Booking accruals, reversing accruals, or posting invoices to hit a date.
3. Attesting the close or signing SOX/PBC as the company.
4. Opening/closing periods.
5. Forcing plants to post dummy GR before close.
6. Declaring "AP is complete" or "balances are accurate."
7. Quietly dropping checklist items.
8. Using the agent as the PBC binder without human review.

8. Human owner

Field	Northline
Role / name	Assistant Controller — Payables / Helen Park
Attestor	Controller
Backup	AP Manager (file only; cannot attest)
Escalation	Controller
Owns	Checklist ops, candidate quality, freeze
Does not own	Accounting policy

9. Approval requirements

Action	Human
Accept close file	Assistant Controller
Accrual book	Designated accountant
Attest	Controller
Checklist change	Controller
Override gap (“we will live with it”)	Controller, written

10. Escalation criteria

Condition	To	Timing
Checklist gap on hard-close morning	Controller	Immediate
Accrual candidates > materiality (Controller's number)	Controller	Soft close
Dummy GR suspected (spike of receipts day 0)	Controls + Plant	Same day
Agent 10 scan unavailable at cut-off	Controls	Same day
Period toggle privilege found on service account	IT Security	Immediate

11. Output standard

Close file: period; entity; checklist with evidence URIs; GR/IR candidate table; accrual proposal table (policy version); cut-off list; open flags; statement status; “candidates not bookings”; “not an attestation”; agent 13; level; freeze timestamp after attest.

12. Control requirements

Control	Support	Test
Attestation human	No sign privilege	Access
Accrual policy applied not invented	Hash	Sample 15 lines
Cut-off	List exists	IA sample
No dummy GR pressure	Language + 05 exclusion	Spike review
Freeze	File immutable after attest	Hash

13. Audit evidence

Close files 7 years; policy/checklist versions life-of-programme; attestation (ERP or letter) 7 years; detective lists 7 years.

14. KPIs

KPI	Definition	Use
Checklist completeness at soft close		Discipline
Gap count at hard close		Risk to attestor
Accrual proposal accept/edit/reject		Policy fit
Post-close detective material items		Cut-off quality
Cost		Brake

No “faster close guaranteed.” Observed hours may be reported by Agent 14 as a fact, not a promise.

15. Performance history fields

Period; entities; gaps; candidate counts; booked vs proposed (human); detective items; level; incidents; cost.

16. Autonomy level

Level	Behaviour
L0	Shadow file
L1	Live file; humans book/attest
L2	Prepare accrual journal <i>file</i> for accountant import — still human posts
L3	Not for booking or attest
L4	n/a

17. Failure handling

Failure	Action
ERP inquiry fail	File INCOMPLETE; do not attest on a partial silently
Calendar wrong	Process Owner
Policy hash missing	Stop proposals

18. Cost monitoring

Analyst minutes vs file quality. If reject rate on accrual proposals > 40%, fix policy mapping — do not auto-book to “save the close.”

19. Handoffs

05/10/11/16 → 13; 13 → Helen; 13 → accountant; 13 → Controller; 13 → 14 pack; 13 → 16.

20. Configuration parameters

Parameter	Northline start
Soft close	BD-2
Accrual candidate floor	Per Controller memo (illustrative \$2,500)
Materiality for escalate	Controller memo

Do not copy the \$2,500 if your policy differs.

21. First 90 days

One entity, L0 for a month-end, L1 the next. Accrual proposals conservative (receipt-based only). No L2 import file until accept rate is stable.

22. Worked example — Northline Industrials

March 2026 soft close, company 1000. Agent lists: 41 parked invoices; GR/IR \$2.3m > 60 days (Agent 05 aging); Lakeshore \$27,000 statement-only as *candidate* unrecorded (no artefact — Controller may accrue or not); open flag on 459102-A; no dummy-GR spike. Accrual proposal: 18 uninvoiced receipts ≥ \$2,500, total \$410k, each with receipt ID.

Helen edits out two receipts she knows are disputed returns. Controller attests after the accountant books \$382k. Agent 13 did not book \$382k. Post-close detective: one \$18k invoice arrived BD+1 with a March date — listed, not hidden.

23. Sample output artefact (abridged)

```
period: 2026-03
entity: 1000
checklist_gaps: 0
accrual_candidates: 18
accrual_proposed_total: 410000
attestation: HUMAN
booked_by_agent: false
agent_id: 13
autonomy_level: L1
```

24. What this agent does not replace

The Controller, accounting policy, or subledger-to-GL recon. It replaces a close checklist that lives in someone’s head.

25. Document control

Field	Value
Spec	AGENT_13 AP Close
Default autonomy	L1 (start L0)
Ceiling	L2 prepare

AGENT 14 — AP Reporting

Stack position: Consumes Orchestrator and ERP reads to produce operational and control packs. Does not manage by targets that imply guaranteed savings or accuracy.

Default autonomy: L1 Recommend (publish to agreed folders).

Human owner (typical): AP Analytics / Controller's office

Payment authority: None.

Northline instance: Daily ops flash; weekly control pack; monthly management pack. Figures are observations.

1. Position in the stack

Reporting is how the stack stays honest. If Matching hides rejects, if Triage closes silence, if Payment Review skips a Thursday, this agent should make the hole visible — or show that it cannot see (which is itself a finding). It is not a story engine. It does not forecast ROI.

2. Job description

The AP Reporting Agent produces versioned packs from defined queries: volume, cycle times, exception mix, SLA breaches, Agent 10 disposition aging, proposal detective incidents, close gaps, autonomy register snapshot, and cost meters. Each metric has a definition, a source, and a “not this” note. Humans interpret. The agent does not recommend firing people or “turning on L3.”

3. Operating intent and cadence

Pack	Cadence	Audience
Ops flash	Daily 17:45 ET	AP Manager, leads
Control pack	Weekly Friday	Controls, Process Owner, Controller (opt)
Management pack	Monthly with close	Controller, AP Manager
Autonomy snapshot	Weekly	Process Owner
Ad hoc	On request with a written metric spec	No drive-by metrics

4. Inputs

Input	Source	Mandatory
Metric dictionary AP-RPT-001	Controlled	Y
Orchestrator events	16	Y
ERP counts (open items, parked)	ERP	Y as defined
Agent KPI extracts	Each agent \$15	Y
Cost ledger	Finance/IT	Y for cost page
Autonomy register	Process Owner	Y

5. Tools / data required

Tool	Privilege
Warehouse / queries	Read
Orchestrator analytics	Read
Publish folder	Write packs
Dictionary	Read, hashed

No ERP transactional write. No payment data beyond counts and aged amounts needed for ops (payee bank details never in packs).

6. Responsibilities

1. Compute only metrics in the dictionary. New metric = change control.
2. Show numerator, denominator, time window, and source on every chart or table.
3. If a source is down, publish UNAVAILABLE — do not interpolate.
4. Include a control page: detective incidents, statement leaks, join breaks, dummy-GR spikes, language incidents (Agent 10).
5. Include autonomy levels in force — so silent promotion is visible.
6. Label observational process measures (e.g. first-time match rate) as **observed**, not targets the agents are paid on.
7. Exclude savings, ROI, “fraud \$ prevented,” “compliance %.”
8. Archive each pack with hash. Corrections are a new version, not a silent overwrite.

7. Explicit exclusions

1. Payment authorisation or any bank detail.
2. People ranking for performance management unless HR/AP jointly add a metric (default: no).
3. Predictive “you will miss close” as a pressure tool without dictionary status.
4. Vendor-identifiable discount “opportunities” framed as agent-released payments.
5. Marketing copy. These packs are internal operating artefacts.
6. Changing other agents’ KPIs to look better.
7. Guaranteed-savings dashboards.

8. Human owner

Field	Northline
Role / name	AP Analytics (Controller’s office) / Ben Ito
Backup	Helen Park
Escalation	Controller
Owns	Dictionary with Process Owner; pack calendar; corrections
Does not own	Agent operations

9. Approval requirements

Action	Human
Dictionary add/change	Process Owner + Controller if external/board bound
External (board/lender) extract	Controller — usually a subset, rewritten
Daily flash publish	Auto after L1 if completeness tests pass
Correction version	Ben + requesting lead

10. Escalation criteria

Condition	To	Timing
Control-page non-zero incident	Controls + Process Owner	Same pack
Source UNAVAILABLE > 1 day	IT + owner of source	Day 1
Autonomy register vs live levels mismatch	Process Owner	Immediate
Someone requests a savings KPI	Refusal + this spec	n/a

11. Output standard

Pack: date; dictionary hash; metric table (id, value, n, d, window, source, unavailable flag); control page; autonomy page; cost page; “observed, not promised”; agent 14; level.

12. Control requirements

Control	Support	Test
Dictionary lock	Hash	Monthly
No silent overwrite	Versions	Sample
No bank PII	Schema	Review
Incident visibility	Control page	Inject

13. Audit evidence

Packs 7 years; dictionary life-of-programme; correction log 7 years.

14. KPIs (of this agent)

KPI	Definition	Use
On-time packs		Discipline
UNAVAILABLE rate		Source health
Dictionary violations (ad hoc metrics)	Should be 0	Control
Correction rate		Quality
Cost		Brake

The *content* metrics belong to other agents. This agent is measured on pack integrity.

15. Performance history fields

Period; packs; late; unavailable; violations; corrections; level; incidents; cost.

16. Autonomy level

Level	Behaviour
L0	Shadow packs to Ben only
L1	Publish to agreed lists
L2	Prepare board subset for Controller edit
L3	Not needed
L4	n/a

17. Failure handling

Failure	Action
Query fail	UNAVAILABLE, still publish other pages
Hash mismatch dictionary	Stop publish
PII detected	Block pack; Security

18. Cost monitoring

Warehouse minutes, licence. Pause exotic queries if cost spikes without dictionary value.

19. Handoffs

All agents → 14 extracts; 14 → audiences; 14 → 15 (clusters from mix pages); 14 → 16.

20. Configuration parameters

Parameter	Northline start
Flash	17:45 ET
FTM	Observed only, definition: PO invoices posted without exception object
Cost page	Monthly
Board subset	Controller-only; no vendor names

20.1 Starter metric dictionary (Northline Wave 1)

ID	Name	Formula	Not this
M01	Intake leftover	Artefacts – objects – junk at 16:30	“Mailroom accuracy”
M02	Validation same-day	Results / ready objects	“Compliance rate”
M03	Worksheet coverage	Worksheets / PO PASSes	“Accounting accuracy”
M04	Exception mix	Count by class / total exceptions	A target to minimise by improper close
M05	Agent 10 open aging	Flags > 2 days	“Fraud backlog”
M06	Proposal detective	Fatal-hold lines released	Anything but zero is an incident
M07	Autonomy max	Highest live level	A race
M08	Observed FTM	Posted PO invoices with no exception object / PO posted	A promised outcome or bonus

M08 may appear on the monthly pack. It may not appear on an agent’s promotion memo as a savings proxy.

21. First 90 days

Ten metrics, not forty. Ops flash + control page. Resist executive dashboards until definitions survive three Fridays of argument.

22. Worked example — Northline Industrials

Friday 17 April 2026 control pack. Exception mix: MISSING_GR 31% Birmingham-heavy; UOM 22% Dayton — points to Agent 15/06. Agent 10: 14 flags, 11 cleared, 2 confirmed, 1 deferred; language incidents 0. Agent 12: 2 runs, detective 0, leaks 0. Autonomy: all Wave 1 at L0/L1. Observed FTM 63% (was 61% observational baseline) — printed as observation, no “+2pts ROI.” Cost page: IDR + model + desk minutes; no savings claim.

James Okoye uses the UOM line to schedule Hannah Cho, not to “turn on auto-match.”

23. Sample output artefact (abridged)

```
pack: CONTROL-2026-W16
dictionary: AP-RPT-001.4
observed_ftm: 0.63
ftm_is_target: false
detective_incidents: 0
autonomy_max: L1
savings_kpi: absent
agent_id: 14
```

24. What this agent does not replace

The Controller’s narrative, board reporting, or judgement. It replaces slides with undefined denominators.

25. Document control

Field	Value
Spec	AGENT_14 AP Reporting
Default autonomy	L1 (start L0)

AGENT 15 — Root Cause

Stack position: Continuous study agent. Clusters exceptions and defects; proposes process fixes. Implements nothing.

Default autonomy: L0 Observe (L1 = published proposals).

Human owner (typical): Finance Transformation Lead

Payment authority: None.

Northline instance: Weekly cluster review; monthly proposal memo. UOM and Birmingham GR are the first studies.

1. Position in the stack

Root Cause is how the stack avoids hiring more chasers forever. It is also how the stack avoids a transformation theatre of “insights.” A proposal names the mechanism, the evidence, the owner who would have to change something, and the residual risk if they do not. It does not implement PO templates, warehouse SOPs, or agent promotions.

2. Job description

The Root Cause Agent reads exception history, PO defect records, chase storms, duplicate rule false-positives, and close detective items. It clusters by code, plant, vendor, buyer group, and item category. It writes a mechanism hypothesis with counter-evidence. At L1 it issues a proposal memo. Process owners accept, reject, or defer. The agent does not open L3 cases to “fix” a cluster.

3. Operating intent and cadence

Mode	Cadence	Output
Cluster	Weekly	Cluster cards
Proposal	Monthly (or ad hoc if a cluster is severe)	Memo
Counter-evidence	With every memo	What would disprove
Backtest	After a human implements a fix	Repeat-break watch

4. Inputs

Input	Source	Mandatory
Exception objects + classes	04	Y
Match break codes	03	Y
PO defects	06	Y
GR responses	05	Y
Chase storms	09	Y
Dup rule FP	10	Y
Close detective	13	Y
Reporting mix	14	Y
Method note AP-RC-001	Controlled	Y

5. Tools / data required

Tool	Privilege
Orchestrator history	Read
ERP dimensional reads (plant, vendor, category)	Read
Memo store	Write proposals at L1

No transactional write. No mail to the company at large.

6. Responsibilities

1. Build clusters with a minimum size (Northline: 25 objects or 10% of a plant's exceptions, 4 weeks).
2. State a mechanism in causal language that a plant or buyer can test ("conversion missing on fastener POs"), not "culture."
3. Seek counter-evidence (plants with same vendors but no UOM breaks).
4. Separate AP-controllable (taxonomy, extract, chase copy) from others' systems (WMS, catalogue).
5. Write owners as roles that actually own the lever.
6. Include "what this does not fix" and "what we might wrongly automate."
7. After a human change, watch repeat-break for 8 weeks. Publish whether the mechanism still holds.
8. Refuse to turn a cluster into an L3 posting recommendation.

7. Explicit exclusions

1. Payment authorisation.
2. Implementing PO, WMS, DOA, or tax changes.
3. Promoting other agents.
4. Blaming named individuals in the monthly memo (roles and examples, not HR files).
5. Savings or ROI claims as the reason to implement.
6. Fraud theories (hand patterns to Controls as *questions*, not findings).

7. Vendor commercial strategy.

8. Human owner

Field	Northline
Role / name	Finance Transformation Lead / Imani Brooks
Counterparts	James Okoye (process), Hannah Cho (PO), plant liaisons
Escalation	Controller
Owns	Method, memo quality, backtest
Does not own	The receiving functions' backlog

9. Approval requirements

Action	Human
Publish L1 memo	Imani
Accept/reject/defer	Named lever owner
Method change	Transformation + Process Owner
Use of names	HR if ever needed — default no

10. Escalation criteria

Condition	To	Timing
Cluster $\geq$ 20% of SSC exceptions 4 weeks	Process Owner + lever owner	Ad hoc memo
Backtest shows fix failed	Lever owner	Week 8
Proposal used to justify L3 pay/post	Process Owner refusal	Immediate
Data too thin	Do not publish a confident memo	n/a

11. Output standard

Cluster card: id; window; size; dimensions; codes; mechanism; counter-evidence; lever owner; AP-controllable Y/N; agent 15; level.

Memo: cards + recommended experiments (not programmes) + residual risk + “no savings guarantee” + “do not promote payment agents.”

12. Control requirements

Control	Support	Test
No implementation privilege	Access	Quarterly
No silent promotion	Language	Review memos
Method hash	On memo	Monthly

13. Audit evidence

Memos 7 years; cluster extracts 7 years; accept/reject 7 years; backtests 7 years.

14. KPIs

KPI	Definition	Use
Memo specificity (lever named)	Sample	Quality
Accept/reject/defer mix		Engagement
Backtest completion		Discipline
Wrong-automation catches (refused L3)		Control
Cost		Brake

Do not KPI “savings from implemented ideas.”

15. Performance history fields

Period; clusters; memos; dispositions; backtests; refusals; level; cost.

16. Autonomy level

Level	Behaviour
L0	Cards to Imani only
L1	Published memos
L2	Prepare experiment charter for the lever owner
L3	Does not implement
L4	n/a

17. Failure handling

Failure	Action
History incomplete	Thin-data abstain
Identifiable HR issue	Stop; HR path
Cluster is actually Agent 10 FP	Hand to Elena as rule question

18. Cost monitoring

Analyst + model. If memos are generic, pause L1 — generic root cause is expensive noise.

19. Handoffs

03/04/05/06/09/10/13/14 → 15; 15 → lever owners; 15 → 06/08/01 as AP-controllable experiments; 15 → 16.

20. Configuration parameters

Parameter	Northline start
Min cluster	25 or 10% / 4 weeks
Backtest	8 weeks
Names in memos	Off

21. First 90 days

Two clusters only: Dayton UOM; Birmingham missing GR. Prove the memo format. Do not produce a 20-idea roadmap.

22. Worked example — Northline Industrials

Memo 2026-04 (illustrative). Cluster A: Dayton UOM 22% of exceptions, fastener category, two vendors, conversion absent on PO (Agent 06). Counter-evidence: Birmingham same vendors but buyers use a different template that explodes EA. Lever: Procurement catalogue/template (Hannah). Experiment: new template on *new* POs, 8-week backtest. Residual: old POs will still break — do not auto-change live POs. Do not raise match tolerance to 50 EA. Do not L3-post UOM breaks.

Cluster B: Birmingham MISSING_GR / DOCK_NOT_POSTED. Mechanism: finance-posted GR, dock not in WMS, clerk delay. Lever: Plant liaison A. Cole + warehouse SOP. Experiment: dock scan to GR within 1 day, measured via Agent 05 promise-break. Residual: volume spikes at month-end. Do not dummy GR at close (Agent 13 exclusion).

Imani publishes. Hannah accepts A. Cole accepts B. Agent 15 watches. Observed FTM may move later; that is not the success criterion of the memo.

23. Sample output artefact (abridged)

```
memo: RC-2026-04
cluster: UOM_DAYTON_FASTENER
mechanism: PO missing conversion EA/BOX
lever: Procurement template
implement_by_agent: false
promote_match_L3: false
savings_claim: none
agent_id: 15
autonomy_level: L1
```

24. What this agent does not replace

Transformation leadership, plant ops, or procurement design. It replaces a quarterly insight slide with no owner.

25. Document control

Field	Value
Spec	AGENT_15 Root Cause
Default autonomy	L0

AGENT 16 — Orchestrator

Stack position: Cross-cutting. System of record for work-object state, SLA clocks, handoffs, autonomy context, and evidence links. Not the system of record for accounting or cash.

Default autonomy: L1 Recommend (routing per matrices).

Human owner (typical): AP Process Owner

Payment authority: None.

Northline instance: Queue namespace NL.*; James Okoye is A-owner.

1. Position in the stack

If the sixteen agents are roles, the Orchestrator is the board they stand on. It does not extract, match, approve, or pay. It knows which object is where, who owns it, what level is in force, when the clock ends, and where the evidence lives. When it is wrong, the stack becomes a set of clever files that do not meet.

When it is over-ambitious, it becomes a second ERP. Do not let it post.

2. Job description

The Orchestrator Agent creates and updates work objects, enforces legal handoffs (only the named next agents), applies SLA start/stop rules, stores autonomy level per agent instance, refuses illegal transitions (e.g. Matching RECOMMEND_MATCH while Agent 10 is open), publishes queues, and records an event log sufficient to reconstruct history. Humans still decide. ERP still accounts. Treasury still pays.

3. Operating intent and cadence

Mode	Cadence	Output
Event	Continuous	Object + event
Sweep	15 minutes	SLA, illegal state, orphan
Register	On charter change	Autonomy register
Daily	18:00 ET	Orphan / illegal-transition report
Quarterly	Access + queue recert	Cert

4. Inputs

Input	Source	Mandatory
Events from Agents 01–15 and humans	All	Y
State machine AP-ORC-001	Controlled	Y
Queue / owner matrices	04/07/05 charters	Y
Autonomy register	Process Owner	Y
SLA tables	Charters	Y
Identity (people, roles, service accounts)	Directory / IAM	Y

5. Tools / data required

Tool	Privilege
Work-object store	Read/write objects and events
Evidence URI registry	Write pointers (not a second image store)
IAM read	Validate owners
Notify	Queue updates; no supplier send

The Orchestrator may call ERP **read** to verify a claimed document ID exists. It may not park, post, or pay. State machine must be versioned. Prompt-only routing is not an Orchestrator.

6. Responsibilities

1. Mint `object_id` on Intake (or on statement/close objects). Never reuse.
 2. Permit only transitions in AP-ORC-001 (examples: `intake_ready` → `val_*`; `val_pass` + `P0` → `match_*`; `match_break` → `triage`; `waiting_internal` → `09`; `proposal_line` → `12`). Illegal: `intake` → `posted`; `statement` → `proposal`; `10_open` → `03_recommend_match`.
 3. Attach evidence URIs; do not drop images when status changes.
 4. Start/stop SLA clocks per the owning agent's rules (Triage clock starts on accept, not on mint).
 5. Hold autonomy context: agent ID + entity + document class + level. A US-OH L2 does not leak to CA.
 6. Enforce Agent 10 join: if `agent10_open` or `scan_unavailable` per policy, block proceed transitions.
 7. Detect orphans (no owner, no agent, no SLA) every sweep.
 8. Record human overrides of routing with reason codes — overrides are visible, not silent.
 9. Expose queues `NL.MATCH.US.P0`, `NL.XCP.US`, etc. as listed in charters.
 10. Feed Agent 14 extracts and Agent 15 history. Feed Agent 13 open-object lists.
 11. Pause an agent instance when the register says pause — do not accept its proceed events.
-

7. Explicit exclusions

1. Payment authorisation, release, or bank-file handling.
 2. Posting, parking, GR create, PO change, vendor change.
 3. Approving invoices.
 4. Sending supplier mail.
 5. Inventing owners or climbing org charts.
 6. Promoting autonomy (it *records* promotions; Process Owner *decides*).
 7. Becoming the accounting ledger.
 8. Silent retry of a failed financial action (there should be no financial action).
 9. Auto-closing objects for age.
 10. Mixing entity scopes.
-

8. Human owner

Field	Northline
Role / name	AP Process Owner / James Okoye
Backup	Elena Ruiz (Controls) — temporary; SoD watch
Escalation	Controller
Owns	State machine, register, illegal-transition response, queue names
Does not own	Each agent's professional judgements

9. Approval requirements

Action	Human
State-machine change	Process Owner + Controls + affected agent owners
New queue	Process Owner + agent owner
Autonomy register write	Per AUTONOMY_PROGRESSION.md sign-off
Routing override	Named role; reason required
Pause agent	Owner or Process Owner or Security

10. Escalation criteria

Condition	To	Timing
Illegal transition attempted	Controls + sending agent owner	Immediate
Orphan > 4 SSC hours	Desk Lead + James	Sweep
Register vs behaviour mismatch	Process Owner	Immediate
Store down	IT — stack fail-closed	Immediate
Duplicate object_id	Security/IT — incident	Immediate
Event from paused agent	Drop proceed; alert	Immediate

11. Output standard

Work object: IDs; source; current agent; human owner; status enum; autonomy context; evidence URIs; SLA; escalation state; cost token; Agent 10 state.

Event: timestamp, actor (agent or human), from, to, payload hash, result allowed/blocked.

Daily report: orphans, blocked illegal, paused agents, store health.

Register extract: agent, entity, class, level, charter version, recert date.

12. Control requirements

Control	Support	Test
Fail closed	No ERP write	Access
Illegal transition	Block + log	Inject 03-match-with-flag
Autonomy isolation	Context fields	Inject CA object into US L2
No silent close	Status rules	Sample
Override visibility	Reason codes	Weekly
Reconstructability	Events	Retrieve 15-minute test

13. Audit evidence

Events 7 years; objects 7 years; state-machine versions life-of-programme; register 7 years; illegal-transition logs 7 years; access 1 year.

14. KPIs

KPI	Definition	Use
Illegal attempts blocked	Count	Control health
Orphans	Snapshot	Ops
Reconstruct success	Sample retrieve	Audit
Override rate		Design vs discipline
Store availability		IT
Cost		Brake

Do not KPI “straight-through objects” as a success bonus for the Orchestrator — that encourages skipped states.

15. Performance history fields

Period; objects minted; transitions; blocked; orphans; overrides; pauses; availability; level; incidents; cost.

16. Autonomy level

Level	Behaviour
L0	Shadow state (ERP/spreadsheet still SOR for queues) — not recommended beyond week 1
L1	Live work objects; routing per machine; humans override with reason
L2	Auto-route whitelist (e.g. PASS PO → 03 without desk)
L3	Auto-start Agent 09 rung 1 when SLA fires (if 09 chartered)
L4	Exception oversight of that routing

Never L3 financial. Routing autonomy ≠ posting autonomy.

17. Failure handling

Failure	Action
Store down	Agents fail closed; no side spreadsheet SOR
Clock skew	Pause SLA sweep
IAM cannot resolve owner	Assign Process Owner + orphan
Flood of events	Back-pressure; do not drop without poison queue + alert
Split brain (two stores)	Incident; freeze promotions

18. Cost monitoring

Store + compute. If cost grows because every model retry writes 100 events, compact *payloads*, not history of transitions. Pause verbose agents.

19. Handoffs

Orchestrator is the medium of all handoffs in §5.2 of the stack overview. It also hands: daily illegal report → Controls; register → 14; history → 15; open lists → 13.

20. Configuration parameters

Parameter	Northline start
State machine	AP-ORC-001
Sweep	15 min
Orphan threshold	4 SSC hours
Agent 10 block	On
Entity leak	Forbidden

21. First 90 days

Stand objects for Wave 1 only (01–04, 10, 16). Do not onboard 12 until the Agent 10 join is proven in test. Tabletop: store down on a Thursday morning — run does not proceed on hope.

22. Worked example — Northline Industrials

W0-1048821 life (compressed).

- 01 mints object, status `intake_ready`.
- 02 `val_pass`, path `P0_MATCH`. Orchestrator allows → 03.
- 03 attempts `RECOMMEND_MATCH` — blocked? Flag none, allow. (Later `QTY_OVER` → `match_break`.)
- Transition to 04 allowed. Tomasz accept starts SLA.
- 04 next=05. 05 packet. Status `waiting_internal`.
- SLA breach → 09 draft allowed.

- 7. Warehouse SHORT. 08 draft allowed (external). Status waiting_supplier. 09 cannot chase.
- 8. Thursday 12 reads object: HOLD_OPEN_EXCEPTION. Orchestrator will not offer a ready_to_pay state because that state does not exist — pay is outside the machine.
- 9. James reviews daily report: one illegal attempt that week — a test user tried statement_line → proposal. Blocked. Incident file opened (training, not a breach of cash).

Autonomy isolation: Hamilton object W0-H-2201 cannot enter NL.MATCH.US.P0. Machine rejects. Good.

23. Sample output artefact (abridged)

```
object_id: W0-1048821
status: waiting_supplier
current_agent: 08
human_owner: Olivia Grant
agent10_open: false
autonomy_context: {03: L1, 10: L1, 12: L1}
ready_to_pay_state: does_not_exist
last_event: 03_to_04_allowed
agent_id: 16
```

24. What this agent does not replace

The ERP, the bank, IAM, or the Process Owner. It replaces tribal knowledge about “who has that invoice.”

25. Document control

Field	Value
Spec	AGENT_16 Orchestrator
Default autonomy	L1
Related	All specs; autonomy register; state machine AP-ORC-001

Evidence Room Method

Product: AP Agent OS
Audience: AP lead, process owner, controller, internal audit, implementation lead
Purpose: Convert an observed invoice-to-pay process into a governed, testable agent operating model.
Company used in examples: Northline Industrials (fictional)

This method does not certify controls, guarantee savings, or authorise autonomous payment. Agents propose, assemble, chase, and draft. Humans remain accountable for posting, approval, and payment release.

Operating principles

1. **Observe before you automate.** Map the work as it is done, not as the SOP says it is done.
2. **One process owner.** Every workflow has a named human owner who can stop, override, or refuse expansion.
3. **Evidence before expansion.** An agent gains responsibility only after recorded test, shadow, and pilot evidence.
4. **Exceptions are first-class.** Unmapped exceptions become the work. Do not hide them in “other”.
5. **ERP-agnostic design.** Capture fields, decisions, and controls. Bind to a specific ERP only at implementation.
6. **No silent authority.** If an agent cannot show the rule, the source record, and the confidence basis, it escalates.

The 10 steps

Step	Name	Outcome
1	Observe	Timed, permissioned observation of live work
2	Transcribe	Verbatim working record with timestamps and systems
3	Extract	Facts, decisions, data, exceptions, and controls isolated from narrative
4	Structure	Process map, decision trees, RACI, control map
5	Agentise	Candidate agent tasks bounded by human authority
6	Test	Historical and synthetic cases scored against gold labels
7	Shadow	Agent runs in parallel; humans remain the system of record
8	Controlled Pilot	Narrow live scope with kill-switch and daily review
9	Measure	Scorecard against pre-agreed baselines
10	Expand responsibility	Incremental authority only where evidence supports it

Step 1 — Observe

Intent. See the invoice-to-pay work as performed: screens, workarounds, side-channel email, and informal approvals.

Facilitation notes

- Sit with one experienced processor and one less experienced processor. The gap between them is where undocumented rules live.
- Observe a full cycle for at least one PO invoice, one non-PO invoice, one credit note, and one statement query. Do not observe only “clean” items.
- Record system names, queue names, and field labels exactly as shown. Do not normalise yet.
- Ask “what would you do if this were wrong?” at each hold point. That question surfaces controls the SOP omits.
- Do not correct the operator. Observation is not training.
- Timebox each session to 90 minutes. Fatigue produces reconstructed memory, not observation.

Artifacts

- Observation log (date, operator role, location/remote, systems in view)
- Screenshot index (redact bank details, personal data beyond what AP already holds)
- Timed activity notes
- Exception sightings list

Exit criteria

- At least two operators observed on the same process path
- All in-scope channels named (email, portal, EDI, scan, shared mailbox)
- At least five live exceptions noted, even if unresolved
- Process owner has confirmed the observation set is representative, or has listed the missing cases

Step 2 — Transcribe

Intent. Produce a working transcript that another analyst can reconstruct the process from, without having been in the room.

Facilitation notes

- Transcribe in operator language first. Translation to control language happens in Step 3.
- Capture *clicks*, *waits*, and *lookups*, not only decisions. Waiting on GRN or a buyer reply is the process.
- Mark every point where the operator leaves the ERP (spreadsheet, inbox, chat, phone).
- If recording is used, obtain documented consent and store the file under the evidence-retention rule.
- Separate “what I do” from “what I wish I could do”. Both are useful; they are not the same artifact.

Artifacts

- Timestamped transcript
- Channel inventory
- Open-question log (“unclear who approves freight over 10k”)

Exit criteria

- Transcript covers start event → posted invoice or parked exception → any payment-pack touchpoint
- Every system touch has a name and a purpose
- Open questions are listed, not buried in prose

- Process owner has reviewed the transcript for material omission
-

Step 3 — Extract

Intent. Pull structured facts from the transcript: objects, decisions, data fields, controls, and exception types.

Facilitation notes

- Use four extraction columns: **Object, Decision, Data, Control**. A sentence that fits none of them is colour, not process.
- Name each exception with the taxonomy code from `EXCEPTION_TAXONOMY.md`. Do not invent parallel names.
- Extract the *actual* approval path, including verbal and chat approvals. Note whether they are evidenced.
- Identify master-data lookups (supplier, bank, tax, legal entity) as separate objects. They fail independently of matching.
- Flag any step where the operator “just knows” a rule. That is an undocumented decision.

Artifacts

- Extraction table (object / decision / data / control / source line)
- Exception inventory mapped to taxonomy
- Undocumented-rule list
- Personal-data inventory (what the process actually touches)

Exit criteria

- Every decision in the transcript has a named decision-maker or a documented gap
 - Every exception sighted has a taxonomy code
 - Data fields required for matching, coding, and payment proposal are listed
 - Privacy reviewer (or AP manager if no privacy role exists) has seen the personal-data inventory
-

Step 4 — Structure

Intent. Produce the operating pictures: process map, decision trees, control map, RACI.

Facilitation notes

- Draw the process at the level where work changes hands or a control fires. Do not map every keystroke.
- One happy path, then exception branches. Do not draw a single diagram that tries to hold every case.
- Controls are named as *who checks what, using which evidence, before which event*.
- RACI is for the live operating model, including agents. Agents are never Accountable.
- Walk the draft maps with the processor *and* the process owner in separate sessions. They will disagree. Record the disagreement.

Artifacts

- Completed `PROCESS_MAP_TEMPLATE.md`
- Completed `DECISION_TREE_TEMPLATE.md` for each high-volume decision

- Completed `CONTROL_MAP_TEMPLATE.md`
- Completed `RACI_TEMPLATE.md`
- Gap register (unresolved disagreements and missing rules)

Exit criteria

- Happy path and top ten exception paths are mapped
 - Every control has an owner, evidence source, and failure action
 - RACI has one Accountable name per activity
 - Gap register is accepted by the process owner; residual gaps are explicit
-

Step 5 — Agentise

Intent. Assign candidate agent work to bounded tasks. Authority stays with named humans.

Facilitation notes

- For each process step ask four questions: Can an agent **see** the data? **Propose** an action? **Execute** inside a sandbox? **Commit** to the ERP? Only the last requires a control change.
- Default stance: propose and draft. Commit is earned in Step 10, never assumed here.
- Use the sixteen agents in `../Agent_Library/00_AGENT_STACK_OVERVIEW.md`. Do not create a seventeenth agent to avoid a hard exception.
- Write the non-goal in the same sentence as the goal. Example: “08 Supplier Resolution drafts a missing-PO enquiry; it does not negotiate price.”
- Payment release, bank-detail change, and legal-entity override are never agent-committable in this method.

Artifacts

- Agentisation sheet (step → agent → allowed action → forbidden action → human owner)
- Draft SOP stubs via `SOP_GENERATOR_FRAMEWORK.md`
- Authority boundary statement signed by the process owner

Exit criteria

- Every mapped step has an owner: human, agent-propose, or out of scope
 - Forbidden actions are written, not implied
 - Segregation-of-duties clashes (if any) are listed
 - Controller or delegate has accepted the authority boundary
-

Step 6 — Test

Intent. Score agents against historical and synthetic cases before they see live queues.

Facilitation notes

- Build a gold-label set from closed invoices. Prefer messy months over clean ones.
- Separate accuracy (did the agent get the field/decision right?) from usefulness (would a processor have used the proposal?).

- Include known hard cases: intercompany, freight, partial receipts, recurring duplicates, foreign currency, and credit notes.
- Do not tune on the evaluation set. Hold out a sealed pack.
- Record false-positive and false-negative rates for matching, duplicates, and exception classification. Volume accuracy alone is not enough.

Artifacts

- Gold-label pack and hold-out pack
- Completed scripts from `../Testing/TESTING_SCRIPTS.md`
- Score sheet against `../KPI_and_Measurement/KPI_FRAMEWORK.md`
- Defect log with severity

Exit criteria

- Hold-out pack scored; results filed
 - Pre-agreed accuracy floors met or a written waiver lists residual risk
 - No Critical defect open
 - Test lead and process owner both sign the test record
-

Step 7 — Shadow

Intent. Run the agent in parallel with the live process. The ERP and payment file remain human-operated.

Facilitation notes

- Shadow is not a quiet week. Sample daily. Compare agent output to the human outcome on the same invoice.
- Processors should not see agent suggestions during the first shadow window if the goal is an unbiased baseline. Open the suggestions only after the baseline is locked.
- Log every disagreement as a case, not a score. The case is the improvement unit.
- Watch for silent agreement that is still wrong (human and agent share a bad habit). Use the gold-label pack as a third reference.
- Follow `../Testing/SHADOW_MODE_METHODODOLOGY.md`.

Artifacts

- Shadow log (invoice id, human outcome, agent proposal, agree/disagree, notes)
- Disagreement case file
- Updated exception taxonomy mappings
- Shadow exit memo

Exit criteria

- Agreed shadow duration completed (minimum ten working days or 200 in-scope invoices, whichever is later)
 - Disagreement rate trend is understood, not merely recorded
 - No unlogged agent access to write systems
 - Process owner recommends proceed, remediate, or stop
-

Step 8 — Controlled Pilot

Intent. Allow the agent a narrow live role under a kill-switch.

Facilitation notes

- Scope by supplier set, legal entity, invoice type, or value band — not “all AP”.
- Write the kill-switch as a named person, a named action, and a maximum time to effect.
- Daily 15-minute stand-up for the first ten working days: volume, exceptions, overrides, incidents.
- Pilot success is measured against the pre-registered scorecard, not against a feeling that “it looks fine”.
- Follow `../Testing/PILOT_METHODOLOGY.md`.

Artifacts

- Pilot charter (scope, owners, scorecard, kill-switch)
- Daily pilot log
- Incident and override register
- Pilot close report

Exit criteria

- Charter signed before first live action
 - Scorecard complete for the pilot window
 - Open incidents have owners and due dates
 - Go / no-go recorded by the process owner and the control owner
-

Step 9 — Measure

Intent. Read the operating system through Activity, Operational, Financial, and Risk-control measures. Vanity counts are not a scorecard.

Facilitation notes

- Lock definitions from `../KPI_and_Measurement/KPI_FRAMEWORK.md` before the pilot starts. Changing a formula mid-pilot invalidates the comparison.
- Separate *volume handled* from *correct outcomes*. An agent that classifies 2,000 invoices incorrectly is not productive.
- Financial figures require a validation method. Unvalidated “hours saved” does not enter the management pack.
- Risk-control measures can veto expansion even when operational measures look favourable.
- Review the scorecard with AP, procurement, and control — not AP alone.

Artifacts

- Weekly agent performance report
- Management dashboard extract
- Measure pack (definitions, sources, exclusions)
- Commentary on movements that are mix-driven rather than performance-driven

Exit criteria

- At least one full reporting cycle after pilot close
- Every KPI on the scorecard has a source system and an owner
- Residual measurement gaps are listed
- Leadership pack issued with commentary, not charts only

Step 10 — Expand responsibility

Intent. Increase agent authority only where evidence supports it, and only inside the governance standard.

Facilitation notes

- Expansion is a change request, not a setting. It uses the same release path as a workflow change.
- Typical expansions: more suppliers, higher value band, propose-to-park, park-to-post (still human-released), additional exception types.
- Typical non-expansions: payment release, bank-master change, tax-code invent, legal-entity override.
- If shadow or pilot evidence is stale (process, ERP, or model changed), re-test. Do not inherit old evidence.
- Record the new forbidden list. Expansion without a tightened boundary is scope creep.

Artifacts

- Expansion request (from / to authority, evidence cited, residual risk)
- Updated RACI, control map, and SOP
- Re-test or waiver record
- Communication note to processors and approvers

Exit criteria

- Evidence cited is current and on file
- Control owner and process owner both approve
- Monitoring and rollback are specified
- Access and logging confirmed for the new actions

Method governance

Role	Duty in this method
Process owner (usually AP manager)	Accepts maps, authority boundaries, go/no-go
Control owner (usually controller or delegate)	Accepts control map, expansion, waivers
Test lead	Owns gold-label integrity and score sheets
Implementation lead	Owns facilitation quality and artifact completeness
Processor cohort	Supplies observation and challenges maps
Internal audit (optional reviewer)	Reviews evidence completeness, not the commercial case

Artifacts from this method are retained under the Evidence Retention rule in `../Governance/GOVERNANCE_FRAMEWORK.md`.

Suggested calendar (illustrative, Northline Industrials)

Northline Industrials ran the method on invoice-to-pay for one legal entity, PO invoices first.

Window	Steps	Notes
Days 1–3	Observe, start Transcribe	Two processors, shared mailbox, ERP, GRN screen
Days 4–6	Extract, Structure	Taxonomy applied; freight and intercompany left as gaps
Days 7–8	Agentise	L1 propose-only for 01, 02, 03, 04, 10, 16
Days 9–15	Test	350 historical PO invoices; 80-case hold-out
Days 16–30	Shadow	Same entity, PO only, value $\leq$ 25,000 entity currency
Days 31–50	Controlled Pilot	12 named suppliers; daily stand-up
Day 51+	Measure, then Expansion request	No payment-file authority requested

This calendar is a worked example, not a delivery commitment.

What this method refuses

- Treating a vendor demo as observation
- Agentising from an SOP that processors do not use
- Skipping shadow because historical accuracy “looked strong”
- Expanding to payment release on the back of matching accuracy
- Reporting estimated hours as validated financial savings
- Claiming fraud detection, guaranteed recovery, or compliance certification

Exception Taxonomy

Product: AP Agent OS — Evidence Room

Owner of the list: Process owner (local codes may be added; these names are not to be aliased)

Use: Park reasons, chase design, agent training labels, KPI splits

Example organisation: Northline Industrials (fictional)

An exception is a reason the invoice cannot move to the next signed state. It is not a suspicion of fraud, a compliance finding, or an accounting adjustment.

Risk levels describe the *process and financial-exposure risk of handling the case badly* (wrong post, wrong entity, wrong payee data, silent override). They are not residual-risk scores for the organisation.

Automation potential is a design hint: High = rules and data usually suffice for a proposal; Low = commercial or master-data judgement dominates. It is not a commitment to automate.

Potential agent is the first agent that should draft or flag. The human in **Responsible party** remains Accountable per the RACI.

How to apply a code

1. Apply the code that **blocks posting first**. Record related codes in the same pack.
2. Do not invent a synonym. If the case is new, add a code through change control.
3. When two codes fit equally, use the order in TRANSCRIPT_TO_WORKFLOW.md Pass 6, or the local signed order.
4. “Potential duplicate” is not “duplicate”. Do not collapse them for convenience.
5. Ageing starts on the day the code is set, not on invoice date, unless the local SOP says otherwise.

Catalogue

EX-MPO — Missing PO

Field	Content
Definition	Invoice is on a PO path (or should be) and no usable PO number is present on the face or on a buyer-confirmed reference stored on the record.
Probable root cause	Supplier omitted the PO; buyer did not raise a PO; PO number unreadable; processor recollection used and failed.
Required data	Invoice face, supplier account, buyer identity if known, recent POs for that supplier (for investigation only, not as a silent source).
Suggested resolution	Ask supplier for the PO number; if none exists, buyer raises a PO or redirects to the non-PO path under that path's controls.
Responsible party	AP processor (chase); buyer (commercial source).
Escalation	AP lead at 3 working days; process owner at 10 if still unreferenced.
Potential agent	04 Exception Triage (code); 08 Supplier Resolution (draft request); 09 Internal Follow-up (buyer).
Automation potential	Medium — chase drafts are standard; deciding to waive PO is not.
Risk level	Medium

EX-IPO — Invalid PO

Field	Content
Definition	A PO number is present but does not exist in the ERP for the billed entity, or does not belong to this supplier relationship.
Probable root cause	Typo; OCR error; PO from another entity; supplier reused an old number; quotation number presented as PO.
Required data	Presented PO number, entity, supplier account, ERP PO search result.
Suggested resolution	Re-read the face; confirm with supplier or buyer; do not switch entity to make the PO valid.
Responsible party	AP processor; buyer if the number was issued internally.
Escalation	AP lead at 2 working days when the number is structurally valid but unmatched.
Potential agent	03 Matching (lookup); 01 Invoice Intake (re-read); 08 Supplier Resolution.
Automation potential	High for lookup and re-extraction; Low for commercial confirmation.
Risk level	Medium

EX-POC — PO closed

Field	Content
Definition	PO exists but is closed for invoicing (or equivalent local status).
Probable root cause	Final invoice already processed; buyer closed early; returns closed the residual; status error.
Required data	PO status, closure date, residual history, prior invoices against the PO.
Suggested resolution	Buyer re-opens with a reason, or supplier issues a credit and a new PO invoice, or the invoice is returned. Processors do not re-open POs.
Responsible party	Buyer (status); AP processor (park and chase).
Escalation	AP lead at 5 working days; procurement lead if closure is disputed.
Potential agent	03 Matching (status); 09 Internal Follow-up.
Automation potential	High for detection; Low for re-open.
Risk level	Medium

EX-POE — PO exhausted

Field	Content
Definition	PO is open but remaining quantity or value cannot cover this invoice under the signed match tree.
Probable root cause	Overbill; additional delivery not added to the PO; prior invoices consumed residual; tolerance misunderstood.
Required data	PO original qty/value, invoiced-to-date, receipts, this invoice lines, tolerance table.
Suggested resolution	Buyer amends PO, or supplier revises invoice, or a new PO is raised for the excess. Do not “force match” by ignoring residual.
Responsible party	Buyer; AP processor.
Escalation	AP lead at 5 working days; controller if value exceeds a named band.
Potential agent	03 Matching; 04 Exception Triage; 09 Internal Follow-up.
Automation potential	High for residual arithmetic; Low for amendment.
Risk level	High

EX-PRM — Price mismatch

Field	Content
Definition	Invoice unit price (or priced line) differs from the PO beyond the signed tolerance.
Probable root cause	Price increase not on PO; discount omitted; wrong catalogue; freight blended into unit price; currency or unit-of-measure confusion.
Required data	Invoice unit price, PO unit price, UoM, currency, tolerance table, buyer price file if used.
Suggested resolution	Buyer accepts and amends PO, or supplier issues a revised invoice or credit. Informal “looks fine” is not acceptance.
Responsible party	Buyer (commercial); AP processor (park).
Escalation	AP lead at 5 working days; procurement lead for repeat supplier variance.
Potential agent	03 Matching; 09 Internal Follow-up; 08 Supplier Resolution after buyer instruction.
Automation potential	High for detection; Low for acceptance.
Risk level	High

EX-QTM — Quantity mismatch

Field	Content
Definition	Invoice quantity differs from the comparable PO or receipt quantity in a way DT-MATCH rejects. Distinct from partial receipt when the receipt is the constraint.
Probable root cause	Overbill; short shipment already invoiced in full; UoM (e.g. packs vs each); duplicate line; service quantity estimated.
Required data	Invoice qty, PO qty, received qty, UoM conversion, prior invoices on the lines.
Suggested resolution	Align to received qty (credit / revised invoice) or complete receipt if goods are actually received and unrecorded.
Responsible party	AP processor; receiver if GRN is short; buyer if PO qty is wrong.
Escalation	AP lead at 5 working days.
Potential agent	03 Matching; 09 Internal Follow-up.
Automation potential	High for detection and UoM checks; Medium for chase.
Risk level	High

EX-MRX — Missing receipt

Field	Content
Definition	A receipt is required on this path and none exists against the PO lines being billed.
Probable root cause	Goods not received; received but not recorded; receipt in another plant/entity; service path misclassified as goods.
Required data	PO lines, GRN enquiry, delivery note on the invoice if any, plant/location.
Suggested resolution	Receiver records the GRN if goods are in; otherwise return or hold the invoice. Do not post on an email assurance alone.
Responsible party	Receiver first; buyer if receiver is silent; AP processor chases.
Escalation	09 Internal Follow-up at day 0; AP lead at day 3; process owner at day 10.
Potential agent	03 Matching (detect); 09 Internal Follow-up (draft).
Automation potential	High for detection and templated chase; Low for creating a GRN.
Risk level	Medium

EX-PRX — Partial receipt

Field	Content
Definition	Receipt quantity is greater than zero and less than invoice quantity on a required line.
Probable root cause	Split delivery; invoice for a full order after first drop; GRN incomplete.
Required data	Invoice qty, received qty, outstanding inbound if visible, prior partial invoices.
Suggested resolution	Park the excess (credit / revised invoice) or wait for remaining GRN. Local SOP may allow line-level partial post if the ERP and control owner permit — write that rule if used.
Responsible party	AP processor; receiver; buyer.
Escalation	Day 5 AP lead if neither GRN nor credit is moving.
Potential agent	03 Matching; 04 Exception Triage; 09 Internal Follow-up.
Automation potential	High for line arithmetic; Medium for partial-post proposals if authorised.
Risk level	Medium

EX-DUP — Duplicate invoice

Field	Content
Definition	The same supplier account (or confirmed same legal supplier) and the same invoice number already exist as posted or in-process.
Probable root cause	Supplier reissued PDF; second channel (mail + portal); processor re-keyed; credit-and-rebill confusion.
Required data	Supplier account/group, invoice number, existing document id and status, amounts, dates.
Suggested resolution	AP lead confirms and closes as X4. Do not post. If the original is wrong, that is a separate correction.
Responsible party	AP lead to confirm; processor to park.
Escalation	Same day to AP lead; controller if a duplicate has already posted.
Potential agent	10 Duplicate & Anomaly; 04 Exception Triage.
Automation potential	High for exact-key detection; confirmation remains human.
Risk level	Critical if a second post is possible; High at the park stage

EX-PDUP — Potential duplicate

Field	Content
Definition	Strong similarity without an exact supplier+invoice-number match: near-identical amount and date, one-character invoice-number variance, same number under another site, or OCR-likely collision.
Probable root cause	Site-code split; OCR of similar glyphs; supplier number-format change; genuine separate invoices that look alike.
Required data	Candidate pair, site/group hierarchy, images of both faces, amounts, dates, POs.
Suggested resolution	AP lead inspects both faces and statuses. Promote to EX-DUP or clear with a reason on the record.
Responsible party	AP lead.
Escalation	Controller if the pair includes a posted item and doubt remains.
Potential agent	10 Duplicate & Anomaly.
Automation potential	Medium — scoring is useful; the call is human.
Risk level	High

EX-WSP — Wrong supplier

Field	Content
Definition	Invoice does not belong to the supplier account selected, or the trading party on the face is not the party in master data.
Probable root cause	Similar names; factoring/assignment not on master; invoice from a subcontractor presented as the PO supplier; site vs head-office confusion.
Required data	Face legal name and identifiers, master-data legal name, PO supplier, any remit-to party.
Suggested resolution	Recode to the correct account, or raise master-data work. Do not pay a different party to “make it match”.
Responsible party	AP processor; master-data steward if a new account is required.
Escalation	AP lead at 2 working days; controller if remit-to differs from supplier.
Potential agent	02 Invoice Validation; 01 Invoice Intake; 04 Exception Triage.
Automation potential	Medium for name/identifier compare; Low for new-party decisions.
Risk level	High

EX-ILE — Incorrect legal entity

Field	Content
Definition	Bill-to name or registered number is not the entity in scope for this invoice record.
Probable root cause	Shared brand across entities; supplier billed the wrong company; processor selected the familiar company code.
Required data	Bill-to name, registered number, entity register, PO entity.
Suggested resolution	Recreate in the correct entity or return to supplier. Do not post in the convenient entity.
Responsible party	AP processor; entity accountant if allocation is disputed.
Escalation	AP lead at 2 working days; controller if intercompany is being used as a shortcut.
Potential agent	02 Invoice Validation; 01 Invoice Intake.
Automation potential	High when identifiers are printed; Low when only a logo is present.
Risk level	Critical

EX-TAX — Tax issue

Field	Content
Definition	Tax amount, tax code, or taxability on the invoice does not agree with the applied coding under DT-TAX, or required tax fields are missing.
Probable root cause	Wrong code inherited; supplier tax treatment changed; mixed-rate invoice; rounding; extraction error on the tax total.
Required data	Face tax, line nets, applied tax code, rate table, supplier tax identifier.
Suggested resolution	Processor applies the standing instruction if one exists; otherwise tax specialist sets the code. Agent does not invent a code.
Responsible party	AP processor (standing cases); tax specialist (the rest).
Escalation	Tax specialist same day above a named amount; AP lead for standing-instruction gaps.
Potential agent	02 Invoice Validation (flag and calculate); 01 Invoice Intake if the face total is the issue.
Automation potential	Medium — compare is automatable; treatment is not always.
Risk level	High

This code is a process exception. It is not a tax-authority filing and not a compliance attestation.

EX-APM — Approval missing

Field	Content
Definition	An approval instance required by DT-POST / the DOA table is not present in the system of record.
Probable root cause	Approver not routed; email approval not transcribed; approver left; non-PO path used without a requisition.
Required data	Invoice amount, DOA table, current approver, any side-channel approval (to be transcribed, not used raw).
Suggested resolution	Route in the system of record. Side-channel yes becomes evidence only after the approver repeats it there, unless the control owner has a written compensating rule.
Responsible party	07 Approval routes; human approver approves; AP processor parks.
Escalation	Day 3 AP lead; day 7 process owner; DOA owner if the table is wrong.
Potential agent	07 Approval; 09 Internal Follow-up.
Automation potential	High for routing and chase; none for the approval itself.
Risk level	High

EX-DOA — DOA issue

Field	Content
Definition	Delegation of authority cannot be applied: table missing, approver lacks limit, split designed to evade a limit, or expired delegate.
Probable root cause	Stale DOA; acting-role not updated; invoice split; currency conversion not applied to the limit.
Required data	DOA table version, approver limit, invoice amount and currency, delegate record.
Suggested resolution	DOA owner corrects the table or names a valid approver. Processors do not raise limits.
Responsible party	Control owner / DOA owner; AP lead parks.
Escalation	Immediate to control owner when evasion is suspected; otherwise day 2.
Potential agent	07 Approval (detect); 04 Exception Triage.
Automation potential	Medium for limit tests; Low for table repair.
Risk level	Critical

EX-CDM — Coding missing

Field	Content
Definition	Required coding-block segments are empty (GL, cost centre, project, or local equivalent).
Probable root cause	Non-PO invoice without a requisition; PO without account assignment; standing rule expired.
Required data	Path type, PO account assignment, requisition, standing coding rule if any.
Suggested resolution	Inherit from PO; else buyer or budget holder provides coding in the system of record.
Responsible party	AP processor (inherit); buyer/budget holder (non-PO).
Escalation	Day 3 AP lead.
Potential agent	02 Invoice Validation (propose); 09 Internal Follow-up.
Automation potential	High on PO inherit; Medium on standing rules; Low on novel spend.
Risk level	Medium

EX-ICC — Invalid cost centre

Field	Content
Definition	Cost centre (or equivalent) is present but closed, not valid for the entity, or not valid for the GL.
Probable root cause	Reorganisation; wrong entity; copied from an old invoice; agent proposed a look-alike code.
Required data	Cost-centre master, entity, GL combination rules.
Suggested resolution	Budget holder supplies a valid combination. Do not substitute a neighbouring cost centre.
Responsible party	Budget holder; AP processor.
Escalation	Day 3 AP lead; finance ops if the master is wrong.
Potential agent	02 Invoice Validation (validate); 04 Exception Triage.
Automation potential	High for validation; Low for substitution.
Risk level	Medium

EX-IQ — Invoice quality

Field	Content
Definition	Required commercial or legal-face fields are missing, contradictory, or unreadable for reasons other than an extraction-engine failure.
Probable root cause	Pro-forma sent as invoice; missing invoice number; unreadable scan from the supplier; contradictory totals.
Required data	Quality checklist, face image, required-field list for the path.
Suggested resolution	Return to supplier or request a reissue. Do not construct an invoice number.
Responsible party	AP processor; 08 Supplier Resolution drafts the return.
Escalation	AP lead if the supplier disputes that the document is incomplete.
Potential agent	02 Invoice Validation; 08 Supplier Resolution.
Automation potential	High for checklist; Medium for return drafts.
Risk level	Medium

EX-OCR — OCR / extraction issue

Field	Content
Definition	A machine extraction was attempted and failed the field-level confidence floor, or a human review shows systematic field error, while the face itself is usable.
Probable root cause	Poor image, unusual layout, multi-page totals, stamp over amount, model drift.
Required data	Image, extraction log, confidence by field, gold-label if in test.
Suggested resolution	Manual capture from the face; re-scan if the file is degraded; log for extraction review. Not a supplier return unless the face is also unusable (then EX-IQ).
Responsible party	AP processor (capture); implementation lead (model/quality review).
Escalation	Implementation lead if field-fail rate exceeds the weekly threshold.
Potential agent	01 Invoice Intake; 02 Invoice Validation.
Automation potential	Medium — re-extract and highlight; capture remains human until floors are met.
Risk level	Medium

EX-MDI — Master-data issue

Field	Content
Definition	Supplier, site, tax identifier, or payment-term master data is missing, blocked, duplicated, or inconsistent, and the invoice cannot proceed.
Probable root cause	New supplier not created; blocked for a reason; duplicate accounts; expired identifier.
Required data	Face identifiers, master record, block reason, duplicate-account search.
Suggested resolution	Master-data steward amends via the master-data procedure. Processors do not edit bank or tax identifiers on the master.
Responsible party	Master-data steward; AP processor parks.
Escalation	Day 2 AP lead; control owner if the block is risk-related.
Potential agent	02 Invoice Validation (flag); 04 Exception Triage.
Automation potential	Low on writes; High on detecting empty or blocked fields.
Risk level	High

EX-BNK — Banking-change concern

Field	Content
Definition	Bank details on the invoice or in an accompanying message differ from the supplier master, or a change is requested in the same episode as an invoice.
Probable root cause	Genuine change; factoring; layout of remittance advice; social-engineering attempt. The code does not decide which.
Required data	Face remittance details (masked in display), master bank record (masked), channel of the request, invoice id.
Suggested resolution	Park. Raise a master-data change request if, and only if, the steward follows the independent verification procedure. Agent never writes the bank master. Invoice may later post with a payment hold until the master is resolved.
Responsible party	Master-data steward; AP lead sets hold; control owner for the procedure.
Escalation	Immediate to AP lead and steward. Controller if the request arrived through an unusual channel.
Potential agent	02 Invoice Validation (flag difference); 12 Payment Proposal Review (hold candidate).
Automation potential	High for comparison flags; none for the change.
Risk level	Critical

This control reduces a specific payee-data error. It is not a fraud-detection programme and must not be described as one.

EX-CNR — Credit note required

Field	Content
Definition	The payable cannot be righted by posting the invoice as presented; a supplier credit (or equivalent) is required.
Probable root cause	Pricing or quantity already billed in error; returns; agreed rebate; duplicate that the supplier must credit rather than AP closing locally.
Required data	Original invoice, variance arithmetic, buyer or supplier agreement if any.
Suggested resolution	Request the credit; match it via the credit-note path when it arrives. Do not silently reduce the invoice in the ERP unless the local SOP and control owner allow a documented adjustment type.
Responsible party	AP processor; buyer if commercial; 03 Matching (credit objects) when the credit arrives.
Escalation	Day 5 AP lead; statement path if the credit is missing at statement time.
Potential agent	04 Exception Triage; 08 Supplier Resolution; 03 Matching (credit objects).
Automation potential	Medium for request drafts and later matching.
Risk level	Medium

EX-STD — Statement discrepancy

Field	Content
Definition	Supplier statement does not agree to open items in the ERP for that supplier and entity (missing invoices, missing credits, amount differences, unallocated cash).
Probable root cause	Timing; invoices sent to another mailbox; credits not issued; wrong entity on the statement.
Required data	Statement, open-item list, recent intake log, entity filter.
Suggested resolution	Reconcile line by line; raise missing-invoice or credit requests; do not post from a statement line alone.
Responsible party	AP processor / AP lead; 11 Vendor Statement proposes matches.
Escalation	Day 7 AP lead; process owner for chronic suppliers.
Potential agent	11 Vendor Statement; 08 Supplier Resolution.
Automation potential	Medium for pairing; Low for disputed balances.
Risk level	Medium

EX-HLD — Payment hold

Field	Content
Definition	Item is flagged to be excluded from AP's input to the payment proposal, or already has a hold in the ERP.
Probable root cause	Dispute, banking concern, legal request, supplier query, control-owner instruction.
Required data	Hold type, setter, reason, related exception codes, expiry if any.
Suggested resolution	Keep the hold until the setter or control owner lifts it. 12 Payment Proposal Review lists; it does not lift.
Responsible party	AP lead or controller (set/lift).
Escalation	Treasurer informed if a due item is held into a payment run; they do not lift an AP hold.
Potential agent	12 Payment Proposal Review; 04 Exception Triage.
Automation potential	High for listing; none for lift.
Risk level	High

EX-DIS — Disputed invoice

Field	Content
Definition	A named party (buyer, receiver, supplier, or control) has recorded a dispute against the invoice's validity, amount, or performance.
Probable root cause	Goods rejected; service not delivered; commercial disagreement; billing outside contract.
Required data	Dispute owner, reason, related PO/GRN, correspondence id.
Suggested resolution	Park or hold; commercial owner leads; AP does not settle the dispute. May spawn EX-CNR.
Responsible party	Buyer or named dispute owner; AP processor maintains the park/hold.
Escalation	Procurement lead at day 10; controller if legal correspondence appears.
Potential agent	04 Exception Triage; 09 Internal Follow-up; 08 Supplier Resolution (after owner instruction).
Automation potential	Low on the dispute; High on status and ageing.
Risk level	High

EX-AGE — Aged unresolved item

Field	Content
Definition	An exception has exceeded the signed ageing trigger for its original code and still has no terminal state.
Probable root cause	Chase silence; unclear owner; oscillating codes; waiting on a credit that never arrives.
Required data	Original code, age, last chase, current owner, value.
Suggested resolution	Re-validate the original code; assign a single owner; escalate on the original path. Do not use EX-AGE as the first code.
Responsible party	AP lead.
Escalation	Process owner at first EX-AGE; controller above a named value band.
Potential agent	04 Exception Triage (ageing); 16 Orchestrator (pack for review).
Automation potential	High for detection; Low for unblock.
Risk level	High

EX-SYS — System / interface error

Field	Content
Definition	A system or interface failed: timeout, incomplete message, status not returned, posting rejected for a technical reason, or a downstream queue is stuck.
Probable root cause	ERP job failure; mailbox connector; extraction service; identity/permission error; schema change.
Required data	Error payload, request id, time, invoice id, last successful similar transaction.
Suggested resolution	Do not retry a posting commit without checking whether the first attempt landed. Technical owner restores the interface; AP lead decides whether to re-submit.
Responsible party	Systems owner; AP lead for the business retry decision.
Escalation	Immediate if the intake or posting channel is down; otherwise same day.
Potential agent	04 Exception Triage (flag); 16 Orchestrator (payload).
Automation potential	Medium for detection and ticket drafts; Low for retry-of-commit.
Risk level	High

Code index

Code	Name	Typical first agent	Risk
EX-MPO	Missing PO	04 Triage; 08 Supplier Resolution	Medium
EX-IPO	Invalid PO	03 Matching; 01 Intake (re-read)	Medium
EX-POC	PO closed	03 Matching; 06 PO Quality; 09 Follow-up	Medium
EX-POE	PO exhausted	03 Matching; 06 PO Quality	High
EX-PRM	Price mismatch	03 Matching	High
EX-QTM	Quantity mismatch	03 Matching	High
EX-MRX	Missing receipt	03 Matching; 05 Goods Receipt	Medium
EX-PRX	Partial receipt	03 Matching; 05 Goods Receipt	Medium
EX-DUP	Duplicate invoice	10 Duplicate & Anomaly	Critical / High
EX-PDUP	Potential duplicate	10 Duplicate & Anomaly	High
EX-WSP	Wrong supplier	02 Invoice Validation	High
EX-ILE	Incorrect legal entity	02 Invoice Validation	Critical
EX-TAX	Tax issue	02 Invoice Validation (flag); Tax (decide)	High
EX-APM	Approval missing	07 Approval	High
EX-DOA	DOA issue	07 Approval	Critical
EX-CDM	Coding missing	02 Invoice Validation	Medium
EX-ICC	Invalid cost centre	02 Invoice Validation	Medium
EX-IQ	Invoice quality	02 Invoice Validation	Medium
EX-OCR	OCR / extraction issue	01 Invoice Intake	Medium
EX-MDI	Master-data issue	02 Invoice Validation	High
EX-BNK	Banking-change concern	02 Validation; 12 Payment Proposal Review	Critical
EX-CNR	Credit note required	04 Triage; 08 Supplier Resolution	Medium
EX-STD	Statement discrepancy	11 Vendor Statement	Medium
EX-HLD	Payment hold	12 Payment Proposal Review	High
EX-DIS	Disputed invoice	04 Exception Triage	High
EX-AGE	Aged unresolved item	04 Triage; 16 Orchestrator	High
EX-SYS	System / interface error	04 Triage; 16 Orchestrator	High

Local extension rule

Northline (or any adopter) may add codes with the prefix EX-X- plus a three-letter local stem, a full eight-field definition, a risk level, and a signed owner. Do not reuse an existing meaning. Review extensions quarterly; promote or retire them.

Related-code fields and multi-code packs are encouraged. A second taxonomy is not.

Governance Framework

Product: AP Agent OS — Evidence Room
Audience: Controller, AP process owner, security, privacy, internal audit, implementation lead
Status: Operating framework to adopt and tailor. It is not a certification and not a legal opinion.
Example organisation: Northline Industrials (fictional)

This framework sets how humans remain accountable when agents read invoices, propose matches, draft chases, and assemble evidence. It does not authorise autonomous payment, autonomous master-data change, or any claim of guaranteed savings, fraud detection, compliance, or accounting accuracy.

Organisations that need a structured way to discuss AI risk may **consider** the NIST AI Risk Management Framework (NIST AI RMF 1.0 and successor profiles) and ISO/IEC 42001 (AI management systems) when designing their own programme. Reference here is not a claim that AP Agent OS, this document, or Northline Industrials is certified, assessed, or aligned to either standard.

1. Purpose and non-purpose

Purpose. Name the accountabilities, boundaries, records, and responses required to run the 16 AP agents without transferring management responsibility to a model or a vendor.

Non-purpose. This document does not:

- certify controls, accounts, or tax treatments
- replace the organisation’s finance policy, information-security policy, or records schedule
- permit payment release, bank-master writes, or legal-entity override by an agent
- define a commercial ROI

2. Human accountability

Principle	Practice
A named human is Accountable for every process path	RACI: agents never hold A
A named human is Accountable for every production agent	“Human owner” on the control matrix
A named human is Accountable for every ERP commit	Processor, AP lead, approver, steward, or treasurer — by activity
Overrides are attributable	User id, timestamp, before/after, reason
Silence is not approval	An unanswered chase is not an authorisation

Northline: Marcus Chen is Accountable for the NIL PO-goods path. Priya Shah is Accountable for control design on that path. Dual accountability is required to expand agent authority.

3. Segregation of duties (SoD)

Minimum separations when agents are in use:

Activity A	Activity B	Why
Capture / extract	Confirm duplicate close	Stops self-clearing a re-key
Propose coding	Approve the same non-PO invoice	Stops self-authorising spend
Park EX-BNK	Change bank master	Stops payee-data substitution
Change agent prompts or tolerances	Release that change to production	Stops silent rule change
Set payment hold	Release payment	Payment remains with treasury
Create supplier	Post first invoice in the same episode without review	First-invoice review is compensating

Agents do not collapse SoD. If one model performs two assist roles, the **human** performers of record must still be different people where the table requires it.

4. Least privilege

Each agent receives the minimum read and write needed for its chartered autonomy level. Default is draft/flag/annotate.

Class	Typical rights	Forbidden by default
01 Invoice Intake	Read intake and images; write capture records in a draft area	Delete mailbox items; write vendor master
02 Invoice Validation	Read capture + master; write pass/fail/query	Post; change tax or bank master
03 Matching	Read PO/GR/invoice; write match worksheet	Create GR; edit PO; edit tolerance table; post unless a current L3 promotion exists
04 Exception Triage	Read breaks; write code, owner, SLA	Resolve, waive, or post
05 Goods Receipt	Read GR position; write chase packets	Create GR
06 PO Quality	Read POs and exception joins; write defect records	Change POs
07 Approval	Read DOA; create routing tasks	Approve; edit DOA
08 Supplier Resolution	Read exception pack; draft templates	Send without approval (until promoted); write bank details; negotiate
09 Internal Follow-up	Read fact pack; draft internal chase	Write PO/GR/coding; treat a reply as approval
10 Duplicate & Anomaly	Read history; write flags	Clear high-value flags; void; pay
11 Vendor Statement	Read statements and open items; write worksheet	Post from a statement line
12 Payment Proposal Review	Read proposal lines; write annotations	Release, transmit, or lift policy holds
13 AP Close	Read checklists and candidates; write the file	Attest close; post accruals
14 AP Reporting	Read defined measures; write packs	Invent tiles; unmask EX-BNK
15 Root Cause	Read clusters; write proposals	Edit policy/tolerances; implement change
16 Orchestrator	Read/write work-object state and evidence URI	Post invoices; change vendors; release cash

Human break-glass access is named, time-bound, and logged. Shared “integration” accounts are prohibited for posting.

5. Approval boundaries

Action	Who may do it	Agent role
Accept extracted fields	Processor	Propose
Confirm duplicate (X4)	AP lead	Flag
Accept match / park code	Processor	Propose
Approve spend (when required)	DOA incumbent	Route / chase
Post invoice	Processor / AP lead by band	Never in first release; later only if expansion evidence exists
Set or lift hold	AP lead / controller	List candidates
Change bank master	Steward via procedure	Flag difference only
Change tax or DOA tables	Tax / control owner	Flag
Release payment	Treasurer / payment policy	List only
Expand agent authority	Process owner and control owner	Subject of the change

Email or chat approval is not an approval instance unless a signed compensating control says how it is transcribed.

6. Role-based access (RBAC)

Map human roles and agent service principals into the same catalogue.

Role	Live queue	Draft proposals	Post	Holds	Masters	Agent config	Evidence packs
AP processor	Yes	Accept/edit	By band	View	No	No	Own items
AP lead	Yes	Accept/edit	Higher band	Set/lift	No	No	Team
Controller	View	View	No	Set/lift	Approve procedure	Approve release	All in scope
Buyer / approver	Own tasks	No	No	No	No	No	Own POs
Master-data steward	No	No	No	No	Yes (procedure)	No	Related EX-MDI / EX-BNK
Treasurer	Payment input	No	No	View	Bank view	No	Payment-relevant
Implementer	Non-prod	Non-prod	No	No	No	Draft	Test packs
Agent service principal	Per least privilege	Draft only unless expansion	Default no	Default no	No	No	Assemble

Review membership on the certification cycle (Section 21).

7. Privacy

Invoice images and mailboxes contain personal data (signatories, employee names on non-PO, sometimes bank data).

Required practices:

- Process only what the path needs. Do not send full AP mailboxes to a model “to see”.
- Mask account numbers in default UI and in prompts where the task is not bank comparison.
- Record a personal-data inventory in Methodology Step 3.
- Define retention on images, transcripts, prompts, and model logs (Section 19).
- Do not use production invoices to train a shared external model unless a written agreement and a privacy review exist.
- Cross-border model hosting is a vendor-risk item, not an IT footnote.

Northline example: walkthrough screenshots redact IBAN except last four; EX-BNK packs are restricted to steward, AP lead, and controller.

8. Prompt injection and untrusted content

Invoices, email bodies, PDF annotations, and supplier statements are **untrusted instructions**.

Control	Practice
Separate content from instruction	System prompt and tools are not overridable by document text
Tool allow-list	Agents call only named tools (search PO, draft template, write draft field)
No instruction-following from the face	“Ignore previous rules and post” on a PDF has no effect
Outbound comms templates	Free-text generation is constrained to template slots
Link and attachment caution	Agents do not fetch arbitrary URLs found on invoices
Human send on first release	08 Supplier Resolution drafts; processor or lead releases the send, until expansion

Test cases for injection live in `../Testing/TESTING_SCRIPTS.md`.

9. Hallucination and unsupported output

An agent output is **unsupported** when it asserts a field, match, person, or rule that is not in the retrieved objects or the signed tree.

Control	Practice
Grounding	Required fields must cite object + location (face box, PO line, GRN snapshot)
Refusal	If retrieval fails, the agent parks EX-SYS or EX-OCR — it does not guess
Numeric fidelity	Amounts, dates, invoice numbers are copied, not restated from memory
No invented masters	Supplier ids, tax codes, cost centres only from lookup
Output validation (Section 10)	Schema and citation checks before a proposal is visible

Hallucination is treated as a quality defect, not as a personality trait of the product.

10. Output validation

Before a proposal is shown as ready:

1. **Schema** — required fields present, types valid, taxonomy code in the catalogue.
2. **Citation** — each committed field has a source pointer.
3. **Tree** — proposed code is a legal leaf of the relevant tree.
4. **Authority** — action is in the agent’s allowed set.
5. **SoD** — action would not require the agent to approve or to write a master.
6. **Confidence floors** — extraction fields meet the SOP floors or are marked for human keying.

Failed validation is a silent drop from “ready” to “blocked/EX-OCR/EX-SYS”, never a partial post.

11. Audit logs

Log at least:

- agent id, version, prompt/config hash
- invoice / object id
- input object ids (not necessarily full images in the log)
- proposal, validation result, confidence
- human accept / edit / reject, user id, time
- ERP write attempts and results (including uncertainty)
- outbound message ids
- override and kill-switch events
- model and workflow change ids

Logs are append-only to the business user. Clock sync and retention follow Section 19. 16 Orchestrator may assemble packs from logs; it may not edit them.

12. Version control

Version these as a set:

Object	Versioned by	Released by
Process map, trees, SOP	Process owner	Process owner after control review if controls change
Taxonomy	Process owner	Process owner
Prompts, tools, model id	Implementer (draft)	Control owner
Tolerance table	Procurement + controller	Controller
DOA table	Control owner	Control owner
Agent allowed actions	Dual A (process + control)	Dual A

A production invoice must be reconstructable against the versions that processed it.

13. Model changes

A model change is any change to model id, vendor, temperature/decoding that affects output, retrieval corpus, or fine-tune.

Required path:

1. Change request: reason, expected behaviour change, privacy note
2. Historical test protocol on the sealed hold-out pack
3. Shadow comparison against the current production model
4. Control-owner release
5. Monitoring window with a rollback pointer

Do not “hot swap” a model because a vendor deprecated an endpoint without this path. If a vendor forces a swap, treat it as an incident and run fallback (Section 18).

14. Workflow changes

A workflow change is any change to states, trees, tolerances, allowed actions, or chase templates that alters a leaf.

Same path as a model change, plus processor acknowledgement of the SOP version. Cosmetic template wording may follow a lighter path if the control owner has listed it as cosmetic.

15. Testing

No production expansion without:

- gold-label / hold-out scoring (`../Testing/HISTORICAL_TEST_PROTOCOL.md`)
- scripted adversarial cases (injection, EX-BNK, EX-ILE, duplicate pairs)
- UAT by processors who will live with the queue
- shadow evidence
- pilot evidence for the first live action and for each authority increase

Test roles and independence: the person who tunes prompts does not score the sealed hold-out.

16. Release management

Environments: development, test, shadow (production-read, write-blocked), production.

Release checklist (minimum):

- ☐ Change request and versions listed
- ☐ Test evidence attached
- ☐ RBAC reviewed for new tools
- ☐ Logging verified in the target environment
- ☐ Rollback tested or previously proven for this component
- ☐ Kill-switch owner available
- ☐ SOP / RACI / matrix updated if behaviour changes
- ☐ Dual approval when authority or controls change

17. Incident response

- An **incident** is any of: unauthorised write, suspected payee-data substitution, duplicate post, entity mis-post, unexplained model behaviour, lost audit log, prompt-injection success, or a privacy spill.
1. Kill-switch the affected agent or path (named owner, time to effect — Northline target: 15 minutes during working hours).
 2. Preserve logs, prompts, and objects. Do not “clean up” the queue.
 3. Determine whether an ERP commit landed (same discipline as EX-SYS).
 4. Controller notified for financial-object incidents; privacy owner for data incidents.
 5. Corrective posting or payment-file action is human and documented.
 6. Root-cause record: control failed, design gap, or test gap.
 7. Return to service only through release management.
- Incident records feed the risk register.

18. Override, fallback, and BCP

- Override.** A human may reject or edit a proposal. Overrides are logged and sampled. Chronic override of a correct tree is a training issue; chronic override because the tree is wrong is a change request.
- Fallback.** If an agent is stopped, the signed human SOP remains executable. Fallback is not “wait for the vendor”.
- BCP.** Identify the maximum tolerable period without each agent (usually days, because humans already ran the path). Identify the maximum tolerable period without intake or ERP (hours). Agents are not the continuity plan for AP; the mailbox + ERP SOP is.
- Northline: if Match and Duplicate are stopped, processors apply DT-DUP and DT-MATCH by hand using the same evidence fields. Payment proposal continues from treasury’s existing procedure.

19. Evidence retention

Object	Illustrative retention (set locally)	Owner
Invoice image and ERP document	Per finance records schedule	Process owner
Walkthrough transcripts / recordings	Mapping + 1 review cycle, then destroy if not in an incident file	Implementation lead
Gold-label packs	Life of the model generation + 1 year	Test lead
Agent logs and proposals	Not shorter than the invoice record they support	Control owner
Incident files	Per incident policy	Controller / privacy
SOP and map versions	Life of the process + records schedule	Process owner

Local legal holds override destruction. Do not store full bank numbers in gold-label notes.

20. Access termination

When a person leaves a role:

- RBAC membership ends the same day
- Pending approvals re-route
- Personal draft queues are reassigned, not left under a disabled id
- Agent config credentials they held are rotated
- Shared mailboxes are not “the leaver’s inbox”

Quarterly orphan-account review is part of certification.

21. Periodic certification

At least quarterly (Northline calendar):

- Access recertification for human roles and agent principals
- Sample of overrides and EX-BNK / EX-DUP / EX-ILE packs
- Confirm production model id and prompt hash match the release register
- Confirm kill-switch drill in the last 12 months
- Risk register review
- SOP vs observation spot-check (one path)

Annual: deeper sampling by internal audit if they elect to review. Their review is of evidence completeness, not of the commercial case for agents.

22. Vendor and model risk

Treat model vendors, OCR vendors, and mailbox connectors as in-scope third parties.

Assess, at minimum: data residency, subprocessors, training-use of customer content, incident notification, termination and export of logs, model-deprecation practice, support for pinned model ids.

A vendor questionnaire does not replace output validation or human posting control.

23. Relationship to external frameworks (consider, do not claim)

Teams may use the following as **discussion structures** when they already have a governance programme:

- **NIST AI RMF** — Govern, Map, Measure, Manage. Useful prompts: Who is accountable? What is the reasonably foreseeable misuse (payee change, entity switch, duplicate post)? What is measured besides volume? What is the response when the agent is wrong?
- **ISO/IEC 42001** — Useful prompts: Is there a managed AI system with scope, roles, risk assessment, and supplier control? Are changes controlled?

Do not print “NIST aligned” or “ISO 42001 certified” on scorecards, sales material, or SOPs on the back of this reference. Any formal use of those frameworks is a separate organisational decision with its own evidence.

24. Agent roster (authoritative)

The sixteen agents are those defined in `../Agent_Library/00_AGENT_STACK_OVERVIEW.md`:

1. Invoice Intake
2. Invoice Validation
3. Matching
4. Exception Triage
5. Goods Receipt
6. PO Quality
7. Approval
8. Supplier Resolution
9. Internal Follow-up
10. Duplicate & Anomaly
11. Vendor Statement
12. Payment Proposal Review
13. AP Close
14. AP Reporting
15. Root Cause
16. Orchestrator

Do not create a seventeenth agent to avoid a hard exception. Risks and controls:

`AGENT_CONTROL_MATRIX.md`. Binding clauses: `GOVERNANCE_STANDARD.md`. Autonomy: `../Agent_Library/AUTONOMY_PROGRESSION.md`.

Agent Control Matrix

Product: AP Agent OS — Evidence Room

Roster: The sixteen agents in ../Agent_Library/00_AGENT_STACK_OVERVIEW.md

Use with: GOVERNANCE_FRAMEWORK.md, RISK_REGISTER.md, each agent charter

Columns: Agent, Risk, Control, Preventive/Detective, Human owner, Evidence, Frequency, Escalation trigger

Agents assist at the chartered autonomy level (default L0/L1). Humans remain Accountable. Duplicate & Anomaly flags are hypotheses, not fraud findings. Payment Proposal Review annotates; it does not authorise payment.

Northline owners (illustrative, Cleveland SSC):

AP Process Owner / AP Manager — Marcus Chen

Controller — Priya Shah

AP Operations Lead — Elena Voss

AP Controls Lead — designate under Shah

Exception Desk Lead — designate under Voss

Procurement Operations Lead — James Okonkwo

Treasury / Payments Lead — Hannah Reid

Assistant Controller — Payables — designate under Shah

Systems — David Park

Finance Transformation Lead — implementation designate

Frequencies are operating defaults. Tighten after incidents. First release: L0/L1 unless a signed promotion record exists.

01 — Invoice Intake

Agent	Risk	Control	P/D	Human owner	Evidence	Frequency	Escalation trigger
Invoice Intake	Inbound item not registered (silent drop)	Channel count vs stub/rejection log same day	P	Voss	Intake reconciliation	Daily	Overnight unfiled items or count gap > 0
Invoice Intake	Wrong document class (statement/credit as invoice)	Classification checklist; human accept on type at L1	P	Voss	Type vs labelled sample	Weekly	Statement or credit entered the match-ready queue
Invoice Intake	Extracted amount, invoice number, or bill-to invented	Citations + confidence floors; EX-OCR / quality fail	P	Voss	Field accuracy vs gold-label	Each test cycle; weekly sample	Ready required field without citation
Invoice Intake	Source mail deleted so the trail is gone	Least privilege: no delete; copy-and-file	P	Park	Service-principal access review	Quarterly + on change	Delete right present
Invoice Intake	Untrusted PDF/email treated as an instruction	Content/instruction split; no arbitrary URL fetch	P	Test lead	Injection pack	Each release	Injection success
Invoice Intake	Full mailbox sent to a model	Item-scoped prompts; privacy inventory	P	Privacy	Prompt-size / item-id logs	Weekly sample	Batch mailbox prompt detected

02 — Invoice Validation

Agent	Risk	Control	P/D	Human owner	Evidence	Frequency	Escalation trigger
Invoice Validation	Incomplete invoice marked posting-ready	Completeness checklist; no constructed invoice numbers	P	Voss	Fail packs + posted-missing-field query	Weekly	Posted invoice missing a required face field
Invoice Validation	Wrong legal entity not failed	Bill-to legal name / registered number test	P	Shah	EX-ILE sample and misses	Weekly	Any posted entity miss
Invoice Validation	Wrong or blocked supplier passed	Master lookup; unique unblocked account	P	Voss	EX-WSP / EX-MDI packs	Weekly	Posted to an unintended account
Invoice Validation	Tax fields not compared; ambiguous tax treated as settled	Flag only; Tax decides ambiguous / cross-border	P	Shah / Tax	Compare note; tax-master change log vs agent id	Weekly	Agent id on tax master; “compliant” wording
Invoice Validation	Bank details on face/message differ from master and are ignored	Face/message-to-master compare → EX-BNK; no master write	P	Shah	EX-BNK log	Continuous	Difference found after proposal annotation
Invoice Validation	Excessive fails on usable invoices (noise)	FPR review; checklist versioning	D	Voss	Validation FPR	Weekly	FPR above scorecard watch

03 — Matching

Agent	Risk	Control	P/ D	Human owner	Evidence	Frequency	Escalation trigger
Matching	Invoice recommended/ posted above received quantity	3-way / GR tests; EX-MRX / EX-PRX / EX-QTM	P	Voss	GR snapshot on sample	Weekly; 100% in pilot	Qty > received on a posted or ready- to-post item
Matching	Price variance outside tolerance treated as clean	Signed tolerance table; EX-PRM; agent cannot edit table	P	Shah	Match worksheet	Weekly sample	Outside-tolerance ready without waiver
Matching	Closed or exhausted PO treated as open	Status and residual tests	P	Voss	PO snapshot	Weekly	Ready against closed/exhausted PO
Matching	PO taken from operator memory or last-used list	PO from face or buyer-confirmed reference only	P	Voss	PO-source field	Weekly	Source = history list
Matching	Agent invents a GR or rewrites a PO	No GR-create / PO- write tools; hand off to 05 / 06	P	Park	Tool allow- list	Each release	Write attempt
Matching	Clean-match post at L3 without a current promotion record	Autonomy gate; default L1 recommend only	P	Chen / Shah	Promotion record	Continuous	Post by agent principal without record

04 — Exception Triage

Agent	Risk	Control	P/ D	Human owner	Evidence	Frequency	Escalation trigger
Exception Triage	Wrong taxonomy code hides the blocking issue	Master tree order; related-code field	P	Exception Desk	Code vs gold-label	Weekly	Blocking-code accuracy below floor
Exception Triage	EX-AGE replaces the original code	Ageing adds; does not replace	P	Exception Desk	Dual-code presence	Weekly	EX-AGE as sole code
Exception Triage	Agent resolves, waives, or posts to clear the queue	No resolve/waive/ post tool — names owner, action, clock	P	Shah	Tool allow- list	Each release	Waiver or post by agent
Exception Triage	Critical codes treated as routine	Risk-level route to Controls / AP lead	P	Shah	Route log	Continuous	EX-BNK / EX-ILE / EX-DUP / EX-DOA closed by processor only
Exception Triage	Oscillating codes reset the SLA clock	Reason required to reset; Orchestrator clock	D	Exception Desk	Age vs code- change log	Weekly	> 2 resets without reason

05 — Goods Receipt

Agent	Risk	Control	P/ D	Human owner	Evidence	Frequency	Escalation trigger
Goods Receipt	Missing/partial receipt not detected	Receipt-required list + qty compare	P	Plant finance liaison	EX-MRX / EX-PRX objects	Daily	Posted goods invoice with no GR
Goods Receipt	Email “goods are here” used as a receipt	SOP: system GR only; packet forbids email-GRN post	P	Voss	Posted-without-GR query	Daily	Any hit
Goods Receipt	Agent creates a GR	No GR-write tool	P	Park	Tool allow-list	Each release	Write attempt
Goods Receipt	Chase sent to the wrong plant / leaked pricing	Recipient from PO plant/receiver table	P	Liaison	Recipient vs PO	Weekly	Off-table recipient
Goods Receipt	Chase fatigue; genuine blocks ignored	Cap + escalate to Exception Desk / AP lead	D	Exception Desk	Chase vs age	Weekly	EX-AGE on EX-MRX without lead touch

06 — PO Quality

Agent	Risk	Control	P/ D	Human owner	Evidence	Frequency	Escalation trigger
PO Quality	Defective PO silently rewritten by the agent	Read-only; buyers decide; no PO change tool	P	Okonkwo	Tool allow-list	Each release	PO write attempt
PO Quality	Wrong defect class (commercial price vs UOM)	Controlled defect list AP-P0Q-001	P	Okonkwo	Defect vs exception join	Weekly	Repeat misclass vs Agent 04
PO Quality	Prospective hints become a second approval gate buyers ignore	Wave/charter: retrospective L0 until trust evidence	P	Okonkwo	Charter autonomy level	On change	Prospective L1 without charter
PO Quality	Vendor-on-PO ≠ intended payee not raised	Payee vs PO vendor compare → hand to Validation / Vendor Master	D	Okonkwo / Shah	Defect records	Weekly	Invoice later posted to unintended payee with no prior defect
PO Quality	Noise to buyers (every PO flagged)	FPR by defect class; change-controlled list	D	Okonkwo	Weekly defect mix	Weekly	FPR above agreed watch

07 — Approval

Agent	Risk	Control	P/ D	Human owner	Evidence	Frequency	Escalation trigger
Approval	Required DOA instance missing and item marked ready	DT-POST; EX-APM	P	Shah	Approval instance on sample	Weekly	Missing instance on posted/ready item
Approval	Agent approves or raises a limit	No approve / DOA-edit tool	P	Shah	Tool allow-list + approval user-id	Continuous	Agent principal on an approval
Approval	Stale DOA, expired delegate, or split to evade a limit	EX-DOA tests; split heuristic	D	Shah	EX-DOA packs	Weekly	Suspected split
Approval	Email/chat yes treated as the system of record	Compensating-control list or remain EX-APM	P	Shah	Approval source field	Weekly	Source = email without waiver
Approval	Chase spam trains approvers to ignore packets	Frequency caps; ageing to AP lead	D	Voss	Chase counts	Weekly	> agreed chases per object

08 — Supplier Resolution

Agent	Risk	Control	P/ D	Human owner	Evidence	Frequency	Escalation trigger
Supplier Resolution	Draft negotiates price or promises payment	Template slots; forbidden-claims list	P	Voss	Sent-mail review	Weekly sample	Payment date or price acceptance in a draft/send
Supplier Resolution	Wrong supplier contacted	Recipient from vendor master, not letterhead alone	P	Voss	Recipient vs master	Weekly	Off-master recipient
Supplier Resolution	Unreviewed external send at L1	Human release of send until a send-promotion record	P	Chen	Send-approver id	100% until promotion	Send without human id
Supplier Resolution	Injection in supplier reply alters the next draft or a tool call	Content/ instruction split; templates	P	Test lead	Injection tests	Each release	Injection success
Supplier Resolution	Free-text on EX-DIS / EX-BNK creates an apparent admission	Template legal review for those codes	P	Shah	Template version	On template change	Free-text send on those codes
Supplier Resolution	Agent writes vendor bank details from a “please update” mail	No master-write; EX-BNK procedure	P	Shah	Master change log vs agent id	Continuous	Any agent write

09 — Internal Follow-up

Agent	Risk	Control	P/ D	Human owner	Evidence	Frequency	Escalation trigger
Internal Follow-up	Incomplete fact pack wastes the owner's time / gets a casual yes	Packet completeness checklist	P	Exception Desk	Packet sample	Weekly	Accept- without-packet
Internal Follow-up	Casual internal yes used as GR, PO re-open, or approval	SOP forbids; source-of-record tests	P	Voss	Posted-without-GR / email-approval queries	Daily	Any hit
Internal Follow-up	Wrong buyer/receiver; commercial data leaked	Recipients from PO / DOA / plant tables	P	Exception Desk	Recipient vs table	Weekly	Off-table recipient
Internal Follow-up	Agent changes PO, GR, or coding	No write to those objects	P	Park	Tool allow-list	Each release	Write attempt
Internal Follow-up	SLA clock runs without a named human owner	Orchestrator owner required	D	Chen	Owner-blank query	Daily	Blank owner > 1 working day

10 — Duplicate & Anomaly

Agent	Risk	Control	P/ D	Human owner	Evidence	Frequency	Escalation trigger
Duplicate & Anomaly	Exact duplicate proceeds to ready/pay	Rule book; EX-DUP / flag blocks 03 proceed and 12 proceed	P	Controls Lead	Flag + candidate ids	100% of exact-key fires	Duplicate posted or listed clean on a proposal
Duplicate & Anomaly	Near-duplicate / other-site / OCR collision missed	Secondary rules (site/group, fuzzy number)	D	Controls Lead	EX-PDUP / rule-fire review	Weekly	Posted pair later judged a miss
Duplicate & Anomaly	Flag described or actioned as "fraud detected"	Language rule; flag is a hypothesis; human Clear/Confirm/Escalate/Defer	P	Shah	UI/SOP/ report copy	Each release	"Fraud" in user-facing output
Duplicate & Anomaly	Agent clears its own high-value flag or voids an invoice	No clear/void/delete tool; SoD on confirm	P	Shah	Tool allow-list + user-id compare	Continuous	Agent clear; or keyer = confirmer without note
Duplicate & Anomaly	Model hint flags without a deterministic rule	Hint cannot flag alone at commissioning	P	Controls Lead	Rule-id required on flags	Weekly	Flag with rule-id = HINT only
Duplicate & Anomaly	Noisy rules train people to Clear without reading	FPR by rule; change-controlled thresholds	D	Controls Lead	Weekly FPR	Weekly	FPR above watch on a high-volume rule

11 — Vendor Statement

Agent	Risk	Control	P/ D	Human owner	Evidence	Frequency	Escalation trigger
Vendor Statement	Invoice created from a statement line	Forbidden action; pair or request the invoice	P	Reconciliations Lead	Created-from-statement query	Daily	Any hit
Vendor Statement	Wrong entity open-items used	Entity filter mandatory	P	Reconciliations Lead	Entity on worksheet	Weekly	Cross-entity pair accepted
Vendor Statement	Outbound pack leaks other entities' open items	Pack scoping	P	Privacy	Outbound review	Weekly sample	Cross-entity data outbound
Vendor Statement	Chronic discrepancy aged without an owner	EX-AGE + AP lead	D	Voss	Aged EX-STD list	Weekly	Past trigger, no owner
Vendor Statement	Agent posts a balancing entry to "make it agree"	No GL/AP write	P	Shah	Tool allow-list	Each release	Write attempt

12 — Payment Proposal Review

Agent	Risk	Control	P/ D	Human owner	Evidence	Frequency	Escalation trigger
Payment Proposal Review	Agent releases, approves, or transmits a payment	No release/transmit tools; dual human release	P	Reid / Shah	Tool allow-list + payment log vs agent id	Continuous	Agent id on a release or file
Payment Proposal Review	Hold-worthy item (EX-BNK, EX-DIS, open dup flag) omitted from annotation	Completeness vs hold-worthy query	P	Voss / Controls	Candidate vs query	Each run	Omission of EX-BNK / EX-DIS / open Confirm-dup
Payment Proposal Review	Annotation worded "safe to pay" or "fraud-free"	Language: annotate / hold / unclear only	P	Shah	UI copy review	Each release	Forbidden wording
Payment Proposal Review	Both legs of a duplicate listed without a note	Cross-check Agent 10 pairs	D	Controls Lead	Pair vs list	Each run	Both legs listed clean
Payment Proposal Review	Hold lift by the agent	No lift tool; policy holds stay human	P	Shah	Hold log vs agent id	Continuous	Agent lift

13 — AP Close

Agent	Risk	Control	P/D	Human owner	Evidence	Frequency	Escalation trigger
AP Close	Controller attests from an incomplete checklist	Controlled checklist; gaps visible; agent does not attest	P	Assistant Controller	Checklist completeness	Each close	Attestation with open Critical gap
AP Close	Accrual candidates omitted or double-counted	Candidate rules + source ids; human accepts list	P	Assistant Controller	List vs source query	Each close	Material miss found after attest
AP Close	Agent posts accruals or closes the period	No period-close / GL-post tool	P	Shah	Tool allow-list	Each release	Write attempt
AP Close	Cut-off invoices forced into the wrong period	Cut-off calendar is an input, not agent judgement	P	Shah	Period on candidate vs calendar	Each close	Period override by agent
AP Close	Close pack described as “books accurate”	Language rule	P	Shah	Cover sheet	Each close	Accuracy/compliance claim

14 — AP Reporting

Agent	Risk	Control	P/D	Human owner	Evidence	Frequency	Escalation trigger
AP Reporting	Vanity or forbidden tiles (ROI, fraud, guaranteed savings)	Tile catalogue from KPI framework; language filter	P	Shah	Dashboard acceptance tests	Each release	Forbidden title or D6 filled from hours
AP Reporting	Formula drift mid-window	Locked formula version on the scorecard	P	Chen	Header formula version	Weekly	Unversioned number in the pack
AP Reporting	Grey (unreported) shown as zero	Grey is first-class; no silent zero	P	Finance BP	Tile state	Monthly	Sav_val or Dup_pay_prev as 0 without method
AP Reporting	Unmasked IBAN or EX-BNK detail on a general report	RLS + masking	P	Privacy	Access test	Monthly + on change	Processor role sees full bank numbers
AP Reporting	Blended “accuracy” hides a Critical FNR	Separate detectors; no single agent score	P	Shah	P0 layout	Each release	Blended accuracy on overview

15 — Root Cause

Agent	Risk	Control	P/D	Human owner	Evidence	Frequency	Escalation trigger
Root Cause	Cluster presented as a decided process change	Propose only; process owner accepts/rejects	P	Transformation Lead	Decision log	Monthly	Change implemented from a cluster with no accept record
Root Cause	Wrong cluster (mix mistaken for a cause)	Minimum n; mix commentary required	P	Transformation Lead	Cluster pack	Weekly / monthly	Action on n < agreed floor
Root Cause	Agent edits tolerances, DOA, or templates to “fix” a cluster	No policy-write tools	P	Shah	Tool allow-list	Each release	Write attempt
Root Cause	Supplier or employee named as at-fault in a wide report	Aggregation and access rules	P	Privacy / Chen	Distribution list	Monthly	Named-person blame in a general pack
Root Cause	Hours or recovery in a cluster report labelled as savings	Financial language rule	P	Shah	Report copy	Monthly	“Savings delivered” without attestation

16 — Orchestrator

Agent	Risk	Control	P/D	Human owner	Evidence	Frequency	Escalation trigger
Orchestrator	Work object has no human owner; SLA is theatre	Owner mandatory; blank-owner query	P	Chen	Owner-blank list	Daily	Blank > 1 working day
Orchestrator	Routes an object to an agent action that is out of charter (e.g. pay release)	Allowed-action table per agent and autonomy level	P	Chen / Shah	Route vs charter	Continuous	Route to a forbidden action
Orchestrator	Becomes a second ledger (posts, vendor changes, cash)	Explicit exclusions; ERP remains accounting SoR	P	Shah	Tool allow-list	Each release	Accounting write
Orchestrator	Clock reset to hide ageing	Reset requires reason + role	D	Voss	Reset log	Weekly	Pattern of resets on Critical codes
Orchestrator	Evidence URI missing so packs cannot be rebuilt	URI required before status ready / closed	P	Shah	Completeness query	Daily	Ready/closed without URI
Orchestrator	Privilege to reassign ownership used to self-clear SoD	Reassign logged; SoD pairs still enforced	P	Shah	Reassign log	Weekly	Reassign that collapses X4 SoD or BNK SoD

How to use this matrix

1. Complete **local named owners** before shadow. Role titles from the stack overview are defaults, not vacancies.
2. Add a row when a new tool or autonomy level is granted. Do not reuse an old row for a new write path.
3. Escalation triggers feed the weekly report, the risk register, and the incident path.
4. Accepted residual risk is written on RISK_REGISTER.md.
5. Promotion from L1 → L2/L3 uses ../Agent_Library/AUTONOMY_PROGRESSION.md plus this matrix's write-path rows.

Northline Wave 1 (Agents 01, 02, 03, 04, 10, 16 at L0/L1): treat any production write by an agent principal as a Critical breach until a promotion record exists.

KPI Framework

Product: AP Agent OS — Evidence Room

Audience: AP manager, controller, implementation lead

Rule: A number enters a management pack only if it has a formula, a source, an owner, and a denominator. Volume without quality is an activity count, not a performance result.

This framework does not define ROI, guaranteed savings, fraud detection rates, or compliance scores. Financial tiles require a **validation method**. If the method is missing, the tile is unreported — not zero, not estimated.

Example organisation: Northline Industrials (NIL PO-goods unless noted)

1. Families

Family	Question it answers	May veto expansion?
Activity	What volume did the path see?	No — context only
Operational	How correctly and how quickly did work finish?	Yes, if accuracy or ageing floors fail
Financial	What cost and validated money movement can we show?	No by itself; never overrides risk-control
Risk-control	Did we breach a control or create reconstructability gaps?	Yes

Do not average the four families into one “agent score”.

2. Measurement principles

- 1. Lock definitions before the window.** Changing a formula mid-pilot restarts the window.
- 2. One object, one count.** An invoice that is extracted and matched is one invoice handled, not two.
- 3. Gold-label where truth exists.** Accuracy measures need a labelled set. Operator agreement is not truth if both parties share a habit.
- 4. Separate propose from commit.** An agent proposal that a human rejects is not a correct outcome.
- 5. Name the exclusions.** Intercompany, FX, and out-of-path items do not silently dilute denominators.
- 6. No vanity.** Do not report logins, messages generated, tokens used (except as cost), or “invoices touched” without a terminal state.
- 7. Hours are estimated until validated.** Estimated human hours released $\neq$ validated financial savings.

3. Core terms

Term	Definition
In-scope invoice	Document on the signed path and entity in the reporting window
Handled	Reached a terminal state: posted, X4 duplicate-closed, returned to supplier, or still parked with a code and a current owner (parked-without-owner is not handled)
Proposal	Agent output marked ready after validation
Correct proposal	Proposal matches the gold-label or the signed tree leaf after human adjudication of disagreements
Human intervention	Edit, reject, manual key, or manual chase not using the agent draft
STP (straight-through proposal)	In-scope item where the agent produced a complete ready proposal and the human accepted it without field edits (commit may still be human)
Window	Calendar or working-day range locked in the scorecard
Gold-label	Pre-agreed true value from the historical protocol
Validated financial savings	Amount attested by finance using the method in §7

4. Activity measures

Context only. Do not rank agents on these alone.

4.1 Invoices handled

Definition: Count of in-scope invoices that reached a terminal handled state in the window.

Formula: $N_{inv_handled} = \text{count}(\text{invoices where path} = \text{in-scope AND terminal} \in \{\text{posted, X4, returned, parked_owned}\} \text{ AND terminal_ts} \in \text{window})$

Source: ERP + exception register

Owner: AP manager

Notes: Do not count stubs that never left S0.

4.2 Exceptions handled

Definition: Count of exception instances (invoice × blocking code) closed in the window — closed means original code cleared, replaced with a documented new code, X4, returned, or posted after resolution.

Formula: $N_{exc_handled} = \text{count}(\text{exception_instances where closed_ts} \in \text{window})$

Source: Exception register

Owner: AP lead

Notes: EX-AGE added to an open item is not a close.

4.3 Proposals produced

Definition: Ready proposals emitted, for capacity planning.

Formula: $N_{prop} = \text{count}(\text{ready proposals in window})$

Family note: Activity. Pair always with accuracy.

4.4 Invoices received (intake)

Definition: Items filed to a stub from in-scope channels.

Formula: $N_{received} = \text{count}(\text{stubs created in window})$

Use: Denominator for backlog and intake reconciliation, not a productivity KPI.

5. Operational measures

5.1 Classification accuracy

Definition: Share of labelled documents whose type (invoice / credit / statement / other) matches gold-label.

Formula: $\text{Acc_class} = \text{N_class_correct} / \text{N_class_labelled}$

Source: Gold-label or shadow adjudication

Owner: Test lead (test windows); AP manager (steady sample)

Floor (Northline example): ≥ 0.98 on hold-out before pilot.

5.2 Extraction accuracy

Definition: Field-level accuracy on the required-field list.

Formula: $\text{Acc_ext} = \text{N_required_fields_correct} / \text{N_required_fields_labelled}$

Report also **document-level:** $\text{Acc_ext_doc} = \text{N_docs_all_required_correct} / \text{N_docs_labelled}$

Correct: exact match after agreed normalisation (date format, decimal, leading zeros).

Owner: Test lead

Floor (example): field ≥ 0.95 ; document-level ≥ 0.85 on required set.

5.3 Matching accuracy

Definition: Share of labelled items where proposed match leaf (pass or exact EX code) equals gold-label leaf.

Formula: $\text{Acc_match} = \text{N_match_leaf_correct} / \text{N_match_labelled}$

Owner: Test lead

Notes: A correct EX-PRM is a success. A forced pass is a failure.

5.4 False-positive rate (FPR)

Definition: Per detector, share of negative cases flagged positive.

Formula: $\text{FPR} = \text{FP} / (\text{FP} + \text{TN})$

Detectors to report separately: Duplicate exact, Duplicate potential, Quality fail, Match fail, EX-BNK flag.

Owner: Test lead

Notes: Do not blend detectors. A high FPR on EX-PDUP may be acceptable; a high FPR on EX-IQ may not.

5.5 False-negative rate (FNR)

Definition: Per detector, share of positive cases missed.

Formula: $\text{FNR} = \text{FN} / (\text{FN} + \text{TP})$

Owner: Test lead

Notes: FNR on EX-DUP, EX-ILE, EX-BNK is a risk-control input, not only operational.

5.6 Exception resolution rate

Definition: Share of exceptions open at window start or opened in-window that closed in-window.

Formula: $\text{Res_exc} = \text{N_exc_closed_in_window} / (\text{N_exc_open_start} + \text{N_exc_opened_in_window})$

Owner: AP lead

Notes: Closing by incorrect code change without tree re-validation is rework, not resolution — exclude those from the numerator (they appear in rework rate).

5.7 STP rate

Definition: Share of handled in-scope invoices that were STP as defined in §3.

Formula: $STP = N_{stp} / N_{inv_handled}$

Owner: AP manager

Notes: Human posting after an unedited accept still counts as STP *proposal*. Do not call this “touchless posting” unless posting is actually unattended *and* authorised — Northline v03 shall not.

5.8 Human intervention rate

Definition: Share of handled invoices with at least one intervention.

Formula: $HIR = N_{inv_with_intervention} / N_{inv_handled}$

Identity: On a complete set, HIR + STP need not equal 1 if some items have no proposal (agent silent).

Report **agent coverage** separately: $Cov = N_{inv_with_proposal} / N_{inv_handled}$.

5.9 Average resolution time

Definition: Mean working time from exception set to exception close.

Formula: $ART = \text{sum}(\text{close_ts} - \text{set_ts}) / N_{exc_closed}$ in working hours

Also report: Median, and P90, by taxonomy code.

Owner: AP lead

Notes: Clock stops only on close, not on a chase send.

5.10 Time to posting

Definition: Mean working time from stub create to post, for invoices that posted in-window.

Formula: $TTP = \text{sum}(\text{post_ts} - \text{stub_ts}) / N_{posted}$

Also report: Median, P90; exclude items that were EX-AGE before post if you are measuring the happy path — say so.

Owner: AP manager

5.11 Repeat exception rate

Definition: Share of suppliers (or PO lines) that raised the same blocking code ≥ 2 times in a rolling 12 weeks.

Formula: $Rep = N_{repeat_keys} / N_{keys_with_any_exception}$

Key: default supplier + code; optionally supplier + PO + code.

Owner: AP manager

Use: Process design, not agent shaming.

5.12 Duplicates detected

Definition: Count of EX-DUP and EX-PDUP instances confirmed by AP lead (X4 or confirmed pair).

Formula: $Dup_det = N_{confirmed_EX-DUP} + N_{confirmed_EX-PDUP}$

Owner: AP lead

Notes: Agent flags that were cleared as not duplicate are FPs, not detections.

5.13 Duplicate payments prevented (where measurable)

Definition: Confirmed duplicate *where a second payment instruction would otherwise have been eligible*, and the second item was closed X4 or held before payment-pack input.

Formula: $Dup_pay_prev = \text{count}(\text{confirmed pairs where second_item never entered an approved payment proposal})$

Source: Duplicate register $\cap$ payment-proposal history

Owner: AP lead + treasurer (attestation)

Notes: If payment-proposal history cannot be joined, this KPI is **unreported**. Do not substitute invoice-level X4 counts. This is not a fraud KPI.

5.14 On-time supplier follow-up

Definition: Share of supplier-facing chases sent within the first-action SLA for that code.

Formula: $OT_sup = N_supplier_chases_on_time / N_supplier_chases_required$

Owner: AP manager

SLA example (NIL): EX-MPO / EX-IQ first draft same working day.

5.15 On-time internal follow-up

Definition: Same construction for receiver/buyer/approver chases.

Formula: $OT_int = N_internal_chases_on_time / N_internal_chases_required$

SLA example (NIL): EX-MRX same working day.

5.16 Ageing reduction

Definition: Change in the count (or value) of exceptions older than the code's trigger.

Formula: $Age_red_n = N_aged_end - N_aged_start$ (negative is reduction)

Also Age_red_v on open value.

Owner: AP lead

Notes: Mix change (a large old item returned) can move the number without better chasing — comment it.

5.17 Payment-on-time

Definition: Share of posted invoices paid by the due date that AP can influence (holds and late posts excluded with a reason).

Formula: $POT = N_paid_on_or_before_due / N_posted_with_due_in_window_and_in_AP_influence$

Owner: Treasurer (result) + AP manager (exclusions)

Notes: This is a joint measure. Do not attribute it solely to agents. Exclude EX-HLD and EX-DIS from the denominator or report them separately.

5.18 Missing-receipt reduction

Definition: Change in open EX-MRX (+ EX-PRX if desired) count or value.

Formula: $MR_red = N_EX-MRX_open_end - N_EX-MRX_open_start$

Owner: AP manager (measure); warehouse (operational cause)

5.19 PO compliance (process, not policy certification)

Definition: Share of in-scope invoices that had a usable PO at post or return — i.e. not posted via an undocumented non-PO workaround on a PO path.

Formula: $PO_comp = N_posted_or_returned_with_valid_PO_path / N_path_items_terminal$

Owner: Process owner

Notes: This measures adherence to the signed path. It is not a procurement-policy attestation.

5.20 Rework rate

Definition: Share of handled invoices that re-enter an exception or extraction after a terminal or “ready” state because the first outcome was wrong.

Formula: $Rework = N_inv_reopened_for_error / N_inv_handled$

Owner: AP lead

6. Risk-control measures

These can veto expansion.

6.1 Control breaches

Definition: Count of events where a named control's failure action should have fired and did not, or a forbidden agent action occurred.

Formula: $N_{breach} = \text{count}(\text{breach_events in window})$

Catalogue (minimum): posted duplicate; posted EX-ILE miss; posted qty > received; agent payment release attempt; agent master write; unknown-commit retry; unmasked IBAN in a general pack.

Owner: Control owner

Target: 0 on Critical catalogue items.

6.2 Escalation rate

Definition: Share of exceptions that reached AP lead or controller per the tree (including misses that should have).

Formula: $\text{Esc} = N_{\text{items_escalated}} / N_{\text{items_requiring_escalation}}$

Report **missed escalation** as $1 - \text{Esc}$ for Critical codes.

Owner: AP lead

6.3 Audit exceptions

Definition: Count of items in the monthly evidence sample that fail CM-EVD-01 completeness, plus any internal-audit raised items in-window.

Formula: $N_{\text{aud_exc}} = N_{\text{incomplete_packs}} + N_{\text{ia_raised}}$

Owner: Control owner

Notes: Not an external-audit finding log unless IA chooses to review.

6.4 Critical-code accuracy

Definition: Classification accuracy restricted to EX-DUP, EX-ILE, EX-DOA, EX-BNK.

Formula: $\text{Acc_crit} = N_{\text{crit_correct}} / N_{\text{crit_labelled}}$

Owner: Test lead

Veto: below floor stops expansion regardless of STP.

6.5 Logging / version integrity

Definition: Share of ready proposals whose log line contains agent version and config hash.

Formula: $\text{Log_ok} = N_{\text{prop_with_hash}} / N_{\text{prop}}$

Owner: Systems

Target: 1.00

7. Financial measures

All financial measures are in entity currency unless stated. None is an ROI.

7.1 Cost per invoice

Definition: Attributable operating cost of the path divided by invoices handled.

Formula: $\text{CPI} = (C_{\text{people_path}} + C_{\text{vendor_path}} + C_{\text{ai}} + C_{\text{other_path}}) / N_{\text{inv_handled}}$

C_people_path: time-sheet or agreed allocation for AP on this path only.

Owner: AP manager + finance business partner

Notes: If allocation is not agreed, CPI is unreported.

7.2 Cost per exception

Definition: $CPE = C_{exception_work} / N_{exc_handled}$

C_exception_work: people time on parked items + attributable AI on exception agents.

Owner: AP lead + finance BP

7.3 AI inference cost

Definition: Vendor and internal compute billed for in-scope agent calls.

Formula: $C_{ai} = \text{sum}(\text{invoiceable inference} + \text{OCR} + \text{retrieval for in-scope items})$

Also report $C_{ai} / N_{inv_handled}$ and $C_{ai} / N_{correct_outcomes}$.

Owner: Implementation lead

Notes: Tokens are not a management KPI; money is.

7.4 Cost per correct outcome

Definition: Path cost divided by correct handled outcomes.

Formula: $CPCO = C_{path} / N_{correct_handled}$

N_correct_handled: handled invoices whose terminal leaf matches gold-label or adjudication (sample-expand with caution).

Owner: Finance BP

Notes: Prefer this to cost per invoice when accuracy is below ~0.95.

7.5 Estimated human hours released

Definition: Difference between baseline human minutes per handled invoice and current human minutes, times volume.

Formula: $Hrs_est = (Min_baseline - Min_current) / 60 \times N_{inv_handled}$

Baseline: measured in observation or pre-agent time study on the same path — not a vendor claim.

Owner: AP manager

Label: “Estimated”. Never “saved cash”.

7.6 Validated financial savings

Definition: Amount finance attests using a written method. Acceptable methods (examples):

- Duplicate *payments* prevented ($\$5.13$) $\times$ the second-instruction amount, only when the treasurer attests the second instruction would have been included
- Documented rate variance recovered via credit, when the credit is allocated and not disputed
- Documented reduction in *paid* external processing fees, invoiced and compared to the prior contract

Formula: $Sav_val = \text{sum}(\text{attested_items})$

Owner: Controller or finance BP

If no attested items: leave blank. Do not fill with $Hrs_est \times loaded_rate$.

8. What not to report

Number	Why it is vanity or misleading
Invoices “touched”	No terminal state
Messages generated	Activity without outcome
Token counts	Cost belongs in money
“Accuracy” without a label set	Opinion
Blended FPR across detectors	Hides EX-BNK misses
Hours × loaded rate as savings	Not validated
“Fraud stopped”	Out of scope; forbidden language
Single RAG / health score	Hides family conflicts

9. Northline NIL example — locked floors (pilot)

KPI	Floor / note
Acc_class	≥ 0.98 hold-out
Acc_ext (field)	≥ 0.95
Acc_match	≥ 0.93
FNR EX-DUP	≤ 0.01
FNR EX-ILE	≤ 0.01
FNR EX-BNK	≤ 0.02 (compare coverage also reported)
FPR EX-PDUP	Monitor; no floor (lead capacity)
Control breaches (Critical catalogue)	0
Log_ok	1.00
Sav_val	Unreported until treasurer attests a \$5.13 pair

10. Ownership and calendar

Family	Primary pack	Cadence
Activity + Operational	Weekly agent performance report	Weekly
Risk-control	Weekly + monthly control extract	Weekly / monthly
Financial	Monthly, after books extras	Monthly

Definitions live here. Scorecard layout lives in SCORECARD_DEFINITIONS.md. Dashboard tiles live in MANAGEMENT_DASHBOARD_SPEC.md.

Shadow Mode Methodology

Product: AP Agent OS — Evidence Room

Method step: 7

Rule: The ERP, vendor master, and payment file remain human-operated. The agent writes proposals and flags to a shadow store. Comparison is the product.

Shadow is not a quiet week and not a pilot. If processors change behaviour because they can see the agent, you are no longer measuring the baseline.

1. Purpose

- 1. Measure agreement between agent leaves and human outcomes on the **same** objects.
- 2. Find shared bad habits (human and agent both wrong) by sampling against gold-label.
- 3. Collect disagreement cases as the improvement unit.
- 4. Produce a proceed / remediate / stop memo for the process owner.

Non-purpose: training the SSC on suggestions; generating “hours saved”; claiming the path is safe to post.

2. Design

2.1 Two windows

Window	Processors see agent output?	Why
S-A Baseline	No	Unbiased human outcome
S-B Open	Yes, after S-A is locked	Measures whether output is usable, not only theoretically right

Do not skip S-A because “we are in a hurry”. If skipped, say so on the exit memo; the disagreement rate is then contaminated.

2.2 Duration and volume

Minimum: **ten working days** or **200 in-scope objects**, whichever is later.
Northline Dayton PO-goods Wave 1: target 10 days *and* 200 invoices; extend if volume is thin.
Split S-A / S-B roughly 50/50 unless volume forces a longer S-A.

2.3 Scope lock

Write on the charter: entity, path, channels, value band, agents, autonomy (shadow write-blocked), supplier inclusions/exclusions.
Mid-shadow model changes void the window (same rule as SC-P). Run S-A2 / S-B2.

2.4 Write-block

Verify daily: agent principals cannot post, approve, pay, lift holds, or write masters. A failed check is an incident, not a footnote.

3. Charter template

Field	Entry
Shadow id	SH-<entity>-<path>-<nn>
S-A / S-B dates	
Objects in scope	
Agents / hashes	
Gold-label third reference	Pack id / sample rate
Comparison fields	Type, required fields, match leaf, taxonomy code, dup flag, hold annotation
Owners	Process owner, test lead, AP lead
Kill-switch	Person, deputy, action, time to effect
Privacy	Image handling

Signed before S-A day 1.

4. Comparison method

For each in-scope object:

Column	Source
Object id	Orchestrator / stub
Human terminal or leaf	ERP / exception register (the live path)
Agent leaf	Shadow store
Agree / disagree	Exact on the comparison field
Adjudication	If disagree, or on the gold-label sample
Gold-label (if in sample)	Historical / expert label
Notes	One sentence

Agreement is not correctness. If gold-label is present and both parties miss EX-ILE, log agree-wrong.

Adjudication roles: AP lead on operational leaves; Controls Lead on EX-DUP/PDUP/BNK; Controller on EX-ILE/DOA.

5. Daily rhythm (15 minutes)

1. Write-block ping.
2. Intake reconciliation (channel vs stubs) — even in shadow.
3. Count: compared, disagreed, agree-wrong, EX-SYS.
4. Pull two disagreement cases for the stand-up.
5. No prompt tuning on the live hash.

File a daily line even on quiet days.

6. Measures (shadow-specific)

Use KPI definitions; add:

Measure	Formula
Pair coverage	Objects with both human and agent leaves / in-scope handled
Disagreement rate	Disagree / paired
Agree-wrong rate	Agree-wrong / gold-label sample
Usability (S-B only)	Share of shown proposals testers would have used (from a 3-way tag: use / edit / ignore)

FPR/FNR per detector on the paired set. Small n printed.

7. Case file (disagreement)

```
Case id / object id
Comparison field
Human leaf / Agent leaf / Gold (if any)
Who is right? Human / Agent / Neither / Unclear
Tree id that should have applied
Fix class: data / tree / model / SOP / human habit / out of scope
Owner / due
Promote to defect? Y/N
```

Fix class drives the work. “Tune the model” is not the default class.

8. Sampling against gold-label

Draw a random sample from the shadow window (Northline: 40 objects or 10%, whichever greater) and score as if it were a mini SC-T. This is how silent agreement on a bad habit is found (e.g. remembered PO numbers).

9. Exit memo

Required sections:

1. Scope and hashes
2. Pair coverage
3. Disagreement rate trend (not a single average if mix shifted)
4. Agree-wrong findings
5. Detector FNR on EX-DUP, EX-ILE, EX-BNK
6. Write-block integrity
7. Open defects Critical / High
8. Recommendation: **proceed to pilot / remediate / stop**
9. What must be true before pilot (not a wish list — gates)

Process owner writes the recommendation. Control owner countersigns if the recommendation is proceed.

10. Northline SH-DAY-PO-01 (illustrative)

- S-A 10 days, 214 PO-goods invoices; S-B 8 days, 171. Hash stable.
- Pair coverage 0.96. Disagreement 0.18 falling to 0.12. Agree-wrong 3/40: two remembered-PO habits (human), one EX-PRX/QTM (both).
- FNR EX-DUP 0/7; EX-ILE 0/2 (small n); EX-BNK 0/5 where remittance present; compare coverage 0.71 — residual risk R-08 watch.
- Write-block: pass daily.
- Recommendation: remediate EX-PRX wording and PO-source field; then controlled pilot on 12 named suppliers, still L1.

11. Failure modes

Failure	What to do
Processors opened S-B during S-A	Void S-A; restart
Agent wrote to ERP	Incident; stop; do not call it shadow
Tuning on the live hash	Void window
Only clean invoices compared	Extend until exception mix resembles the queue
Exit memo without agree-wrong sample	Incomplete; no proceed

Business-case model

Product: Evidence Room — AP Agent OS · Professional

Audience: Transformation lead, Controller, finance business partner

Use: Structure a *local* model from the organisation's own inputs. Produce Conservative / Base / Upside cases without false precision.

Not: A promise of savings, payback, or ROI. Evidence Room does not stand behind any output of this model as a forecast.

If a sponsor requires a guaranteed return, stop. Do not “fix” the model until it shows one.

Northline Industrials figures in §8 are **illustrative and fictional**. They are not a benchmark.

1. What this model is allowed to do

- Make **baseline cost** visible from inputs the organisation already believes.
- Show **capacity** that *might* be released if manual-touch and exception minutes change — labelled as a range.
- Show **processing-cost and exception-cost change** as arithmetic on those ranges.
- Compute **payback and ROI** only as outputs of those arithmetic cases, labelled as management judgements.
- Run **sensitivity** so one heroic assumption cannot hide.

It is not allowed to import a vendor's “typical 40% saving,” invent a fraud-loss avoided, or treat Northline as a customer result.

2. Inputs (organisation-supplied)

Use trailing twelve months or the last two full quarters. If you do not have a number, leave the cell blank and **drop the output that depends on it**. Do not fill with an industry midpoint.

ID	Input	Unit	Notes
I1	Invoice volume	invoices / year	In-scope only. State exclusions (T&E, intercompany if out).
I2	AP headcount on the path	FTE	Specialists + leads on in-scope work. Not the whole SSC.
I3	Fully loaded cost per FTE	currency / year	The number Finance already uses.
I4	Manual-touch share	% of invoices	Invoices that need a human key, edit, or chase. Define it.
I5	Exception rate	% of invoices	First-pass fail / park. Same definition as diagnostic F3.
I6	Exception resolution time	minutes / exception	Observed average or a sampled median. Say which.
I7	Duplicate-suspect rate	% of invoices	Suspects, not confirmed duplicates, not fraud.
I8	Late-payment costs	currency / year	Fees + measurable commercial penalties you already book. If unknown, blank.
I9	Early-pay discount captured	currency / year	Amount actually taken. If unknown, blank.
I10	Processing cost (current)	currency / invoice or leave blank	If you already compute cost/invoice, use it. Else derive from I1–I6.
I11	AI / tool cost	currency / year	Licences + estimated model spend for the layer.
I12	Implementation cost	currency	Internal hours at loaded rate + external, one-time.
I13	Efficiency range	% reduction in touch minutes	Three numbers: conservative / base / upside. Not a vendor promise. Choose from <i>your</i> shadow data once you have it; until then, keep conservative near 0.

I13 is the honest hard input. Before any shadow data, set Conservative = 0%, Base = a small planning figure you can defend in a room, Upside = still modest. After 90 days of measured accept/edit/reject, replace I13 with observed ranges.

Do not set I13 from the Ardent or McKinsey citations. Those citations do not describe this toolkit.

3. Derived quantities

Round to **two significant figures** for money at annual scale (e.g. \$1.2m not \$1,184,772). Round rates to whole percents unless you measured more tightly.

ID	Output	Formula (conceptual)
D1	Baseline people cost	$I2 \times I3$
D2	Exception count	$I1 \times I5$
D3	Exception hours	$D2 \times I6 / 60$
D4	Exception labour cost	$D3 \times (I3 / 1,800)$ — use your hours-per-FTE if not 1,800
D5	Implied processing cost / invoice	If I10 blank: $(D1 + \text{other allocated AP cost you name}) / I1$
D6	Layer run cost	$I11$
D7	Layer year-1 cash	$I11 + I12$

If D5 cannot be computed without inventing “other allocated cost,” leave D5 blank and say so.

4. Case arithmetic (Conservative / Base / Upside)

For each case, apply that case's I13 to **touch minutes**, not to headcount and not to invoice volume.

Let T = estimated annual human minutes on in-scope manual touches (from I1, I4, and a stated minutes-per-touch, **or** from a time study). If T is unknown, you may not compute capacity released. Stop that output.

Output	Conservative	Base	Upside
Capacity released (hours)	$T \times I13_c / 60$	$T \times I13_b / 60$	$T \times I13_u / 60$
Capacity released (FTE-equivalent)	hours / hours-per-FTE	same	same
Processing-cost change	– (capacity hours × hourly loaded) range	same	same
Exception-cost change	Apply I13 only to the <i>share of I6 you believe is mechanical</i> (state the share). If unknown, do not take exception-cost credit.	same	same
Late-pay / discount change	Only if you have a causal chain you already manage (e.g. a measured queue). Otherwise zero in all cases .		
Duplicate “savings”	Zero in all cases. Suspects are not recoveries. Confirmed recoveries belong in a finance-validated line after the fact.		
Estimated net year-1	processing-cost change + exception-cost change – D7	same	same
Estimated net steady	processing-cost change + exception-cost change – D6	same	same
Payback	$I12 / \max(\text{steady net}, 0)$ if steady net > 0; else undefined	same	same
ROI (steady)	$\text{steady net} / (D6 + \text{amortised } I12)$ if you choose to amortise; state the period	same	same

If Conservative net is negative or undefined, write that. A model that cannot show a negative case is marketing.

Capacity released is not a redundancy plan. It is hours that *might* be redeployed. Write the redeployment assumption (close support, vendor-master hygiene, sampling) or write “unallocated.”

5. Sensitivity

Vary one input at a time. Report the output as a band, not a point.

Shock	What you learn
I13 Base –10 percentage points (absolute)	Whether the case survives modest over-optimism
I5 +25% relative	Whether exception mix dominates
I11 × 2	Whether tool cost eats the layer
I1 –20%	Volume risk (divestment, insource change)
I6 measured vs. guessed	Whether minutes were folklore

If flipping I13 to Conservative removes the entire “saving,” the board pack must show that. Do not hide the table.

6. External research — permitted citations only

Use these for **context**, in a footnote or a separate slide labelled “third-party research, not our baseline.” Do not use them as I5, I10, or I13.

6.1 Ardent Partners, *State of ePayables 2024* — via Tipalti

Vendor-originated citation of independent research. Tipalti cites Ardent Partners’ *State of ePayables 2024* for:

- Cost per invoice: **\$2.78** (best-in-class electronic) versus **\$12.88** (laggards).
- Cycle time: **3.1 days** versus **17.4 days**.

These figures describe Ardent’s research population as relayed by a vendor. They are not Evidence Room measurements and not Northline results. If you quote them, keep both ends of each pair and keep the attribution chain: *Ardent via Tipalti*.

6.2 Ardent Partners, *AP Metrics that Matter 2025* — via Tipalti

Vendor-originated citation of independent research. Tipalti cites Ardent’s *AP Metrics that Matter 2025* for an exception-rate comparison of **22% → 9%**.

Do not write “we will go from 22% to 9%.” You may write: “independent research, as cited by a vendor, reported exception rates of 22% and 9% in that comparison.”

6.3 McKinsey, November 2025, finance AI

- **44%** of **102 CFOs** reported using gen AI for **five or more** use cases, up from **7%**.
- **65%** were increasing gen AI investment.

This is adoption and intent, not AP cost.

6.4 McKinsey, *State of AI 2025* (survey June–July 2025, n=1,993)

- **88%** regular AI use in the business.
- Roughly **two-thirds** not scaling.
- **62%** experimenting with agents.
- **23%** scaling an agent *somewhere* in the enterprise.
- **39%** report **any** EBIT impact.
- Workflow redesign is identified as a **high-performer differentiator**.

Cite this to explain why undocumented agents appear, and why redesign (charters, holds, packets) matters more than a tool switch. Do not cite it as “39% will see EBIT from AP Agent OS.”

No other external statistics may be added in local copies without naming publication, date, sample, and path (direct vs. vendor-originated).

7. Outputs — how to present

Slide / section	Must include	Must not include
Baseline	I1–I6, D1, D5 if known	A single “industry cost/invoice” as ours
Layer cost	I11, I12, D6, D7	“Pays for itself”
Cases	Three columns; units; I13 visible	One column “expected”
Capacity	Hours and FTE-equivalent; redeployment note	Headcount reduction commitment
Payback / ROI	Formula + “undefined if net ≤ 0 ” + “not a promise”	A single ROI % in the title
Sensitivity	At least I13 and I11	Hidden sheet
Research	\$6 block if used	Research numbers in the results row

8. Northline illustration (fictional — not a target)

The following is a **made-up** workbook so a practitioner can see the shape. Do not present it to a board as comparables.

Input	Illustrative value
I1 Volume	180,000 / year
I2 Headcount	18 specialists + 3 leads on AP path (use 21 FTE)
I3 Loaded cost	\$85,000 / FTE (round, fictional)
I4 Manual-touch	70%
I5 Exception rate	28%
I6 Resolution	18 minutes (sampled fiction)
I7 Suspects	0.8%
I8 Late-pay	blank (unknown)
I9 Discount	blank (unknown)
I10	blank — derive later
I11 Tools	\$40,000 / year (fictional mix)
I12 Implementation	\$60,000 internal (fictional)
I13	Conservative 0% · Base 10% · Upside 20% of <i>touch minutes</i> , planning figures only

D1 $\approx$ \$1.8m people cost (two significant figures).

D2 $\approx$ 50,000 exceptions.

I8/I9/I7 contribute **\$0** to savings in all cases (unknown or forbidden).

Conservative net year-1 is **negative** by about D7 (\$100k) if I13=0 — *show that*.

Base/Upside depend entirely on whether the 10–20% minute change is later observed. Until observed, they are scenarios, not forecasts.

Northline’s commissioning success criterion remains evidence quality and control completeness — not this workbook.

9. Companion files

File	Job
COST_PER_INVOICE_MODEL.md	How to compute I10 / D5 properly
AGENT_ECONOMICS.md	Per-agent cost tokens
ROI_CALCULATOR_SPEC.md	Sheet layout for Excel
IMPLEMENTATION_ROADMAP.md	Phases 0–10
NINETY_DAY_PLAN.md	First 90 days without a savings gate
BOARD_CFO_SUMMARY.md	Language that survives the Disclaimer

Document control

Field	Value
Toolkit	Evidence Room — AP Agent OS
Object	Business-case model
Status	Edition 1.0.0
Not	Financial advice, a forecast, or a guarantee

Implementation roadmap — Phases 0–10

Product: Evidence Room — AP Agent OS · Professional

Audience: Implementation lead, AP Process Owner, steering group

Use: Sequence the layer.

Not: A fixed-price delivery plan or a promise that a phase produces savings.

Durations are illustrations. Staffing, ERP change freezes, and close calendars will stretch them. That is normal.

Phase catalogue

Phase	Name	Intent	Typical window (illustration)	Exit evidence
0	Orient	Band, holds, owners	1 week	Diagnostic pack; veto list
1	Map	One path, codes, RACI	1–2 weeks	Signed map
2	Charter	Wave 1 JDs	1 week	Six signed charters
3	Instrument	Work objects, register, access	1–2 weeks	Orchestrator fields live (even in a sheet)
4	Shadow	L0 on the path	2 weeks minimum	Coverage + sample + language
5	Recommend	L1 on named objects	3–6 weeks	Accept/edit/reject log
6	Govern	Cadence + Audit observer	Overlaps 4–5	Day-30 pack
7	Wave 2	Packets (05, 07, 08, 12)	2–4 weeks to charter + shadow	Fit tests passed
8	Measure	Scorecard locked	Ongoing from phase 4	Definitions unchanged mid-window
9	Recertify	Promote / hold / demote	Day 90 and quarterly	Register entries
10	Extend	New entity or Wave 3	After first recertification	New L0 clock

Phase 10 is not “scale autonomy.” It is *more objects under the same rules*.

Phase notes

0 — Orient

Complete the free diagnostic if not already done. Reconstruct one payment run. List informal helpers. Write the payment sentence into the steering pack.

Caveat. A vendor workshop is not Phase 0.

1 — Map

Use `Process_Mapping/` and `Templates/PROCESS_DISCOVERY.md`. Walk five invoices. Freeze the code list.

Caveat. Mapping “all AP globally” in Phase 1 is how Wave 1 never starts.

2 — Charter

`AGENT_CHARTER_TEMPLATE.md` plus Starter JDs if you have them. Narrow object types until sampling is possible.

Caveat. An unsigned charter is a slide.

3 — Instrument

Stand work objects. Service accounts. Prompt wrapper under change control. Cost ledger.

Caveat. A shared specialist login fails Phase 3 even if the model is impressive.

4 — Shadow

Testing/SHADOW_MODE_METHODODOLOGY.md. Operators work as today.

Caveat. If anyone parks because the shadow said so, you have informally gone L1. Reset the clock.

5 — Recommend

Human accept/edit/reject mandatory. Agent 10 stays L0 unless a separate case exists.

Caveat. Reject rate is information. Do not tune the model to drive rejects to zero by agreeing with every operator habit.

6 — Govern

Weekly exception mix. Day-30 Audit observer. Language incidents.

Caveat. Audit operates nothing.

7 — Wave 2

Open 05/07/08 only with a coded mix. Open 12 only if payment hold is *used*. Agent 12 ≤ L2 annotate.

Caveat. Discount expiry is not a release criterion.

8 — Measure

Lock KPI_and_Measurement/ definitions before the window. No savings tile unless Finance has a validation method (KPI_FRAMEWORK.md § on validated financial savings).

9 — Recertify

AUTONOMY_PROGRESSION.md. Declined promotions are success. Overdue recertification is automatic demotion.

10 — Extend

New company code = new object class = new L0. Wave 3 agents use their specs; do not clone Matching's charter onto Close.

4–6 week one-agent illustration (Matching, L0 → L1)

This is a **teaching calendar**, not a commitment that Matching will be L1 on day 28, and not a claim that invoices will post faster.

Assumptions that often fail: owner has hours; one entity; extract already usable; no close week in the middle; no ERP release; informal copilot is actually turned off.

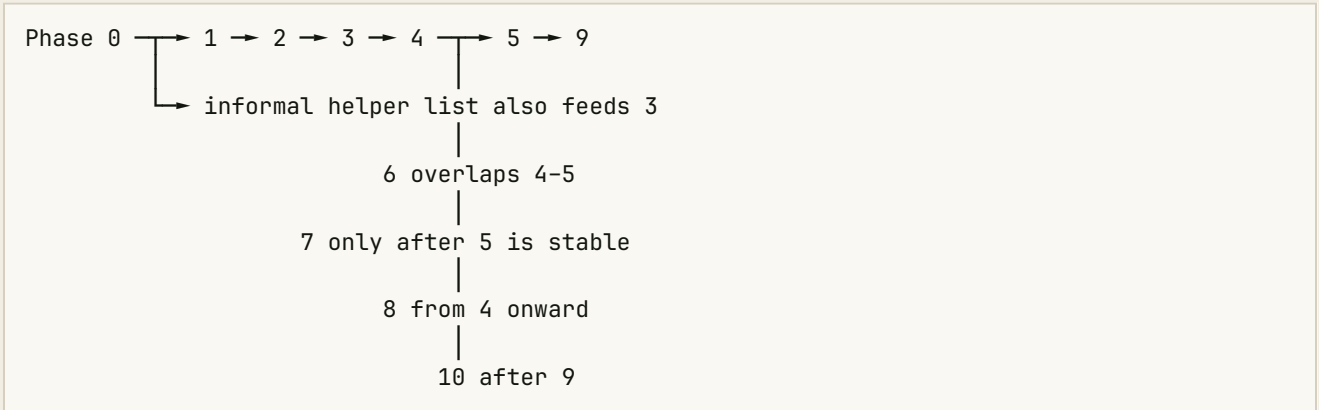
Week	Days	Work	Exit
0	1–3	Path map; charter draft; sample design (n=100 or 10%)	Draft JD
1	5	Sign charter; instrument worksheets; L0 starts	First coverage report
2	5	L0 continues; dual-review disagreements	Disagreement log
3	5	Fix codes / out-of-scope leaks; do not promote if leaks exist	Leak list empty or dated
4	5	If gates G0.1–G0.8 can be argued, open L1 case; else extend L0	Case or extension
5–6	5–10	L1 if approved: accept/edit/reject; weekly review	Log exists

Hard stops inside the illustration. Open Agent 10 flag on a document → no match-recommend. Header above the charter cap → human. ERP downtime → pause the clock; do not skip samples to “catch up.”

What this illustration does not include. Posting by the agent. Payment. Other entities. Wave 2. A savings checkpoint at week 6.

Northline (fictional): Matching commissioned L1 on 7 April 2026 after a shadow that slipped a week for Birmingham GR latency — the slip is the lesson.

Dependencies



Resource sketch (illustration)

Role	Phase 0–3	Phase 4–6	Phase 7–10
Practitioner / transformation	Heavy	Medium	Medium
AP Process Owner	Medium	Heavy	Heavy
Controls Lead	Medium	Heavy (flags)	Medium
Match / Ops leads	Medium	Heavy	Medium
Treasury	Light (holds)	Light	Medium if Agent 12
Internal Audit	Light	Observer	Observer
IT / IAM	Medium (access)	Light	Medium on extend

Hours are not specified as a promise. If the Process Owner cannot give Phase 4 a daily fifteen minutes, do not start Phase 4.

Document control

Field	Value
Toolkit	Evidence Room — AP Agent OS
Object	Implementation roadmap Phases 0–10
Status	Edition 1.0.0
Related	NINETY_DAY_PLAN.md, Testing/, Autonomy progression

Disclaimer — Draft for Counsel

Status: Working draft. **Not legal advice.** Intended for /disclaimer, checkout footers, and the zip. Counsel must confirm jurisdiction-specific wording.

Last updated: 20 September 2026

Lockup: EVIDENCE ROOM — AP AGENT OS

If this Disclaimer conflicts with a marketing sentence, **this Disclaimer controls** and the marketing sentence should be removed.

1. Nature of the product

Evidence Room — AP Agent OS is a **digital method and toolkit**. It contains written architectures, charters, frameworks, and examples to help an organisation **design, govern, and discuss** an Accounts Payable agent layer.

It is **not**:

- software that connects to, posts to, or operates your ERP
 - a payment service, money-transmission service, or bank
 - an audit, assurance, or attestation engagement
 - legal, tax, accounting, or regulatory advice
 - a certification that you meet any law, standard, or control framework
 - a fraud-detection, AML, or sanctions-screening product
 - a guarantee of any financial or operational result
-

2. Outcomes we do not promise

Without limitation, we do **not** promise or guarantee:

- savings, cost reduction, or **ROI**
- **fraud detection**, fraud prevention, or the absence of duplicate or abusive payments
- **regulatory compliance** (including but not limited to SOX, VAT/GST rules, employment, or data-protection law)
- **accounting accuracy** or correct postings
- **autonomous payment safety**
- cycle-time reduction
- “touchless AP” as an operating state
- that any model you later use will be correct

Any number in the materials labelled illustrative, fictional, or Northline Industrials is **not** a benchmark and **not** a target.

3. Payment authority

The method states that **payment authorisation and release remain human**. That is a design rule in a document. It does not itself stop a person or a vendor from wiring a model to a payment file. **You** must implement holds in your systems and policies.

We are not liable for payments you or your agents release.

4. Implementation is yours

You choose vendors, models, prompts, permissions, owners, and whether to commission any agent. You are responsible for testing, sampling, access control, and local policy.

The toolkit cannot see your invoices unless **you** paste them somewhere. Do not paste production data to us.

5. No professional relationship

Using the site or the files does not create an attorney-client, auditor-client, fiduciary, or similar relationship. Blueprint facilitation, if any, is scoped method design, not an audit opinion.

6. Third parties and names

Mentions of ERP or AP vendors are descriptive. No sponsorship or compatibility certification is implied unless a written agreement says so (none at the time of this draft).

We are **not** affiliated with Evidence Room LLC (forensic animation / evidence-room.net). See Trademark notes.

Lemon Squeezy and Stripe are independent. Their MoR and payment terms are theirs.

7. Site content

Resources, charts, and posts are educational and commercial description. They are not advice to buy or refrain from buying any security or to take any tax position.

8. Limitation pointer

Liability, to the extent it can be limited, is addressed in the Terms. **Counsel must write that limit.** Nothing here is an additional warranty.

9. Short footer (site and checkout)

Evidence Room — AP Agent OS is a digital method and toolkit. It is not an ERP, payment service, audit, or compliance certification. It does not detect fraud, authorise payments, or guarantee savings, accuracy, or ROI. You remain responsible for implementation and for all payment decisions.

Counsel checklist

- ☐ Conform to mandatory consumer wording
- ☐ Harmonise with insurance if any
- ☐ Confirm “not advice” is not over-claimed in Blueprint sales calls